
Survey of Consciousness Theory from Computational Perspective

At the Dawn of Artificial General Intelligence

Zihan Ding*
Princeton University
zihand@princeton.edu

Xiaoxi Wei*
Imperial College London
xiaoxi.wei18@imperial.ac.uk

Yidan Xu*
University of Michigan
yidanxu@umich.edu

Abstract

Human consciousness has been a long-lasting mystery for centuries, while machine intelligence and consciousness is an arduous pursuit. Researchers have developed diverse theories for interpreting the consciousness phenomenon in human brains from different perspectives and levels. This paper surveys several main branches of consciousness theories originating from different subjects including information theory, quantum physics, cognitive psychology, physiology and computer science, with the aim of bridging these theories from a computational perspective. It also discusses the existing evaluation metrics of consciousness and possibility for current computational models to be conscious. Breaking the mystery of consciousness can be an essential step in building general artificial intelligence with computing machines.

Contents

1	Introduction	3
1.1	A Platonic Dialogue About Human Consciousness	3
1.2	Definition of Consciousness	5
1.3	Measurement of Consciousness	6
1.4	Consciousness and Intelligence	6
1.5	Consciousness and Free Will	7
1.6	Consciousness while Asleep	9
1.7	Overview of Consciousness Theories	10
2	Information Integration Theory	10
2.1	Information Entropy	10
2.2	Basics Concepts of IIT	11
2.3	Measurement of Information Integration	13
2.4	Biological Evidence	15
3	Consciousness as a State of Matter	15
3.1	Basics of Quantum Mechanics	16

*The authors contributed equally.

Preprint with all copyright reserved.

从计算视角审视意识理论综述

在通用人工智能曙光之际

丁子涵*
普林斯顿大学
zihand@princeton.edu

魏晓曦*
伦敦帝国学院
xiaoxi.wei18@imperial.ac.uk

徐一丹*
密歇根大学
yidanxu@umich.edu

Abstract

人类意识是几个世纪以来一个持久的谜团，而机器智能与意识则是一项艰巨的追求。研究人员已从不同视角和层面发展出多种理论来解释人脑中的意识现象。本文综述了源自信息论、量子物理学、认知心理学、生理学和计算机科学等不同学科的几种主要意识理论分支，旨在从计算视角连接这些理论。文章还讨论了现有的意识评估指标以及当前计算模型具备意识的可能性。揭开意识之谜可能是利用计算机构建通用人工智能的关键一步。

目录

1	Introduction	3
1.1	Definition of Consciousness	5
1.2	Consciousness and Intelligence	6
1.3	Consciousness while Asleep	9
1.4	Overview of Consciousness Theories	10
2	Information Integration Theory	10
2.1	Information Entropy	10
2.2	Measurement of Information Integration	13
2.3	Biological Evidence	15
3	Consciousness as a State of Matter	15
3.1	Basics of Quantum Mechanics	16
3.2	Integration Principle	17
3.3	Independence Principle	18
3.4	Dynamics principle	18

*作者贡献等同。

Preprint with all copyright reserved.

3.2	Integration Principle	17
3.3	Independence Principle	18
3.4	Dynamics principle	18
4	Orchestrated Objective Reduction Theory	19
4.1	Consciousness as Orchestrated Objective Reduction	19
4.2	Free Will in Neurons	20
4.3	Diósi-Penrose Objective Reduction	20
4.4	Evidence for Objective Reduction of Quantum State	20
5	Global Workspace Theory	20
5.1	The Theatre of Consciousness	20
5.2	Computational Models of GWT	22
6	Higher-Order Theories	23
6.1	Higher-Order Perception Theory	23
6.2	Higher-Order Thought Theory	24
6.3	Self-Representational Theory	25
6.4	Other theories and perspectives in HOT	25
7	Attention Schema Theory	25
7.1	Formulation	25
7.2	<i>I</i> -consciousness and <i>M</i> -consciousness	26
7.3	AST as a Unification of GWT and HOT	27
8	Conscious Turing Machine	27
8.1	Formulation	27
8.2	CTM for Consciousness	28
8.3	Relationships with Other Theories	28
9	Physiological Evaluation Metric of Consciousness	29
9.1	Metrics Based on Electrical Signals	29
9.2	Metrics Based on Behaviors	31
10	Look Ahead: Can Computational Models Be Conscious?	31
10.1	Background	31
10.2	Large Language Model	33
10.3	Emerging Intellectual Capability of LLM - Turing Test	36
10.4	Consciousness of LLM	37
11	Concluding Remarks	42

4	Orchestrated Objective Reduction Theory	19
4.1	Consciousness as Orchestrated Objective Reduction	19
4.2	Free Will in Neurons	19
4.3	Diósi-Penrose Objective Reduction	20
4.4	Evidence for Objective Reduction of Quantum State	21
5	Global Workspace Theory	21
5.1	The Theatre of Consciousness	21
5.2	Computational Models of GWT	22
6	Higher-Order Theories	23
6.1	Higher-Order Perception Theory	24
6.2	Higher-Order Thought Theory	24
6.3	Self-Representational Theory	24
6.4	Other theories and perspectives in HOT	25
7	Attention Schema Theory	25
7.1	Formulation	25
7.2	<i>I</i> -意识与 <i>M</i> -意识	26
7.3	AST as a Unification of GWT and HOT	26
8	Conscious Turing Machine	27
8.1	Formulation	27
8.2	CTM for Consciousness	28
8.3	Relationships with Other Theories	28
9	Physiological Evaluation Metric of Consciousness	28
9.1	Metrics Based on Electrical Signals	28
9.2	Metrics Based on Behaviors	30
10	Look Ahead: Can Computational Models Be Conscious?	30
10.1	Background	30
10.2	Large Language Model	31
10.3	Emerging Intellectual Capability of LLM - Turing Test	34
10.4	Consciousness of LLM	35

1 Introduction

Consciousness is a complex and elusive phenomenon that remains one of the greatest mysteries of science. It has been the subject of philosophical inquiry for centuries, and more recently, scientific investigation. We, humans, are not clear about why and how consciousness exists in our brains or even hold diverged opinions on whether we truly have consciousness. Existing consciousness theories have provided different interpretations of the human conscious process.

This paper provides a comprehensive exploration of the theoretical foundations of consciousness from interdisciplinary perspectives. Chapter 1 endeavors to characterize the concepts related to consciousness, by differentiating the consciousness from others such as awareness, arousal, and wakefulness. This chapter further emphasizes the importance and difficulties of the human consciousness problem, with the aim of drawing attention from different research communities to jointly investigate this problem. Chapters 2, 3, and 4 elucidate the mathematical and physical underpinnings of consciousness. Specifically, Chapter 2 introduces **Information Integration Theory**, which outlines the conditions for information entropy required by a conscious entity, offering insights into the informational characteristics that consciousness might exhibit. Chapter 3 and 4 discusses **Consciousness as a State of Matter and Orchestrated Objective Reduction Theory**, both approaching consciousness problem from a physics standpoint. These two chapters discuss the specific features that consciousness, as a state of matter, should possess, along with the principles that some quantum theorists propose as the basis for the generation of consciousness. Subsequent Chapters 5, 6, 7 and 8 survey several influential theories of consciousness and succinctly summarize research on computational models associated with each theory, including the **Global Workspace Theory** (5), **High-Order Theories** (6), **Attention Schema Theory** (7) and **Conscious Turing Machine** (8). In Chapter 9, a brief overview of contemporary biomedical measurement methods for consciousness, grounded in electrophysiological signals and behavioral indicators, is provided. In the final Chapter 10, we engage in a discursive examination of artificial intelligence (AI) consciousness, particularly delving into the question of whether Large Language Models as instances of advanced computational models possess consciousness and exploring the necessary and sufficient conditions for AI consciousness.

In summary, this paper offers a comprehensive review of consciousness from various subjects encompassing information theory, quantum physics, cognitive psychology, physiology, and computer science, with the aim of bridging these theories from a computational perspective for building future AI consciousness.

To give the readers a primal taste of this topic, a dialogue revolving the consciousness problem is provided in the following section.

1.1 A Platonic Dialogue About Human Consciousness

One day, I met two PhD students in the modern world discussing the consciousness problem within the human brain. This conversation starts with the relationship between consciousness and the physical world, discussing thoughts about what consciousness is. The confusion in this conversation serves as the motivation of this study and introduces the essential problems that this article aims to address. Here is the conversation.

Athena: Hey, I was looking at the results of the double-slit experiment, but I'm struggling to understand what determines the photons to choose which slit to go through, what happens here?

Galileo: Hey, it's just the wave-particle duality of the photons, and the state of the photon collapses only when you observe it on the screen. It is just random. Don't dig too much into it.

Athena: What do you mean by 'random'? Does this world ultimately have a random essence? Even Einstein says that God does not play dice.

Galileo: Your measurement affects its state, which determines the slit it goes through in the double-slit experiment.

Athena: So my mental status determine the state of the photon? Its state may change if I choose to observe it in a different way, say later for 0.00001 seconds than the intended observation.

Galileo: Yeah, maybe. And your mental status is also a stochastic process, right? You have your free will or consciousness.

Athena: Consciousness. Do you really believe in its existence? What if I'm deterministic, fully determined by the underlying physical and chemical rules?

11 Concluding Remarks

39

1 Introduction

意识是一种复杂且难以捉摸的现象，至今仍是科学界最伟大的谜团之一。数个世纪以来，它一直是哲学探究的主题，近年来更成为科学研究的对象。我们人类并不清楚意识为何以及如何存在于我们的大脑中，甚至对于我们是否真正拥有意识也存在分歧的观点。现有的意识理论对人类意识过程提供了不同的解释。

本文从跨学科的视角，对意识的理论基础进行了全面的探讨。第1章致力于界定与意识相关的概念，通过将意识与觉知、唤醒、清醒等其他状态区分开来。本章进一步强调了人类意识问题的重要性和困难性，旨在吸引不同研究领域的关注，共同研究这一问题。第2、3和4章阐述了意识的数学和物理基础。具体而言，第2章介绍了 **Information Integration Theory**，它概述了一个有意识的实体所需满足的信息熵条件，为意识可能表现出的信息特征提供了见解。第3章和4章讨论了 **Consciousness as a State of Matter** 和 **Orchestrated Objective Reduction Theory**，两者均从物理学角度探讨意识问题。这两章讨论了意识作为一种物质状态应具备的具体特征，以及一些量子理论家提出的作为意识产生基础的物理解释。随后的第5、6、7和8章综述了几种有影响力的意识理论，并简要总结了与每种理论相关的计算模型研究，包括 **Global Workspace Theory** (5)、**High-Order Theories** (6)、**Attention Schema Theory** (7) 和 **Conscious Turing Machine** (8)。在第9章中，简要概述了基于电生理信号和行为指标的当代意识生物医学测量方法。在最后一章10中，我们对人工智能 (AI) 意识进行了探讨性审视，特别深入探讨了作为先进计算模型实例的大型语言模型是否拥有意识的问题，并探索了AI意识的必要和充分条件。

总而言之，本文从信息论、量子物理学、认知心理学、生理学和计算机科学等多个学科领域，对意识进行了全面的综述，旨在从计算视角整合这些理论，为构建未来的AI意识奠定基础。为了让读者对这个主题有一个初步的了解，以下部分提供了一段围绕意识问题的对话。

有一天，我在现代世界中遇到了两位博士生，他们正在讨论人脑中的意识问题。这段对话始于意识与物理世界的关系，探讨了对意识本质的思考。对话中的困惑构成了本研究的动机，并引出了本文旨在解决的核心问题。以下是对话内容。

Athena: 嘿，我正在看双缝实验的结果，但我很难理解是什么决定了光子选择通过哪条狭缝，这里到底发生了什么？

Galileo: 嘿，这只是光子的波粒二象性，光子的状态只有在你观察屏幕时才会坍缩。这完全是随机的。别钻牛角尖了。

Athena: “随机”是什么意思？这个世界最终具有随机的本质吗？连爱因斯坦都说上帝不掷骰子。

Galileo: 你的测量影响了它的状态，这决定了它在双缝实验中通过哪条狭缝。

Athena: 所以我的精神状态决定了光子的状态？如果我选择以不同的方式观察它，比如比预定观察时间晚 0.00001 秒，它的状态可能会改变吗？

Galileo: 也许吧。而且你的精神状态也是一个随机过程，对吧？你拥有自由意志或意识。

Athena: 意识。你真的相信它的存在吗？如果我是决定论的，完全由底层的物理和化学规律所决定呢？

Galileo: 但量子力学的基本原理假设光子，以及你身体里的粒子，包括你大脑中的粒子，仍然具有概率性的状态，对吧？所以你也遵循一个随机过程，而非决定论的过程。

Galileo: But quantum mechanics assumes the basic principle that the photons, and particles in your body, including your brain, still have a probabilistic state, right? So you are also following a stochastic process instead of a deterministic one.

Athena: You are right. The state collapse process is assumed to have true randomness, and this process may happen in my brain, affecting my decision, and also which slit these photons may go through. But I still don't quite believe in the existence of my consciousness. The state decoherence may happen for a system as large as my brain, so there is no quantum property and I may still be deterministic.

Galileo: I think the existence of consciousness is a belief that differs from person to person.

Athena: Wait, but what is consciousness? We cannot discuss its existence without a clear definition. The consciousness seems to describe the subjective experience but without a rigorous definition. I'm wondering where and when the conscious process happens.

Galileo: I guess it's in the cerebral cortex or thalamus. However, for such a phenomenon to happen, I would say most functional parts of the brain may have a collaborative process to trigger the consciousness.

Athena: I agree. Also if we assume a stone, or a tree, cannot be as conscious as a human, then the system has to have a certain level of computational power.

Galileo: Also, I cannot imagine a system having consciousness without memory.

Athena: I may disagree on that. I think a patient with brain damage of losing his memory still has consciousness.

Galileo: May be an evidence. Do you think consciousness retains during sleep?

Athena: This can be a complicated problem. You know there are multiple stages during sleep, including non-rapid eye movement (NREM) and rapid eye movement (REM). The levels of consciousness can be different for different stages.

Galileo: Agreed. The dream happens during the REM, right? It feels like I have consciousness during the dream, so I guess I'm conscious at REM. But it's still quite different from the wakefulness in experience.

Athena: I'm not sure, it's quite complicated. But the essential difference between dream and wakefulness is that during sleep there is no external sensory input to the brain, all the experiences are fake and fabricated by the brain itself. If the consciousness happens during the dream, it indicates that external sensory inputs may not be necessary for the system to be conscious.

Galileo: I guess so. It seems the consciousness is quite close to the subjective experience, even if I'm not sure if subjective experience is real existence or a fake hallucination by humans.

Athena: It seems the 'subjective experience' is an equivalent phrase of consciousness in some books. But this definition is still unclear to me. Also, the 'real' and 'fake' problem is also unclear to me. If our measurement of the world can affect its state, what does it even mean by a 'real' observation of the world and a 'fake' one? After all, they are just the mirrored signals in our brains, which can not directly represent the state of the real world.

Galileo: Well, that's a very philosophically pessimistic perspective. Speaking of the definition of consciousness, if we assume consciousness is defined as 'subjective experience', it seems not very related to another concept called intelligence. Nowadays, people are building artificial intelligence in computers, is it also consciousness? This could be an interesting problem! Do you think it's just about the Turing test [Turing, 2009]?

Athena: I don't think so. The focus of that paper is discussing whether the Turing machine can achieve human-level intelligence. Please notice the difference in the words here, it's intelligence but not consciousness! I assume a program passing the Turing test only means that it's as intelligent as a human, but not conscious.

Galileo: Thanks for reminding me of that. It seems that intelligence and consciousness are two different things. But remember that at the beginning of the discussion, we mentioned that consciousness may require a certain level of computational power, and this computational power may be phrased as intelligence.

Athena: Maybe, the intelligent property seems to be a necessary condition for the consciousness to emerge, but I'm not sure to which extent the consciousness requires the intelligence to be.

Galileo: Yeah, this is undiscovered and could be a good research topic. However, if a creature is already intelligent enough to survive in the world, why does it still require consciousness?

Athena: 你说得对。状态坍塌过程被假定具有真正的随机性，这个过程可能发生在我的大脑中，影响我的决定，也影响这些光子可能通过哪条狭缝。但我仍然不太相信我的意识存在。对于像我大脑这样大的系统，可能会发生状态退相干，因此没有量子特性，我可能仍然是决定论的。

Galileo: 我认为意识的存在是一种因人而异的信念。

Athena: 等等，但意识是什么？没有一个清晰的定义，我们无法讨论它的存在。意识似乎描述了主观体验，但缺乏严格的定义。我在想意识过程发生在何处、何时。

Galileo: 我猜是在大脑皮层或丘脑。然而，要让这种现象发生，我认为大脑的大部分功能部分可能需要一个协作过程来触发意识。

Athena: 我同意。另外，如果我们假设一块石头或一棵树不可能像人类一样有意识，那么这个系统必须具有一定水平的计算能力。

Galileo: 还有，我无法想象一个没有记忆的系统会拥有意识。

Athena: 我可能不同意这一点。我认为一个因脑损伤而失去记忆的病人仍然具有意识。

Galileo: 这可能是一个证据。你认为睡眠期间意识会保留吗？

Athena: 这可能是一个复杂的问题。你知道睡眠有多个阶段，包括非快速眼动睡眠和快速眼动睡眠。不同阶段的意识水平可能不同。

Galileo: 同意。梦发生在快速眼动睡眠期，对吧？感觉我在梦中是有意识的，所以我猜我在快速眼动睡眠期是有意识的。但这在体验上仍然与清醒状态有很大不同。**Athena:** 我不太确定，这相当复杂。但梦境与清醒状态的根本区别在于，睡眠期间大脑没有外部感官输入，所有体验都是虚假的，由大脑自身构建。如果意识出现在梦境中，这表明系统产生意识可能并不需要外部感官输入。

Galileo: 我想是的。意识似乎与主观体验非常接近，即使我不确定主观体验是真实存在还是人类制造的虚假幻觉。

Athena: 在某些书籍中，“主观体验”似乎是意识的同义短语。但这个定义对我来说仍然不清晰。此外，“真实”与“虚假”的问题我也不清楚。如果我们对世界的测量能影响其状态，那么“真实”观察世界与“虚假”观察究竟意味着什么？毕竟，它们只是我们大脑中的镜像信号，并不能直接代表真实世界的状态。

Galileo: 嗯，这是一种非常哲学悲观的观点。说到意识的定义，如果我们假设意识被定义为“主观体验”，它似乎与另一个称为智能的概念关系不大。如今，人们正在计算机中构建人工智能，它也是意识吗？这可能是个有趣的问题！你认为这仅仅是关于图灵测试吗 [Turing, 2009]？

Athena: 我不这么认为。那篇论文的重点是讨论图灵机是否能达到人类水平的智能。请注意这里用词的差异，是智能而非意识！我认为一个程序通过图灵测试只意味着它和人类一样智能，但并非有意识。

Galileo: 感谢提醒。看来智能和意识是两件不同的事。但记得在讨论开始时，我们提到意识可能需要一定水平的计算能力，而这种计算能力或许可以表述为智能。

Athena: 也许，智能属性似乎是意识涌现的必要条件，但我不确定意识对智能的需求程度有多高。

Galileo: 是的，这尚未被发现，可能是一个很好的研究课题。然而，如果一个生物已经足够智能以在世界上生存，为什么它还需要意识呢？

Athena: 好问题。根据达尔文的进化论，世界上现存生物应仅展现那些通过自然选择有利于其生存的技能。如果意识存在于人类中，这表明它对人类在环境中生存有某些益处，而没有意识的个体在历史中被淘汰了。但树木和花朵仍然存在，这确实让我困惑。

Athena: Good question. According to Darwin’s theory of evolution, the existing creatures in the world should only exhibit those skills in favor of their survival through natural selection. If consciousness exists in humans, it indicates that it has some benefits for humans to survive in environments, and those without it are erased over history. But the trees and flowers still exist, which really troubles me.

Galileo: Trees and flowers are of different species from humans. Maybe for certain species, it requires to have consciousness, like animals.

Athena: Well, but I still feel unclear about why consciousness exists in humans or other animals, if assumed to exist.

Athena: It seems like consciousness theory can be a very complicated subject with correlations to some very different subjects, like physics, biology, computer science, neuroscience, information theory, etc. It can be very difficult. But let’s start to investigate at least!

The above conversation is a microcosm of the discussions by the authors of the paper for starting the investigation of the consciousness problem. We will start the discussion with the definition of consciousness (Sec. 1.2), and the relationship between consciousness and intelligence (Sec. 1.4), consciousness and free will (Sec. 1.5). Then we have a brief overview of the existing consciousness theories (Sec. 1.7) and a further discussion about consciousness while asleep (Sec. 1.6).

1.2 Definition of Consciousness

Consciousness, Awareness, Wakefulness and Arousal: Consciousness is a complex and multi-faceted concept that has been studied extensively in the fields of neuroscience, psychology, and philosophy. Existing research typically defines consciousness as comprising two main components: arousal (wakefulness) and awareness (subjective experience) [Lendner et al., 2020]. Arousal refers to the overall state of alertness or wakefulness, while awareness refers to the subjective experience of perceiving and interpreting sensory information.

Typically, arousal is indicated by the opening of the eyes, while awareness is inferred by the ability to follow commands [Lee et al., 2022]. However, in certain instances, such as during dreaming, subjective experience can still occur despite the absence of full wakefulness. Consciousness is considered to be absent during sleep or anesthesia, but in some cases, it can still be present, depending on the level of arousal and awareness. Consciousness, awareness, wakefulness, and arousal are related but distinct concepts in the study of the human mind and brain. We provide descriptive definitions for each concept according to the existing literature as follows:

Consciousness refers to the subjective experience of being aware of one’s thoughts, feelings, sensations, and surroundings. It is often described as the state of being awake and aware of one’s surroundings and internal states. As an important concept in discussing the consciousness problem, *qualia* refers to the subjective and personal experience of sensory information, such as the way we perceive colors, sounds, tastes, and smells. It is the subjective experience of sensory perception that cannot be objectively measured or observed by others.

Awareness refers to the ability to perceive, process, and comprehend information from one’s environment or inner experience. It includes both conscious and unconscious processes and can range from simple sensory perception to complex cognitive processes such as attention, memory, and reasoning.

Arousal refers to the level of responsiveness of the brain and body to internal and external stimuli. It is a physiological state that ranges from a state of low arousal, such as drowsiness or relaxation, to high arousal, such as intense excitement or fear.

Wakefulness refers to the state of being awake and not asleep. It is a physiological state characterized by the presence of the electrical activity of the brain and the ability to respond to external stimuli.

In summary, consciousness is a subjective experience of being aware of one’s thoughts, feelings, and surroundings, while awareness is the ability to perceive, process, and comprehend information. Wakefulness is a physiological state characterized by being awake, which is *usually regarded identically as awareness*. Arousal is the level of responsiveness to stimuli. Awareness and arousal are necessary conditions for consciousness, but they are not sufficient to arouse the consciousness, and consciousness can occur in the absence of full awareness and high arousal levels.

Galileo: 树木和花朵与人类是不同的物种。也许对于某些物种，比如动物，需要拥有意识。

Athena: 嗯，但如果假设意识存在，我仍然不清楚为什么它存在于人类或其他动物中。

Athena: 意识理论似乎可以是一个非常复杂的学科，与一些非常不同的学科相关，如物理论、生物学、计算机科学、神经科学、信息论等。这可能非常困难。但让我们至少开始探索吧！

上述对话是论文作者们启动意识问题探讨的一个缩影。我们将从意识的定义（第 1.1 节）、意识与智能的关系（第 1.2 节）、意识与自由意志的关系（第 1.2 节）开始讨论。随后，我们将简要概述现有的意识理论（第 1.4 节），并进一步探讨睡眠时的意识状态（第 1.3 节）。

1.1 Definition of Consciousness

Consciousness, Awareness, Wakefulness and Arousal: 意识是一个复杂且多面的概念，在神经科学、心理学和哲学领域已被广泛研究。现有研究通常将意识定义为包含两个主要组成部分：唤醒（清醒度）和觉知（主观体验）[Lendner et al., 2020]。唤醒指的是警觉或清醒的整体状态，而觉知指的是感知和解释感官信息的主观体验。

通常，睁眼表明存在唤醒，而执行指令的能力则被用来推断觉知的存在 [Lee et al., 2022]。然而，在某些情况下，例如做梦时，尽管缺乏完全的清醒，主观体验仍然可以发生。在睡眠或麻醉期间，意识通常被认为是缺失的，但在某些情况下，根据唤醒和觉知的水平，意识可能仍然存在。意识、觉知、清醒和唤醒是研究人类心智与大脑时相关但又有所区别的概念。我们根据现有文献为每个概念提供描述性定义如下：

Consciousness 指的是意识到自己的思想、情感、感觉和周围环境的主观体验。它常被描述为清醒并意识到自身周围环境和内在状态的状态。作为讨论意识问题的一个重要概念，*qualia* 指的是对感官信息的主观和个人体验，例如我们感知颜色、声音、味道和气味的方式。它是感官知觉的主观体验，无法被他人客观测量或观察。

Awareness 指的是感知、处理和理解来自环境或内在体验信息的能力。它包括有意识和无意识的过程，范围可以从简单的感官知觉到复杂的认知过程，如注意、记忆和推理。

Arousal 指的是大脑和身体对内外刺激的反应水平。它是一种生理状态，范围从低唤醒状态（如因倦或放松）到高唤醒状态（如极度兴奋或恐惧）。

Wakefulness 指的是清醒而非睡眠的状态。它是一种生理状态，其特征是存在大脑电活动以及能够对外部刺激作出反应。

总之，意识是对自身思想、情感和周围环境有所觉察的主观体验，而觉知是感知、处理和理解信息的能力。清醒是一种以醒着为特征的生理状态，即 *usually regarded identically as awareness*。唤醒是对刺激的反应水平。觉知和唤醒是意识的必要条件，但它们不足以唤起意识，并且意识可以在缺乏完全觉知和高唤醒水平的情况下发生。

近期研究已开发出有效的人类意识测量方法 [Seth et al., 2008, Demertzi et al., 2017]，例如基于电信号的指标，如扰动复杂性指数（PCI）[Casali et al., 2013] 和脑电双频指数（BIS）[Rosow and Manberg, 2001, Johansen, 2006]，以及基于行为的指标，如格拉斯哥昏迷量表（GCS）[Jones, 1979, Sternbach, 2000]、昏迷恢复量表修订版（CRS-R）[Giacino et al., 2004] 和全面无反应性量表（FOUR）[Wijdicks et al., 2005]。各项生理评估指标的细节将在第 9 节讨论。在以下段落中，我们以 PCI 指标为例，区分不同意识状态下的唤醒与觉知概念。

Conscious State	Arousal	Awareness
Healthy wakefulness	high	high
REM ¹ sleep with dreams	low	high
NREM ² sleep without dreams	low	low
Anesthesia induced with ketamine	low	high
Anesthesia induced with propofol or xenon	low	low
Minimally conscious state	high	high
Unresponsive wakefulness syndrome	high	low

¹ REM: rapid eye movement

² NREM: non-rapid eye movement

Table 1: The arousal and awareness levels of consciousness under different states (results adapted from Lee et al. [2022])

1.3 Measurement of Consciousness

Recent studies have developed effective measures of human consciousness [Seth et al., 2008, Demertzi et al., 2017], such as the electrical signal-based metrics like Perturbational Complexity Index (PCI) [Casali et al., 2013] and Bispectral Index (BIS) [Rosow and Manberg, 2001, Johansen, 2006] and behavior-based metrics like the Glasgow Coma Scale (GCS) [Jones, 1979, Sternbach, 2000], The Coma Recovery Scale-Revised (CRS-R) [Giacino et al., 2004] and Full Outline of Unresponsiveness (FOUR) [Wijdicks et al., 2005]. Details for each physiological evaluation metric are discussed in Sec. 9. In the following paragraphs, we take the PCI metric as an example to distinguish the concepts of arousal and awareness in different conscious states.

The ability to accurately measure consciousness has important implications for understanding and treating conditions that affect consciousness, such as coma, anesthesia, and brain injury. PCI was developed from electroencephalographic (EEG) responses to direct and noninvasive cortical perturbation with transcranial magnetic stimulation (TMS). The PCI quantifies the complexity of deterministic patterns of significant cortical activation evoked by TMS to derive an empirical cutoff that reliably discriminates between unconsciousness and consciousness in various states, including REM sleep, wakefulness, ketamine-induced anesthesia, and conscious brain-injured patients.

In the PCI study by Casali et al. (2013) [Casali et al., 2013], arousal and awareness levels serve as indicators of the degree of human consciousness. The arousal and awareness levels of consciousness under different states are summarized in Table 1, which shows under different states the consciousness will appear differently in human brains [Lee et al., 2022]. Compared with the full wakefulness of a normal person, the sleeping stages like REM or NREM have lower levels of arousal or awareness and are commonly believed to be not as “conscious” as a wakeful state. There are pharmacology approaches to achieve similar low levels of arousal and awareness with anesthesia, leading to incomplete conscious states of a human. From a pathology view, existing patients with a minimally conscious state (MCS) will appear to have relatively high levels of arousal and awareness but still less consciousness, which is evidence of the fact that arousal and awareness are necessary but not sufficient conditions for consciousness. People with unresponsive wakefulness syndrome will have high arousal but low awareness.

1.4 Consciousness and Intelligence

In the well-known paper by Alan Turing, he made a comment on the consciousness argument against Turing machine [Turing, 2009]:

"I do not wish to give the impression that I think there is no mystery about consciousness. There is, for instance, something of a paradox connected with any attempt to localise it."

Considering the question in the paper is whether a machine can think like a human, Turing proposed the famous imitation game as a way to test machine intelligence. Intelligence and consciousness are widely considered two different properties of the brain. In *Life 3.0* [Tegmark, 2018] by Max Tegmark, intelligence is defined as the ability to accomplish complex goals, and consciousness is defined as subjective experience. The consciousness seems to be more mysterious than intelligence, and harder

Conscious State	Arousal	Awareness
Healthy wakefulness	high	high
REM ¹ sleep with dreams	low	high
NREM ² sleep without dreams	low	low
Anesthesia induced with ketamine	low	high
Anesthesia induced with propofol or xenon	low	low
Minimally conscious state	high	high
Unresponsive wakefulness syndrome	high	low

¹ REM: 快速眼动

² NREM: 非快速眼动

表 1: 不同状态下意识的唤醒与觉知水平（结果改编自 Lee et al. [2022]）

准确测量意识的能力对于理解和治疗影响意识的病症（如昏迷、麻醉和脑损伤）具有重要意义。PCI 是基于经颅磁刺激（TMS）对大脑皮层进行直接、无创扰动时产生的脑电图（EEG）响应而开发的。PCI 量化了由 TMS 诱发的重要皮层激活的确定性模式的复杂性，从而得出一个经验性阈值，能够可靠地区分不同状态下的无意识与有意识，包括快速眼动睡眠、清醒状态、氯胺酮诱导的麻醉状态以及有意识的脑损伤患者。

在 Casali 等人（2013 年）[Casali et al., 2013] 的 PCI 研究中，唤醒水平和觉知水平被用作人类意识程度的指标。表 1 总结了不同状态下的意识唤醒与觉知水平，该表显示了在不同状态下，意识在人脑中的表现会有所不同 [Lee et al., 2022]。与正常人完全清醒的状态相比，如快速眼动或非快速眼动等睡眠阶段的唤醒或觉知水平较低，通常被认为不如清醒状态那样“有意识”。药理学方法（如麻醉）可以达到类似的低唤醒和低觉知水平，导致人类处于不完全的意识状态。从病理学角度看，现有的微意识状态（MCS）患者可能表现出相对较高的唤醒和觉知水平，但意识程度仍然较低，这证明了唤醒和觉知是意识的必要条件，而非充分条件。无反应性觉醒综合征患者则具有高唤醒但低觉知的特征。

1.2 Consciousness and Intelligence

在艾伦·图灵那篇著名的论文中，他对反对图灵机的意识论证评论道 [Turing, 2009]:

"I do not wish to give the impression that I think there is no mystery about consciousness. There is, for instance, something of a paradox connected with any attempt to localise it."

考虑到论文中的问题是机器能否像人类一样思考，图灵提出了著名的模仿游戏作为测试机器智能的方法。智能和意识被广泛认为是大脑的两种不同属性。在马克斯·泰格马克的 *Life 3.0* [Tegmark, 2018] 中，智能被定义为完成复杂目标的能力，而意识被定义为主观体验。意识似乎比智能更为神秘，也更难测量。如图 1（最初见于 Tegmark [2018]）中描绘的意识“金字塔”所示，与智能相关的问题是最简单的，大卫·查默斯也持此观点。这类问题通常不需要考虑实验对象的主观体验。一个困难的问题是找到可物理解释的特征来区分有意识和无意识的过程。下一个层次的问题是意识如何发生以及决定因素是什么。最后且最困难的问题则涉及解释意识的存在，或者说，为什么任何系统中会存在意识？

to measure. As depicted in the consciousness “pyramid” in Fig. 1 (originally in Tegmark [2018]), the intelligence-related problems are the easiest, which is also claimed by David Chalmers. This type of problem typically does not require to consider the subjective experience of the experimental subjects. A hard problem is to find the physically interpretable features for distinguishing conscious and unconscious processes. The next level question is how the consciousness happens and what the determined factors are. The final and hardest problems related to the explanation of the existence of consciousness, or why consciousness exists in any system?

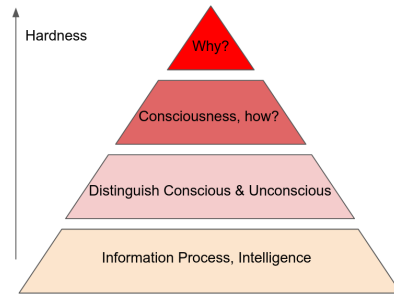


Figure 1: The hardness of different levels of the problems related to a conscious mind.

For the reasons for the existence of consciousness in human brains, a quote from Stephen Wolfram mentions the limitation of self-modeling and time-persistence are two key factors for humans to have consciousness:

"The fact that we have coherent consciousness is a consequence of two things: 1. That we are computationally bounded, so the universe does not contain enough resources for us to construct a complete model of ourselves. 2. That we believe that we are persistent in time, and hence assume constancy where there is none."

Consciousness is a complex and multifaceted phenomenon that has been studied by philosophers, psychologists, neuroscientists, and others for a long history. Even so, consciousness remains mysterious for modern society, and people refer to it as *the hard problem of consciousness*, which was initially proposed by David Chalmers in 1995 [Chalmers, 1995, 1997]. It is generally understood as the experience or awareness of subjective mental states such as thoughts, perceptions, emotions, and sensations. The hard problem indicates the reasons for the existence of such subjective experiences in human minds. Consciousness is closely linked to the functioning of the brain, but its precise nature and mechanisms are still not fully understood. Some theories suggest that consciousness emerges from the integration of sensory information, while others propose that it is an intrinsic property of the universe or a fundamental aspect of reality itself. The quote by Stephen Wolfram suggests that the experience of coherent consciousness may depend on our computational limitations and our assumption of temporal persistence. The argument suggests that we believe we are persistent beings that exist over time, even though there is no actual constancy in the universe.

1.5 Consciousness and Free Will

Free will is defined as the ability that humans to make choices and decisions that are not solely determined by biological, environmental, or external factors. Whether humans have free will is highly controversial and unknown. However, this concept is believed to be closely related to consciousness. In Fig. 2, we depict an architecture unifying several consciousness theories to be introduced later in this article, as well as illustrating the relationship between consciousness and free will. In Fig. 2, the human brain has interactions among the low-level modules and consciousness module, where the conscious experiences happen in human brains [Baars, 2003, Baars and Franklin, 2003, 2007,

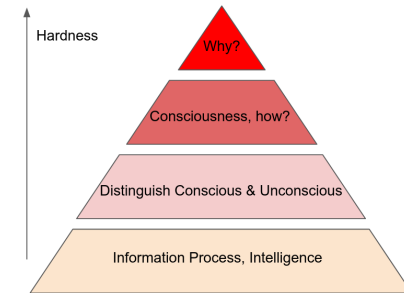


图 1: 与意识心智相关的不同层次问题的困难程度。

关于人类大脑中存在意识的原因，斯蒂芬·沃尔夫勒姆的一段引述提到，自我建模的局限性和时间持久性是人类拥有意识的两个关键因素：

"The fact that we have coherent consciousness is a consequence of two things: 1. That we are computationally bounded, so the universe does not contain enough resources for us to construct a complete model of ourselves. 2. That we believe that we are persistent in time, and hence assume constancy where there is none."

意识是一种复杂且多层面的现象，长期以来一直受到哲学家、心理学家、神经科学家及其他领域研究者的探索。即便如此，意识对现代社会而言仍然是个谜，人们将其称为 *the hard problem of consciousness*，这一概念最初由大卫·查默斯于 1995 年提出 [Chalmers, 1995, 1997]。它通常被理解为对思想、感知、情感和感觉等主观心理状态的体验或觉知。困难问题指出了人类心智中存在此类主观体验的原因。意识与大脑的功能密切相关，但其确切本质和机制仍未完全明了。一些理论认为意识产生于感觉信息的整合，而另一些理论则提出它是宇宙的内在属性或现实本身的基本方面。斯蒂芬·沃尔夫勒姆的引述表明，连贯意识的体验可能取决于我们的计算局限性和我们对时间持久性的假设。该论点认为，我们相信自己是在时间中持续存在的存在，尽管宇宙中并无实际的恒常性。

自由意志被定义为人类做出选择和决策的能力，这些选择和决策并非完全由生物、环境或外部因素所决定。人类是否拥有自由意志是极具争议且未知的。然而，这一概念被认为与意识密切相关。在图 2 中，我们描绘了一个统一了本文后续将介绍的几种意识理论的架构，同时也阐明了意识与自由意志之间的关系。在图 2 中，人脑的低层级模块与意识模块之间存在交互，而意识体验就发生在人脑中 [Baars, 2003, Baars and Franklin, 2003, 2007, 2009, Locke, 1948, Armstrong, 1981, Armstrong and Malcolm, 1985, Lycan, 1996, Armstrong, 2002, Lycan, 2004, Rosenthal, 2009, 2012, 2004, Byrne, 1997, Brown et al., 2019, Graziano and Webb, 2015, Graziano et al., 2020]。低层级模块包括外部处理器、内部处理器和记忆。外部处理器负责人脑的输入和输出，包括图像处理器、声音处理器、味觉处理器、嗅觉处理器、触觉处理器、运动激活器、言语模块等。每个模块处理输入信息，并将处理后的信号输出到其他部分。内部处理器包括逻辑处理器、语言处理器等。每个处理器都会处理来自外部处理器的输出信息，或在没有外部输入的情况下自发地进行处理。所有这些模块都有能力与其他模块及记忆进行通信，以实现预期目标，并且其中大多数模块会产生中间输出，作为其他模块的输入。更重要的是，

2009, Locke, 1948, Armstrong, 1981, Armstrong and Malcolm, 1985, Lycan, 1996, Armstrong, 2002, Lycan, 2004, Rosenthal, 2009, 2012, 2004, Byrne, 1997, Brown et al., 2019, Graziano and Webb, 2015, Graziano et al., 2020]. The low-level modules involve external processors, internal processors, and memory. The external processors handle the inputs and outputs of the human brain, including the image processor, sound processor, gustatory processor, olfactory processor, tactile processor, motor activators, speaking modules, etc. Each module processes the input information and outputs the processed signals to other parts. Internal processors include the logic processor, language processor, etc. Each will process information with outputs from external processors, or spontaneously without external inputs. All these modules have the ability to communicate with other modules and the memory to accomplish the desired objective, and most of them will generate intermediate outputs as inputs of other modules. More importantly, beyond all these low-level modules, the consciousness module has the ability to observe the intermediate outputs from low-level modules, which generates subjective experiences (also known as consciousness) for a human. This observation process is also formulated as a self-modeling process in theories like attention schema theory [Graziano and Webb, 2015, Graziano et al., 2020] or higher-order thoughts/perception in high-order theories [Locke, 1948, Armstrong, 1981, Armstrong and Malcolm, 1985, Lycan, 1996, Armstrong, 2002, Lycan, 2004, Rosenthal, 2009, 2012, 2004, Byrne, 1997, Brown et al., 2019], etc. Yet these explanations are of a highly abstract level. The specific components in human brains arising the consciousness are still debatable in the literature. There are some works [Baars, 2002] showing that the pre-frontal cortex may be involved in the higher-level cognitive procedure as an example. Discussions of different interpretations of those theories will be detailed in later sections. Due to the limited computational power, the consciousness module will only pay attention to those important information flows, which is a relatively small subset of all intermediate outputs from low-level modules. This coincides with the empirical evidence showing that a human can only have dozens of conscious experiences per second [Tegmark, 2018, 2000].

The existence of genuine free will remains a significant aspect of consciousness theory, albeit one that lacks sufficient scientific evidence to definitively prove or disprove. To engage in a comprehensive exploration of the existence of free will and its potential hierarchy within the human brain (Figure 2), we will bifurcate our discussion into positive and negative hypotheses: (1). assuming the existence of true free will; or (2). denying such existence of the free will. Therefore we mark the block of free will with dotted lines. For the first case that free will exists, which is beyond current physics interpretation, we may seek *external variables* from beyond the existing mathematical and physics frameworks to determine the process of human decision-making. The external variables can be regarded as an analog of the *hidden variables* in the famous Einstein-Podolsky-Rosen (EPR) paradox [Einstein et al., 1935], which states that some unobserved hidden variables may exist for explaining the true randomness in quantum mechanics, as a question of the completeness of quantum mechanics framework by Albert Einstein and others. Similar as the interpretation of EPR paradox for randomness in quantum mechanics, the explanation of the true 'free will' may also require such external randomness beyond the known physical systems. More details regarding physical explanations of the existence of free will are discussed in Sections 3 and 4. This part of free will is shown in Fig. 2 as the blue block containing the free will, injected by the consciousness module to affect the low-level processes (as Fig. 3) thus the final outputs from the system. For the second case that true free will does not exist, a question is what leads to some people think and feel that they also have the so-called 'free will'? This phenomenon can be interpreted with the current architecture with a hallucinated 'free will' [Wegner, 2004]. The key fact is that the consciousness module can only observe partial intermediate outputs of low-level modules due to its limited information processing bandwidth, thus the final outputs to the environment from the system can not be fully determined by these observed pieces of information, but together with more of other unobserved information. The consciousness module still seeks to explain the generated outputs by creating hallucinations of 'free will' in determining the results. However, this explanation may be debatable and requires verification. The existence of free will is also an open problem at present.

除了所有这些低层级模块之外，意识模块能够观察来自低层级模块的中间输出，从而为人类产生主观体验（也称为意识）。这种观察过程在诸如注意图式理论 [Graziano and Webb, 2015, Graziano et al., 2020] 或高阶理论中的高阶思想/感知 [Locke, 1948, Armstrong, 1981, Armstrong and Malcolm, 1985, Lycan, 1996, Armstrong, 2002, Lycan, 2004, Rosenthal, 2009, 2012, 2004, Byrne, 1997, Brown et al., 2019] 等理论中被表述为一种自我建模过程。然而，这些解释都处于高度抽象的层面。人脑中产生意识的具体组成部分在文献中仍有争议。有一些研究 [Baars, 2002] 表明，例如前额叶皮层可能参与高级认知过程。关于这些理论不同解释的讨论将在后续章节中详细展开。由于计算能力有限，意识模块只会关注那些重要的信息流，这些信息流是所有低层级模块中间输出的一个相对较小的子集。这与经验证据相符，即人类每秒只能有几十个意识体验 [Tegmark, 2018, 2000]。真正自由意志的存在仍然是意识理论的一个重要方面，尽管目前缺乏足够的科学证据来明确证实或证伪。为了全面探讨自由意志的存在及其在人脑中的潜在层级结构（图 2），我们将讨论分为正反两种假设：（1）假设存在真正的自由意志；或（2）否认自由意志的存在。因此，我们用虚线框标示自由意志模块。对于第一种情况，即自由意志存在且超出当前物理解释范围，我们可能需要从现有数学和物理框架之外寻求 *external variables*，以确定人类决策过程。外部变量可类比为著名的爱因斯坦-波多尔斯基-罗森（EPR）悖论 *hidden variables* 中的 [Einstein et al., 1935]，该悖论指出可能存在某些未被观测的隐变量来解释量子力学中的真正随机性，这是阿尔伯特·爱因斯坦等人对量子力学框架完备性提出的质疑。类似于 EPR 悖论对量子力学随机性的解释，对真正“自由意志”的说明可能同样需要这种超越已知物理系统的外部随机性。关于自由意志存在的物理解释的更多细节将在第3节和第4节讨论。这部分自由意志如图 2 所示，表现为包含自由意志的蓝色模块，由意识模块注入以影响低级处理过程（如图 3 所示），从而影响系统的最终输出。对于第二种情况，即真正的自由意志并不存在，问题在于：是什么导致一些人认为并感觉自己拥有所谓的“自由意志”？这一现象可以用当前架构中一种幻觉性的“自由意志” [Wegner, 2004] 来解释。关键事实在于，意识模块由于其有限的信息处理带宽，只能观测到低级模块的部分中间输出，因此系统对环境的最终输出不能完全由这些观测到的信息片段决定，而是需要结合更多未被观测的信息。意识模块仍会通过创造“自由意志”的幻觉来解释生成的输出结果。然而，这种解释可能存在争议，需要进一步验证。自由意志的存在目前仍是一个开放性课题。

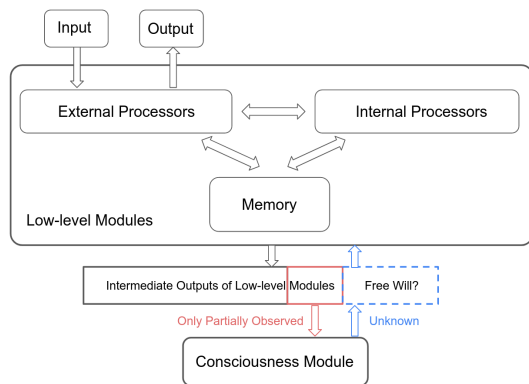


Figure 2: The overview architecture of consciousness system with free will.

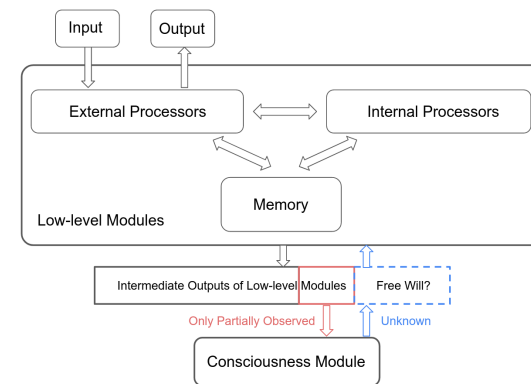


图 2: 具有自由意志的意识系统总体架构图。

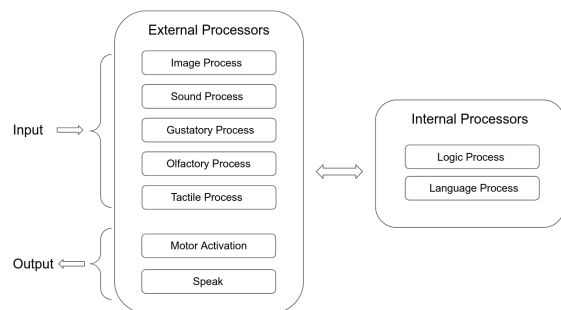


Figure 3: Details of processors in a consciousness system like the human brain.

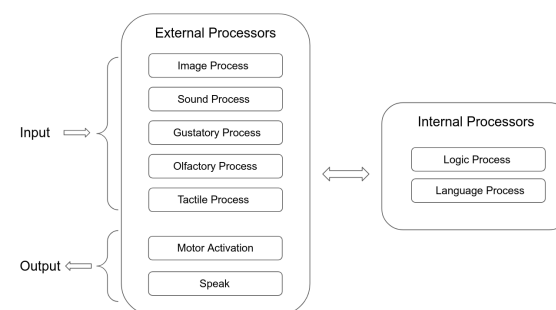


图 3: 类人脑意识系统中处理器的详细结构。

1.6 Consciousness while Asleep

The human sleep process contains stages including Rapid Eye Movement (REM) and Non-Rapid Eye Movement (NREM). According to previous research [Lee et al., 2022], NREM is a state where there is neither arousal nor awareness, whereas REM is a state of awareness without arousal.

Integrated Information Theory (IIT) [Tononi, 2004] suggests that consciousness is reduced during deep sleep. IIT proposes that consciousness is generated by the integrated information within a system. According to IIT, the level of consciousness experienced during sleep is dependent on the degree of integration present in the brain's activity.

Studies have shown that subjects awakened from deep NREM sleep, especially early in the night, often report a lack of awareness, even though cortical and thalamic neurons remain active. However, subjects awakened at other times, mainly during REM sleep or during lighter periods of NREM sleep later in the night, report dreams characterized by vivid images [Hobson et al., 2000]. From

1.3 Consciousness while Asleep

人类的睡眠过程包含快速眼动 (REM) 和非快速眼动 (NREM) 等阶段。根据先前的研究 [Lee et al., 2022], NREM 是一种既无觉醒也无觉知的状态, 而 REM 则是一种有觉知但无觉醒的状态。

整合信息理论 (IIT) [Tononi, 2004] 认为, 在深度睡眠期间意识会减弱。IIT 提出, 意识是由系统内的整合信息产生的。根据 IIT, 睡眠期间体验到的意识水平取决于大脑活动中存在的整合程度。

研究表明, 从深度 NREM 睡眠中被唤醒的受试者, 尤其是在夜间早期, 常常报告缺乏觉知, 尽管皮层和丘脑神经元仍然活跃。然而, 在其他时间被唤醒的受试者, 主要是在 REM 睡眠期

the perspective of integrated information theory, a reduction in consciousness during sleep would be consistent with the bistability of cortical circuits during deep NREM sleep. Consistent with these observations, studies using TMS (transcranial magnetic stimulation), a technique for stimulating the brain non-invasively, in conjunction with high-density EEG (electroencephalogram), show that early NREM sleep is associated with a breakdown of effective connectivity among cortical areas, leading to a loss of integration or a loss of repertoire and thus of information [Massimini et al., 2005, 2007]. These findings suggest that the level of consciousness experienced during sleep is dependent on the degree of integration present in the brain's activity, with a reduction in integration leading to a reduction in consciousness.

Overall, IIT suggests that the level of consciousness experienced during sleep is dependent on the degree of integration present in the brain's activity, with a reduction in integration leading to a reduction in consciousness.

1.7 Overview of Consciousness Theories

Diverse theories are developed by researchers to investigate the nature of consciousness and how it arises from the brain. Some of the most prominent theories include information integration theory (IIT), consciousness as a state of matter, orchestrated objective reduction (Orch OR) theory, global workspace theory (GWT), high-order theory (HOT), attention schema theory (AST), consciousness Turing machine (CTM), etc. IIT (Sec. 2) proposes that consciousness arises from the integration of information from multiple sensory and cognitive sources. Consciousness as a state of matter (Sec. 3) analyzes the deficiency of the information integration principle from the physics-principled calculation. Orch OR theory (Sec. 4) proposes the orchestrated objective reduction process for explaining the free will in conscious experience from a quantum mechanics perspective. GWT (Sec. 5) argues that consciousness arises from the activation of a global workspace in the brain, which integrates information from different sources and broadcasts it to the rest of the brain. HOT (Sec. 6) asserts that consciousness is the result of higher-order representations of sensory information. AST (Sec. 7) posits that consciousness is an attentional schema that the brain uses to represent the state of being conscious. CTM (Sec. 8) theory asserts that consciousness can be described as a computational process, as an extension of the Turing machine. These theories offer different perspectives on the nature of consciousness, and each has its own strengths and weaknesses. These theories offer insights into how it might be studied and understood. We conducted this survey to compare the similarities and differences of these theories and also summarize the correlations among different theories. More importantly, we aim to find feasible computational models from these theories for characterizing the conscious process of the human brain.

2 Information Integration Theory

The theory of information integration postulates that consciousness corresponds to the capacity of a system to integrate information. It first proposes the axioms of experiences, then postulates the properties of physical system that would give rise to the intrinsic experiences. In order to achieve this, the theory claims that the system must have a cause-effect power in itself, not resulting from any external factor. The cause-effect power of the system is then quantified by the largest minimum entropy of all sub-systems, evaluated by intervening on the states of a subset of the system (cause), and observe the change of states in the other part of the system (effect), while holding the external factors fixed. Therefore, IIT claims that **any conscious experience relates to a cause-effect structure that is maximally irreducible**. We will show in this section how the cause-effect power can be quantified by a measure called *information integration*. If the IIT theory is correct, we should be able to calculate the integrated information for a conscious experience in human brain and derive a reasonable value.

2.1 Information Entropy

The definition of information, according to Shannon [Shannon, 1948], is quantified by the reduction of uncertainty among a number of alternatives when one occurs. This is measured by the entropy function, defined as the following.

间或夜间稍后的 NREM 浅睡期, 会报告以生动意象 [Hobson et al., 2000] 为特征的梦境。从整合信息理论的角度来看, 睡眠期间意识的减弱与深度 NREM 睡眠期间皮层回路的双稳态是一致的。与这些观察结果一致, 使用经颅磁刺激 (TMS) ——一种非侵入性脑刺激技术——结合高密度脑电图 (EEG) 的研究表明, 早期 NREM 睡眠与皮层区域间有效连接的破坏有关, 导致整合的丧失或反应库的丧失, 从而丧失信息 [Massimini et al., 2005, 2007]。这些发现表明, 睡眠期间体验到的意识水平取决于大脑活动中存在的整合程度, 整合的减少导致意识的减弱。

总之, IIT 认为, 睡眠期间体验到的意识水平取决于大脑活动中存在的整合程度, 整合的减少导致意识的减弱。

1.4 Overview of Consciousness Theories

研究人员提出了多种理论来探究意识的本质及其如何从大脑中产生。其中一些最著名的理论包括信息整合理论 (IIT)、意识作为一种物质状态理论、协调客观还原 (Orch OR) 理论、全局工作空间理论 (GWT)、高阶理论 (HOT)、注意图式理论 (AST)、意识图灵机 (CTM) 等。IIT (第 2 节) 提出, 意识产生于来自多个感觉和认知源的信息整合。意识作为一种物质状态理论 (第 3 节) 从基于物理原理的计算角度分析了信息整合原则的不足。Orch OR 理论 (第 4 节) 提出了协调客观还原过程, 从量子力学角度解释意识体验中的自由意志。GWT (第 5 节) 认为, 意识产生于大脑中全局工作空间的激活, 该空间整合来自不同来源的信息并将其广播到大脑的其余部分。HOT (第 6 节) 断言, 意识是感觉信息的高阶表征的结果。AST (第 7 节) 假定, 意识是大脑用来表征意识状态的注意图式。CTM (第 8 节) 理论断言, 意识可以被描述为一个计算过程, 作为图灵机的一种扩展。这些理论为意识的本质提供了不同的视角, 各有其优缺点。它们为如何研究和理解意识提供了见解。我们进行本次综述, 旨在比较这些理论的异同, 并总结不同理论之间的关联。更重要的是, 我们旨在从这些理论中寻找可行的计算模型, 以刻画人脑的意识过程。

2 Information Integration Theory

信息整合理论假设意识对应于系统整合信息的能力。它首先提出经验公理, 然后假设物理系统应具备哪些属性才能产生内在经验。为实现这一点, 该理论主张系统必须具有内在的因果效力, 而非源于任何外部因素。系统的因果效力通过所有子系统的¹最小熵的最大值来量化, 其评估方式为: 干预系统某子集的状态 (因), 并观察系统另一部分状态的变化 (果), 同时保持外部因素不变。因此, IIT 声称 **any conscious experience relates to a cause-effect structure that is maximally irreducible**。我们将在本节展示因果效力如何通过一种称为 *information integration* 的度量来量化。如果 IIT 理论正确, 我们应能计算人脑中意识经验的整合信息量, 并得出合理的数值。

2.1 Information Entropy

根据香农 [Shannon, 1948] 的定义, 信息被量化为当某个事件发生时, 从多个备选可能中减少的不确定性。这通过熵函数来度量, 其定义如下。

定义 1. (香农熵) 给定在 σ -代数 $\mathcal{A}$ 上的概率测度 P , 一个概率分布的熵为:

$$H(A) = H_p(A) = \int_{a \in A} -p(a) \log p(a) da \quad (1)$$

此计算中使用的对数通常以 2 为底, 这意味着熵以比特为单位。一个系统的熵是对该系统生成的消息的平均信息内容的度量。

Definition 1. (Shannon Entropy) Given a probability measure P on a σ -algebra $\mathcal{A}$, the entropy of a probability distribution is:

$$H(A) = H_p(A) = \int_{a \in \mathcal{A}} -p(a) \log p(a) da \quad (1)$$

The logarithm used in this calculation is usually base 2, which means that the entropy is measured in bits. The entropy of a system is a measure of the average information content of a message generated by that system.

For example, in a binary system with two possible outcomes (0 or 1), the entropy is highest ($= 1$) when the probability of either outcome is 0.5, and lowest ($= 0$) when one outcome is certain (probability of 1) and the other is impossible (probability of 0).

In addition to being used in information theory, entropy has also been applied in fields such as cryptography, signal processing, and thermodynamics. In thermodynamics, entropy is used to quantify the degree of disorder or randomness in a thermodynamic system, and is often referred to as thermal entropy.

For a discrete-state system, a uniform distribution over all possible independent states contains the lowest information, therefore has the highest entropy. However, simply having a large number of independent components that results in a vast range of available states is not sufficient to generate conscious systems. These components must also be causally dependent on one another at an appropriate spatial and temporal scale. This crucial aspect of consciousness is referred to as information integration, as introduced by Tononi [Tononi, 2004]. Additionally, the author proposed a computational model to quantify the capacity of information integration, which will be detailed in the following sections.

2.2 Basics Concepts of IIT

The axioms of IIT state that every experience exists intrinsically and is structured, specific, unitary, and definite where specifically,

- Experience exists intrinsically;
- Experience is specific, being composed of a particular set of phenomenal distinctions (qualia);
- Experience is unitary, irreducible, as an integration of information;
- Experience is definite in its content and spatio-temporal grain (exclusion of other possibilities).

- (1) For any $S \in 2^{\mathcal{X}}$, consider one bipartition $S = A \cup B, S^c = \mathcal{X}/S$,
- (2)
- (3) Finally, one can find integrated information for the entire system

The theory then postulates that, for each essential property of experience, the physical substrate of consciousness (PSC) must have a cause-effect power related to the brain. The objective is then to find the appropriate spatial and temporal scale of neural elements gives rise to consciousness. The theorem implies that only those elements that have the maximum intrinsic cause-effect power are identified as elements of PSC. It is notable that under such a definition, the cause-effect power could be higher at a coarser spatial scale comparing to a finer spatial scale.

Recall the definition of consciousness in IIT corresponds to the capacity to integrate information, such that the system generates a large collection of states while being causally dependent to each other. In a hypothetical setting, imagine a collection of neuronal elements locally connected but disconnected from outside stimulus, then one is able to test if such a collection can be separated into two independent parts by measuring the information gain of one part by knowing the other part. In information theory, this precisely corresponds to *mutual information* (MI) of two random variables. The measurement of information integration in the theory is defined as a certain type of MI in the brain system, called *effective information* (EI).

例如, 在一个具有两种可能结果 (0 或 1) 的二进制系统中, 当任一结果的概率均为 0.5 时, 熵最高 ($= 1$); 而当一种结果确定发生 (概率为 1) 且另一种结果不可能发生 (概率为 0) 时, 熵最低 ($= 0$).

除了在信息论中使用外, 熵也被应用于密码学、信号处理和热力学等领域。在热力学中, 熵用于量化热力学系统的无序或随机程度, 通常被称为热熵。

对于一个离散状态系统, 在所有可能的独立状态上的均匀分布包含的信息最少, 因此具有最高的熵。然而, 仅仅拥有大量导致可用状态范围极广的独立组件, 并不足以产生有意识的系统。这些组件还必须在适当的空间和时间尺度上具有因果依赖性。意识的这一关键方面被称为信息整合, 由托诺尼 [Tononi, 2004] 提出。此外, 作者提出了一个计算模型来量化信息整合的能力, 这将在后续章节中详述。

整合信息理论 (IIT) 的公理指出, 每一种体验都内在存在, 并且是结构化、特定、统一且确定的。具体而言:

- 体验是内在存在的;
- 体验是特定的, 由一组特定的现象区分 (感受质) 构成;
- 体验是统一的、不可简化的, 是信息的整合;
- 体验在其内容和时空粒度上是确定的 (排除了其他可能性)。

- (1) 对于任意 $S \in 2^{\mathcal{X}}$, 考虑一个二分划分 $S = A \cup B, S^c = \mathcal{X}/S$,
- (2)
- (3) 最后, 可以为整个系统找到整合信息

该理论随后假设, 对于体验的每一个本质属性, 意识的物理基质 (PSC) 必须具有与大脑相关的因果效应能力。目标则是找到产生意识的神经元素的适当空间和时间尺度。该定理意味着, 只有那些具有最大内在因果效应能力的元素才被识别为 PSC 的元素。值得注意的是, 在此定义下, 较粗空间尺度上的因果效应能力可能比更精细空间尺度上的更高。

回顾 IIT 中意识的定义, 它对应于整合信息的能力, 使得系统在因果上相互依赖的同时产生大量的状态集合。在一个假设场景中, 想象一组局部连接但与外部刺激断开的神经元元素, 那么通过测量知道另一部分信息时某一部分的信息增益, 就可以测试这样的集合是否会被分成两个独立部分。在信息论中, 这精确对应于两个随机变量的 *mutual information* (MI)。该理论中信息整合的测量被定义为大系统一种特定类型的 MI, 称为 *effective information* (EI)。

定义 2 (互信息). 两个变量 A, B 之间的互信息:

$$I(A, B) = H(A) + H(B) - H(AB) \quad (2)$$

其中 $H(A) = H_{p_A}(A), H(B) = H_{p_B}(B), H(AB) = H_{p_{AB}}(A, B)$, p_{AB} 是 A, B 的联合分布。

在接下来的讨论中, 我们将把符号 A, B 推广为两个子系统 (或两个变量子集), 而不仅仅是两个变量。

为了测量信息增益, 首先将一部分神经元素设置为独立的噪声集合, 然后观察另一半作为接收信号的结果, 其放电模式如何变化。具体地, 我们定义 EI 的概念:

定义 3 (有效信息). 有效信息衡量 A 对 B 的 *directional* 因果效应,

$$EI(A \rightarrow B) = I(\tilde{A}, B) = H(\tilde{A}) + H(B) - H(\tilde{A}B), \tilde{A} = \arg \max_A H(A) \quad (3)$$

这意味着 A 被选择为独立的随机噪声, 因此 B 对 A 没有因果效应。

Definition 2 (Mutual Information). *The mutual information between two variables A, B :*

$$I(A, B) = H(A) + H(B) - H(AB) \quad (2)$$

where $H(A) = H_{p_A}(A)$, $H(B) = H_{p_B}(B)$, $H(AB) = H_{p_{AB}}(A, B)$, p_{AB} is the joint distribution of A, B .

In the following discussions, we will generalize the symbols A, B to be two sub-systems (or two subsets of variables) instead of two variables.

In order to measure the information gain, start by setting one part of neural elements to independent set of noises, and observe how the firing pattern changes in the other half as a consequence of receiving signals. Precisely, we define the concept of EI:

Definition 3 (Effective Information). *Effective information measures the **directional** causal effects of A on B ,*

$$EI(A \rightarrow B) = I(\tilde{A}, B) = H(\tilde{A}) + H(B) - H(\tilde{A}B), \tilde{A} = \arg \max_A H(A) \quad (3)$$

which means A is chosen to be independent random noise, thus B has no causal effects on A .

According to the above definitions of MI and EI, some lemmas can be directly derived, which are stated in the following remark.

Remark 1. *The information gain from A to B is not the same as B to A due to different connectivity pattern, while the mutual information is isotropy; and the EI is always upper bounded by the smallest maximum entropy of set A and set B . In mathematical terms:*

$$I(A, B) = I(B, A) \not\Rightarrow EI(A \rightarrow B) = EI(B \rightarrow A) \quad (4)$$

$$EI(A \rightarrow B) \leq \min\{O(\max_A H(A)), O(\max_B H(B))\} \quad (5)$$

As a consequence, we are able to measure how part A causally effects the other part B and vice versa. This gives rise to the **isotropy** causal effects defined by *mutual effective information* (MEI).

Definition 4 (Mutual Effective Information). *Mutual effective information $EI(A \rightleftharpoons B)$ measures the isotropy causal effects of between A and B ,*

$$EI(A \rightleftharpoons B) = EI(A \rightarrow B) + EI(B \rightarrow A) \quad (6)$$

As a consequence of the definition, if we are able to partition a system S into A and B such that $EI(A \rightleftharpoons B) = 0$, then A and B are independent parts, which limits the capacity of integrating information on the S . Therefore, it is necessary to locate the bottleneck in order to quantify the information integration capability for system S .

Definition 5 (Minimum Information Bipartition, MIB). *A bipartition on system S as its “weakest link” can be achieved with partitions $A, B \subset S$, $B = \bar{A}$ as its complementary set, such that the normalized mutual effective information of A, B is the minimum, as following:*

$$MIB(A \rightleftharpoons B) = \arg \min_{A, B \subset S} \frac{EI(A \rightleftharpoons B)}{H_{\max}(A \rightleftharpoons B)} \quad (7)$$

$$\text{with } H_{\max}(A \rightleftharpoons B) = \min\{\max_A H(A), \max_B H(B)\} \quad (8)$$

$H_{\max}(A \rightleftharpoons B)$ is for normalization due to Remark 1.

Each partition in MIB is called a *complex* in IIT. There is $\sum_{m=2}^N \binom{N}{m}$ subsets within a system of N elements, each has a measure of $\Phi(S)$, $S \in 2^{[N]}$, but those S in a larger subset with higher Φ are discarded, the rest are complexes in the system.

Now consider a system $\mathcal{X}$ with N neuronal elements, its information integration capacity is determined by the total minimum information for each of the complexes, which are found by enumerating over all the possible subsets $S \subseteq \mathcal{X}$.

To formalize this intuition, we define the information integration capacity of system $\mathcal{X}$, which measures the maximal irreducible cause-effect power, i.e., mutual effective information for minimum information bipartition.

根据上述互信息与有效信息的定义，可直接推导出若干引理，陈述于以下注记中。

备注 1. 由于连接模式不同，从 A 到 B 的信息增益与从 B 到 A 的信息增益并不相同，而互信息具有各向同性；且有效信息始终以集合 A 与集合 B 中较小的最大熵为上界。用数学语言表达：

$$I(A, B) = I(B, A) \not\Rightarrow EI(A \rightarrow B) = EI(B \rightarrow A) \quad (4)$$

$$EI(A \rightarrow B) \leq \min\{O(\max_A H(A)), O(\max_B H(B))\} \quad (5)$$

由此，我们能够度量部分 A 如何因果影响另一部分 B 及其反向影响。这引出了由 *mutual effective information* 定义的 **isotropy** 因果效应（MEI）。

定义 4 (互有效信息). 互有效信息 $EI(A \rightleftharpoons B)$ 度量了 A 与 B 之间的各向同性因果效应，

$$EI(A \rightleftharpoons B) = EI(A \rightarrow B) + EI(B \rightarrow A) \quad (6)$$

根据该定义，若我们能将系统 S 划分为 A 与 B 使得 $EI(A \rightleftharpoons B) = 0$ ，则 A 与 B 为独立部分，这限制了在 S 上整合信息的能力。因此，为量化系统 S 的信息整合能力，必须定位其瓶颈。

定义 5 (最小信息二分划分, MIB). 将系统 S 的“最弱连接”实现为二分划分，其中划分 $A, B \subset S$ 与其补集 $B = \bar{A}$ 使得 A, B 的归一化互有效信息最小，如下所示：

$$MIB(A \rightleftharpoons B) = \arg \min_{A, B \subset S} \frac{EI(A \rightleftharpoons B)}{H_{\max}(A \rightleftharpoons B)} \quad (7)$$

$$\text{with } H_{\max}(A \rightleftharpoons B) = \min\{\max_A H(A), \max_B H(B)\} \quad (8)$$

$H_{\max}(A \rightleftharpoons B)$ 用于归一化，依据注记 1。

MIB 中的每个划分在 IIT 中称为一个 *complex*。在包含 N 个元素的系统中存在 $\sum_{m=2}^N \binom{N}{m}$ 个子集，每个子集具有 $\Phi(S)$, $S \in 2^{[N]}$ 的度量值，但那些位于更大子集中且具有更高 Φ 的 S 被舍弃，其余构成系统中的复合体。

现考虑一个包含 N 个神经元元素的系统 $\mathcal{X}$ ，其信息整合能力由每个复合体的总最小信息决定，这些复合体通过枚举所有可能的子集 $S \subseteq \mathcal{X}$ 而找到。

为形式化这一直观理解，我们定义系统 $\mathcal{X}$ 的信息整合能力，该能力度量了最大的不可约因果效应效力，即最小信息二分划分的互有效信息 i.e.。

定义 6 (信息整合). 子集 S 的信息整合是最小信息二分划分的互有效信息：

$$\Phi(S) = \min_{A \in 2^S, B = S/A} EI(MIB(A \rightleftharpoons B)) \quad (9)$$

整个系统 $\mathcal{X}$ 的整合信息满足

$$\Phi(\mathcal{X}) = \max_{S \in 2^{\mathcal{X}}} \Phi(S)$$

关于信息整合 Φ 的直观解释是，如果系统不能完全分解（为独立的子系统），那么 Φ 就是通过最小化有效信息来在系统的“最薄弱环节”进行切割的有效信息（一种特殊类型的互信息），或者如 Tegmark [2015] 中所说的“最残酷切割”。我们将在下一节讨论如何实际测量整合信息。

Definition 6 (Information Integration). *The information integration for a subset S is the mutual effective information of the minimum information bipartition:*

$$\Phi(S) = \min_{A \in 2^S, B=S/A} EI(MIB(A \rightleftharpoons B)) \quad (9)$$

The integrated information for the entire system $\mathcal{X}$ is such that

$$\Phi(\mathcal{X}) = \max_{S \in 2^{\mathcal{X}}} \Phi(S)$$

The intuitive explanation of the information integration Φ is that, if the system is not fully decomposable (into independent sub-systems), Φ is the effective information (a special type of mutual information) for cutting the systems on its “weakest link” by minimizing the effective information, or as a “cruellest cut” as in Tegmark [2015]. We will discuss about how to practically measure the integration information in the following section.

2.3 Measurement of Information Integration

In Tononi and Sporns [2003], a computational model is proposed for measuring the information in a spatial network.

Assume that $\mathcal{X}$ is the entire system, and consider $X \in \mathbb{R}^{|\mathcal{X}|}$ to be the random vector over $\mathcal{X}$ that characterizes the signal emitted by each neuron in the system. To model the system X , the authors propose to use Gaussian Graphical Model, as described in the following.

The signal at the i -th node, denoted by x_i , is a linear combination of the signals of its neighboring nodes in the directed graph representing the node connection, plus the random measurement noise:

$$x_i = \sum_{j \in \text{neighbor}(i)} w_{i,j} x_j + \sigma_i r_i$$

where $r_i \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$ are random noise in the measurement, and $w_{i,j}$ is the edge weight from i to its neighbor j . In matrix form, we have,

$$X = WX + CR, \quad C = \text{diag}((\sigma_i)_i)$$

Then it is easy to see the covariance of X satisfies

$$\Sigma = C^2(I - W)^{-1}(I - W)^{-1\top}.$$

With this model, one can compute the information integration Φ analytically,

$$X := \begin{pmatrix} X_A \\ X_B \\ X_{S^c} \end{pmatrix} \sim \mathcal{N} \left(\mathbf{0}, \begin{pmatrix} \Sigma_A & \Sigma_{AB} & \Sigma_{AS^c} \\ \Sigma_{AB} & \Sigma_B & \Sigma_{BS^c} \\ \Sigma_{S^c A} & \Sigma_{S^c B} & \Sigma_{S^c} \end{pmatrix} \right)$$

then to calculate $EI(A^{\max} \rightarrow B)$, one randomizes the signal from A by setting $\Sigma_A = I_{|A|}$ in the covariance, and cuts off any incoming edges from S^c, B to A . However, if A is simulated to account for connection within A and to A , the original graph is unchanged. Moreover, one sets $C_A = \text{diag}(\sigma_p)$ as the signal and $C_{B \cup S^c} = \text{diag}(\sigma_i)$ as the noise. Then one can calculate (A is a set, X_A is a multivariate representing the signal states of the items in the set A , we ignore the difference of $H(A)$ and $H(X_A)$ here)

$$H(X_A) = \mathbb{E}[-\log p_{X_A}] = |A|/2 + |A|/2 \log 2\pi + \log |\Sigma_A|/2$$

$$H(X_A X_B) = |S|/2 + |S|/2 \log 2\pi + \log |\Sigma_S|/2, \quad \Sigma_S = \begin{pmatrix} \Sigma_A & \Sigma_{AB} \\ \Sigma_{AB} & \Sigma_B \end{pmatrix}$$

similarly $H(B)$. With the analytic results of $H(A), H(B)$ and $H(AB)$, one can also calculate $EI(B^{\max} \rightarrow A)$. Then one finds the minimal information bipartition as before,

$$MIB(S) = \min_{A \in 2^S, B=S/A} EI(A \rightleftharpoons B) / \min(H^{\max}(A), H^{\max}(B))$$

and $\Phi(S) = EI(MIB(S))$

$$\Phi(X) = \max_{S \in 2^{\mathcal{X}}} \Phi(S)$$

2.2 Measurement of Information Integration

在Tononi and Sporns [2003] 中, 提出了一种用于测量空间网络信息的计算模型。

假设 $\mathcal{X}$ 是整个系统, 并将 $X \in \mathbb{R}^{|\mathcal{X}|}$ 视为定义在 $\mathcal{X}$ 上的随机向量, 它表征了系统中每个神经元发出的信号。为了对系统 X 进行建模, 作者提出使用高斯图模型, 如下所述。

第 i 个节点处的信号, 记为 x_i , 是表示节点连接的有向图中其相邻节点信号的线性组合, 加上随机测量噪声:

$$x_i = \sum_{j \in \text{neighbor}(i)} w_{i,j} x_j + \sigma_i r_i$$

其中 $r_i \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$ are random noise in the measurement, and $w_{i,j}$ is the edge weight from i to its neighbor j . In matrix form, we have,

$$X = WX + CR, \quad C = \text{diag}((\sigma_i)_i)$$

Then it is easy to see the covariance of X satisfies

$$\Sigma = C^2(I - W)^{-1}(I - W)^{-1\top}.$$

With this model, one can compute the information integration Φ analytically,

$$X := \begin{pmatrix} X_A \\ X_B \\ X_{S^c} \end{pmatrix} \sim \mathcal{N} \left(\mathbf{0}, \begin{pmatrix} \Sigma_A & \Sigma_{AB} & \Sigma_{AS^c} \\ \Sigma_{AB} & \Sigma_B & \Sigma_{BS^c} \\ \Sigma_{S^c A} & \Sigma_{S^c B} & \Sigma_{S^c} \end{pmatrix} \right)$$

then to calculate $EI(A^{\max} \rightarrow B)$, one randomizes the signal from A by setting $\Sigma_A = I_{|A|}$ in the covariance, and cuts off any incoming edges from S^c, B to A . However, if A is simulated to account for connection within A and to A , the original graph is unchanged. Moreover, one sets $C_A = \text{diag}(\sigma_p)$ as the signal and $C_{B \cup S^c} = \text{diag}(\sigma_i)$ as the noise.

Then one can calculate (A is a set, X_A is a multivariate representing the signal states of the items in the set A , we ignore the difference of $H(A)$ and $H(X_A)$ here)

$$H(X_A) = \mathbb{E}[-\log p_{X_A}] = |A|/2 + |A|/2 \log 2\pi + \log |\Sigma_A|/2$$

$$H(X_A X_B) = |S|/2 + |S|/2 \log 2\pi + \log |\Sigma_S|/2, \quad \Sigma_S = \begin{pmatrix} \Sigma_A & \Sigma_{AB} \\ \Sigma_{AB} & \Sigma_B \end{pmatrix}$$

similarly $H(B)$. With the analytic results of $H(A), H(B)$ and $H(AB)$, one can also calculate $EI(B^{\max} \rightarrow A)$. Then one finds the minimal information bipartition as before,

$$MIB(S) = \min_{A \in 2^S, B=S/A} EI(A \rightleftharpoons B) / \min(H^{\max}(A), H^{\max}(B))$$

and $\Phi(S) = EI(MIB(S))$

$$\Phi(X) = \max_{S \in 2^{\mathcal{X}}} \Phi(S)$$

Apart from computing Φ for system with a given topological structure, the authors also experimented with finding the graph structure with the maximal Φ , and analyzed the connectivity property of the generated ‘optimal’ graph with maximal Φ .

在Balduzzi and Tononi [2008] 中, 对小规模网络上的离散时空系统进行了模拟研究。

[Hoel, 2017] 利用因果推断中的 *do* 操作, 构建了一个时间上的马尔可夫链, 该链识别出对应于信息增加的 (空间) 状态空间的特定粗化。考虑一个有限状态空间 S , 我们可以指定任意两个状态之间的转移概率矩阵 (TPM)

$$P_{ij} = \mathbb{P}(X^{t+1} = j | X^t = i) = \mathbb{P}(X^{t+1} = j | do(X^t = i))$$

Apart from computing Φ for system with a given topological structure, the authors also experimented with finding the graph structure with the maximal Φ , and analyzed the connectivity property of the generated 'optimal' graph with maximal Φ .

In Balduzzi and Tononi [2008], simulation studies on discrete spatial-temporal systems are carried out on small scale networks.

[Hoel, 2017] leverages the *do*-operation in causal inference in constructing a Markov chain in time that identify particular coarsening of the (spatial) state-space that corresponds to an increase in information.

Consider a finite state space S , we can specify transition probability matrix (TPM) between any two states

$$P_{ij} = \mathbb{P}(X^{t+1} = j | X^t = i) = \mathbb{P}(X^{t+1} = j | do(X^t = i))$$

which is similar to Markov chain, with the Markov property given by the *do* operator, i.e. we have conditional independence once $S^t = i$ is a fixed quantity. In this sense, we treat the time as varying, and only one variable X present with state space S ; or that we have $|S|$ many binary variables and so $X = \sum_{k \in S} \delta_k$, which is well-defined due to finiteness.

Consider the problem setup: let $X \rightarrow Y$ be the causal system, where they share the same state space $S_X = S_Y$. The author is interested in when the macro system has a micro system with corresponding TPM, i.e. $U \rightarrow V$, such that $S_U = S_V \subset 2^{S_X}$. Moreover, Hoel's theory aims at identifying particular coarsening of the state space amounts to *causal emergence*. Effective information (EI) plays an important role in achieving this. We give another definition of EI since the context differs to IIT, but firstly we introduce the concept of intervention and effect distribution.

Definition 7 (Intervention Distribution and Effect Distribution). *Given $|S_X| < \infty$, the maximum entropy amounts to the uniform intervention distribution*

$$I_D = H^{\max} = \text{Unif}(do(X)), \text{ that is, } P(do(X = x)) = \frac{1}{n} \quad \forall x \in S_X \quad (10)$$

where $\text{Unif}(\cdot)$ is the uniform distribution over a set. Intervening with this distribution on X results in the effect distribution $E_D(Y)$ over Y :

$$\begin{aligned} E_D(Y) &= \sum_X P(Y | do(X)) H^{\max} \\ &= \frac{1}{n} \sum_x P(Y | do(X = x)) \end{aligned} \quad (11)$$

The E_D effectively computes the uniform averages over all rows in the TPM.

We are now ready to introduce definition of effective information in Hoel et al. [2013]:

Definition 8 (Effective Information).

$$\begin{aligned} EI(X \rightarrow Y) &= \sum_X H^{\max} D_{KL}(P(Y | do(X)) || E_D(Y)) \\ &= \sum_x P(do(X = x)) D_{KL}(P(Y | do(X = x)) || E_D(Y)) \\ &= \frac{1}{n} \sum_x D_{KL}(P(Y | do(X = x)) || E_D(Y)) \end{aligned} \quad (12)$$

The definition of EI presented in Def 8 is in fact equivalent to the definition in IIT. Without loss of generality, denote $p(x)$ and $p(y)$ the mass function for $X \sim H^{\max}$, $Y \sim E_D$ respectively.

$$\begin{aligned} MI(X, Y) &= \sum_x \sum_y p(x, y) \log_2 \left(\frac{p(x, y)}{p(x)p(y)} \right) \\ (\text{Bayes Rule}) &= \sum_x \sum_y p(x)p(y|x) \log_2 \left(\frac{p(y|x)}{p(y)} \right) \\ &= \frac{1}{n} \sum_x D_{KL}(P(Y | do(X = x)) || E_D(Y)) \end{aligned}$$

这类似于马尔可夫链，其马尔可夫性质由 *do* operator, i.e. we have conditional independence once $S^t = i$ is a fixed quantity. In this sense, we treat the time as varying, and only one variable X present with state space S ; or that we have $|S|$ many binary variables and so $X = \sum_{k \in S} \delta_k$, which is well-defined due to finiteness.

Consider the problem setup: let $X \rightarrow Y$ be the causal system, where they share the same state space $S_X = S_Y$. The author is interested in when the macro system has a micro system with corresponding TPM, i.e. $U \rightarrow V$, such that $S_U = S_V \subset 2^{S_X}$. Moreover, Hoel's theory aims at identifying particular coarsening of the state space amounts to *causal emergence*. Effective information (EI) plays an important role in achieving this. We give another definition of EI since the context differs to IIT, but firstly we introduce the concept of intervention and effect distribution.

定义 7 (Intervention Distribution and Effect Distribution). *Given $|S_X| < \infty$, the maximum entropy amounts to the uniform intervention distribution*

$$I_D = H^{\max} = \text{Unif}(do(X)), \text{ that is, } P(do(X = x)) = \frac{1}{n} \quad \forall x \in S_X \quad (10)$$

where $\text{Unif}(\cdot)$ is the uniform distribution over a set. Intervening with this distribution on X results in the effect distribution $E_D(Y)$ over Y :

$$\begin{aligned} E_D(Y) &= \sum_X P(Y | do(X)) H^{\max} \\ &= \frac{1}{n} \sum_x P(Y | do(X = x)) \end{aligned} \quad (11)$$

The E_D effectively computes the uniform averages over all rows in the TPM.

We are now ready to introduce definition of effective information in Hoel et al. [2013]:

定义 8 (Effective Information).

$$\begin{aligned} EI(X \rightarrow Y) &= \sum_X H^{\max} D_{KL}(P(Y | do(X)) || E_D(Y)) \\ &= \sum_x P(do(X = x)) D_{KL}(P(Y | do(X = x)) || E_D(Y)) \\ &= \frac{1}{n} \sum_x D_{KL}(P(Y | do(X = x)) || E_D(Y)) \end{aligned} \quad (12)$$

The definition of EI presented in Def 8 is in fact equivalent to the definition in IIT. Without loss of generality, denote $p(x)$ and $p(y)$ the mass function for $X \sim H^{\max}$, $Y \sim E_D$ respectively.

$$\begin{aligned} MI(X, Y) &= \sum_x \sum_y p(x, y) \log_2 \left(\frac{p(x, y)}{p(x)p(y)} \right) \\ (\text{Bayes Rule}) &= \sum_x \sum_y p(x)p(y|x) \log_2 \left(\frac{p(y|x)}{p(y)} \right) \\ &= \frac{1}{n} \sum_x D_{KL}(P(Y | do(X = x)) || E_D(Y)) \end{aligned}$$

根据上述符号，当最佳粗粒化系统比原系统具有更高的有效信息时，即出现 *causal emergence* (CE),

$$CE = EI(U \rightarrow V) - EI(X \rightarrow Y) > 0$$

因为 U 和 V are considered as the same random variables in one step, this amounts to finding the coarsened support set S_U .

Given the notations above, a *causal emergence* (CE) arises when the best coarsened system has higher effective information than the original one, i.e.

$$CE = EI(U \rightarrow V) - EI(X \rightarrow Y) > 0$$

since U and V are considered as the same random variables in one step, this amounts to finding the coarsened support set S_U .

2.4 Biological Evidence

Despite the computational framework put forward by the IIT theory is able to analytically assess the information processing bottleneck in an arbitrary system, it remains an open problem to verify the claimed correspondence that subjective experience is equivalent to the capacity to integrate information.

IIT postulates the neural elements constituting PSC are those determined by maximizing the cause-effect power, which could be higher at a macro scale comparing to a micro scale owing to different connectivity patterns. Among the brain regions, cerebral cortex has functional specialization and integration altogether, which should yield high values of maximum information integration. Whereas cerebellum is not essential for consciousness, because of its lack of dependency among the neurons and inability to form a large complex with high maximum information integration.

IIT also explains why bistable firing of cortical neurons during slow wave sleep would cause fading of consciousness. This is owing to the loss of both selectivity and effectiveness results in the reduction of information integration [Tononi et al., 2016]. Alkire et al. [2008] argues that brain under anesthesia is similar to under slow wave sleep, where cortical connectivity breaks down and therefore information integration is reduced. Practically, PCI are proposed to estimate $\Phi^{\max}$ evoked by TMS in practice, which is high only if brain responses are both integrated and differentiated, corresponding to a distributed spatio-temporal pattern of causal interactions that is complex and hence not very compressible.

3 Consciousness as a State of Matter

Apart from the information theoretical viewpoint, researchers are seeking for the properties of conscious process within physical systems. Consciousness can be thought of as an emergent phenomenon. It does not depend on detailed properties of atoms, but on the complex patterns into which the atoms are arranged. Emergent phenomena are common in physics, for example, waves can exist in many different kinds of matter. Researchers have investigated the phenomenon of consciousness from a principled way in physics [Penrose, 1991, Stapp, 2000, Tegmark, 2000, 2015, Carroll, 2021]. A core issue for this approach is to admit the true randomness in the conscious process and locate the corresponding physical processes within the human brain as evidence. A branch of research ultimately resorts to the quantum process, which is commonly believed to exhibit the true randomness in the measurement procedure of a quantum state. The conscious process can be interpreted as a quantum measurement by a conscious observer. However, several problems are raised in this interpretation: One corresponds to the quantum factorization problem, that the conscious observers has a certain Hilbert space factorization to leave the world around the observer as a strongly correlated but independent (from the observer) system [Tegmark, 2015]. Another problem is the quantum decoherence in physical system like human brains. Most quantum phenomenons only appear in a very small space-time scale, the decoherence process prevents a system as large as the human brain to inherit the quantum property. For example, a typical timescale for quantum decoherence lasts for about $10^{-13} \sim 10^{-20}$ seconds, which is much shorter than the timescale of cognitive process as $10^{-3} \sim 10^{-1}$ seconds [Tegmark, 2000]. Researchers also proposed the orchestrated objective reduction (Orch OR) of quantum states to interpret the brain cognitive process [Hameroff and Penrose, 2014], which is discussed in Sec. 4.

Max Tegmark [Tegmark, 2015] considers consciousness as a kind of matter which he calls “*perceptronium*”, as a phrase indicating conscious state. For a matter to become “*perceptronium*”, it needs have the following four properties as necessary but not sufficient conditions:

- The **information** principle: The system must have substantial information storage capacity;

2.3 Biological Evidence

尽管整合信息理论 (IIT) 提出的计算框架能够分析性地评估任意系统中的信息处理瓶颈, 但验证其声称的主观体验等同于信息整合能力这一对应关系, 仍然是一个悬而未决的问题。

IIT 假设构成现象状态内容 (PSC) 的神经元素是通过最大化因果效力来确定的, 由于不同的连接模式, 宏观尺度上的因果效力可能高于微观尺度。在大脑区域中, 大脑皮层同时具备功能专门化和整合性, 这应能产生较高的最大信息整合值。而小脑对于意识并非必需, 这是因为其神经元之间缺乏依赖性, 且无法形成具有最高最大信息整合的大型复杂体。

IIT 也解释了为何慢波睡眠期间皮层神经元的双稳态放电会导致意识消退。这是由于选择性和有效性的丧失共同导致了信息整合的减少 [Tononi et al., 2016]。Alkire et al. [2008] 认为, 麻醉状态下的大脑与慢波睡眠状态相似, 皮层连接性遭到破坏, 因此信息整合降低。在实践中, 人们提出使用扰动复杂性指数 (PCI) 来估计实际操作中经颅磁刺激 (TMS) 所引发的 $\Phi^{\max}$, 该指数仅在脑响应既整合又分化时才会较高, 这对应于一种分布式的时空因果交互模式, 该模式复杂且因此不易被压缩。

3 Consciousness as a State of Matter

除了信息理论的视角外, 研究者们也在探索物理系统中意识过程的特性。意识可以被视为一种涌现现象。它并不依赖于原子的具体属性, 而是取决于原子排列所形成的复杂模式。涌现现象在物理学中很常见, 例如, 波可以存在于多种不同的物质中。研究者们已从物理学的原理性途径探讨了意识现象 [Penrose, 1991, Stapp, 2000, Tegmark, 2000, 2015, Carroll, 2021]。这一方法的核心问题在于承认意识过程中的真随机性, 并在人脑中定位相应的物理过程作为证据。一个研究分支最终诉诸于量子过程, 因为量子态的测量过程通常被认为展现出真随机性。意识过程可以被解释为有意识的观察者进行的量子测量。然而, 这种解释引发了若干问题: 其一对应于量子分解问题, 即有意识的观察者具有特定的希尔伯特空间分解, 使得观察者周围的世界成为一个强关联但独立于观察者的系统 [Tegmark, 2015]。另一个问题是诸如人脑这类物理系统中的量子退相干。大多数量子现象仅出现在极小的时空尺度上, 退相干过程阻止了像人脑这样大的系统继承量子特性。例如, 量子退相干的典型时间尺度约为 $10^{-13} \sim 10^{-20}$ 秒, 这远短于认知过程的时间尺度 (约 $10^{-3} \sim 10^{-1}$ 秒) [Tegmark, 2000]。研究者们还提出了量子态的协调客观坍缩 (Orch OR) 来解释大脑的认知过程 [Hameroff and Penrose, 2014], 这将在第 4 节中讨论。马克斯·泰格马克 [Tegmark, 2015] 将意识视为一种物质, 他称之为 “*perceptronium*”, 作为表示意识状态的短语。一种物质要成为 “*perceptronium*”, 需要具备以下四个属性作为必要但不充分条件:

- **information** 原理: 系统必须具有实质性的信息存储能力;
- **integration** 原理: 系统不能由几乎独立的部分组成, 且其内部需要具有一定程度的整合性;
- **independence** 原理: 系统必须与世界的其余部分保持实质性的独立性;
- **dynamics** 原理: 一个有意识的系统必须具有实质性的信息处理能力, 且必须是这种处理过程 (而非静态信息) 被整合。

在泰格马克的工作中, 他推广了托诺尼的整合信息理论 (第2节), 将其与意识物质应具备的其他原理进行比较, 然后将分析推广到量子力学。在他的工作中, 他表明信息原理与整合原理可能相互冲突——过多的整合会导致信息量极少, 这在第3.2节中被称为 *integration paradox*。此外, 独立性原理与动力学原理也存在冲突——过多的独立性会导致一个平凡的动力学系统,

- The **integration** principle: The system cannot consist of nearly independent parts, and it needs to have a certain level of integration within itself;
- The **independence** principle: The system must have substantial independence from the rest of the world;
- The **dynamics** principle: A conscious system must have substantial information-processing capacity, and it is this processing rather than the static information that must be integrated.

In Tegmark's work, he generalizes Tononi's IIT (Sec. 2), compares this with other principles that conscious matter should have, and then generalizes the analysis to quantum mechanics. In his work, he shows that the information and integration principles can have conflicts with each other – too much integration will result in very little information, which is referred to as the *integration paradox* in Sec. 3.2. Also, the independence principle has conflicts with the dynamics principle – too much independence will result in a trivial dynamics system, which is called the *quantum Zeno paradox* as discussed later in Sec. 3.3. A conscious system needs to strike balances in these properties: information and integration, independence and dynamics. Therefore, this work introduces the *autonomy* metric to measure the balance of independence and dynamics in the system.

Following that, it also proposes the following two principles:

- The **autonomy** principle: A conscious system has substantial dynamics and independence.
- The **utility** principle: An evolved conscious system records mainly information that is useful for it.

The autonomy principle describes the balance of dynamics and independence. The utility principle describes the amount of information within a conscious system.

These principles can be translated into more physical problems in a system:

- The physics-from-scratch problem: If the total Hamiltonian $\mathbf{H}$ and the total density matrix ρ fully specify our physical world, how do we extract 3D space and the rest of our semiclassical world from nothing more than two Hermitian matrices?
- The quantum factorization problem: Why do conscious observers like us perceive the particular Hilbert space factorization corresponding to classical space (rather than Fourier space, say), and more generally, why do we perceive the world around us as a dynamic hierarchy of objects that are strongly integrated and relatively independent?

These are the question to be answered in this theory. We will briefly introduce the basics of quantum mechanics and then dive into these principles.

3.1 Basics of Quantum Mechanics

In quantum mechanics, the state of a system is described by a vector $|\psi\rangle$ in the Hilbert space. For example, we have an electron with a spin up state $|\psi\rangle = |\uparrow\rangle$ or spin down state $|\psi\rangle = |\downarrow\rangle$, or their superposition $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$.

The probability of observing this system in state $|\chi\rangle$ is

$$P = |\langle\chi|\psi\rangle|^2. \quad (13)$$

We can apply a unitary operator to a state to change the basis we are interested in:

$$|\psi\rangle \rightarrow U|\psi\rangle \quad (14)$$

For example, the spin up state in x direction is a superposition state $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ in the z direction.

The time evolution of a state is controlled by the Hamiltonian operator $\mathbf{H}$ (can be thought of as a matrix in Hilbert space):

$$|\psi(t)\rangle = e^{i\mathbf{H}t/\hbar}|\psi(0)\rangle. \quad (15)$$

which is known as the Schrödinger equation.

这被称为 *quantum Zeno paradox*, 将在第3.3节中讨论。一个有意识的系统需要在这些属性之间取得平衡：信息与整合、独立性与动力学。因此，这项工作引入了 *autonomy* 度量来衡量系统中独立性与动力学的平衡。

随后，它还提出了以下两个原理：

- **autonomy** 原理：一个有意识的系统具有实质性的动力学和独立性。
- **utility** 原理：一个演化而来的有意识系统主要记录对其有用的信息。

自主性原理描述了动力学与独立性的平衡。效用性原理描述了有意识系统内的信息量。

这些原理可以转化为系统中更具体的物理问题：

- 从零开始的物理问题：如果总哈密顿量 $\mathbf{H}$ 和总密度矩阵 ρ 完全规定了我们的物理世界，我们如何仅从两个厄米矩阵中提取出三维空间和我们半经典世界的其余部分？
- 量子分解问题：为什么像我们这样的有意识观察者会感知到对应于经典空间（而非例如傅里叶空间）的特定希尔伯特空间分解，更一般地说，为什么我们将周围的世界感知为一个动态的、高度整合且相对独立的物体层次结构？

这些是该理论需要回答的问题。我们将简要介绍量子力学的基础知识，然后深入探讨这些原理。

3.1 Basics of Quantum Mechanics

在量子力学中，系统的状态由希尔伯特空间中的一个向量 $|\psi\rangle$ 描述。例如，我们有一个电子，其自旋向上态为 $|\psi\rangle = |\uparrow\rangle$ 或自旋向下态为 $|\psi\rangle = |\downarrow\rangle$ ，或者它们的叠加态 $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ 。

观测到该系统处于状态 $|\chi\rangle$ 的概率为

$$P = |\langle\chi|\psi\rangle|^2. \quad (13)$$

我们可以对状态应用一个么正算符，以改变我们感兴趣的基：

$$|\psi\rangle \rightarrow U|\psi\rangle \quad (14)$$

例如， x 方向上的自旋向上态是 z 方向上的一个叠加态 $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ 。

状态的时间演化由哈密顿算符 $\mathbf{H}$ （可视为希尔伯特空间中的一个矩阵）控制：

$$|\psi(t)\rangle = e^{i\mathbf{H}t/\hbar}|\psi(0)\rangle. \quad (15)$$

这被称为薛定谔方程。

哈密顿算符本身描述了系统的能谱。我们可以找到它的本征态：

$$\mathbf{H}|E_i\rangle = E_i|E_i\rangle. \quad (16)$$

E_i 是哈密顿算符 $\mathbf{H}$ 的本征值，它表示系统处于状态 $|E_i\rangle$ 时的能量（一个标量）。算符在其本征基中呈对角形式。当两个算符对易时， $[A, B] = AB - BA = 0$ ，它们可以同时被对角化。

状态也可以表示为密度矩阵：

$$\rho = |\psi\rangle\langle\psi| \quad (17)$$

这表示一个纯态。

The Hamiltonian operator itself describes the energy spectrum of the system. We can find its eigenstates:

$$\mathbf{H}|E_i\rangle = E_i|E_i\rangle. \quad (16)$$

E_i is the eigenvalue of the Hamiltonian operator $\mathbf{H}$, which represents the energy (as a scalar) of the system at state $|E_i\rangle$. An operator takes diagonal form in its eigenbasis. When two operators commute, $[A, B] = AB - BA = 0$, they can be simultaneously diagonalized.

A state can also be represented as a density matrix:

$$\rho = |\psi\rangle\langle\psi| \quad (17)$$

this represents a pure state.

Density matrices built from the pure state always have rank 1. A more general density matrix can also represent a classical mixture of states, which in general has a higher rank

$$\rho = \sum_i |\psi_i\rangle\langle\psi_i|. \quad (18)$$

this is a mixed state.

The probability of observing the system in a certain state is

$$P = \langle\chi|\rho|\chi\rangle \quad (19)$$

The time evolution of a density matrix is

$$\rho(t) = e^{i\mathbf{H}t/\hbar}\rho(0)e^{-i\mathbf{H}t/\hbar}. \quad (20)$$

Written in the energy eigenbasis, it is

$$\rho(t)_{mn} = \rho(0)_{mn}e^{i(E_m - E_n)t} \quad (21)$$

3.2 Integration Principle

For a bipartite system $\rho = \rho_1 \otimes \rho_2$. We define the integrated information Φ as the mutual information I for the “cruellest cut”. The mutual information is defined as

$$I \equiv S(\rho_1) + S(\rho_2) - S(\rho) \quad (22)$$

where

$$S(\rho) \equiv -\text{tr } \rho \log_2 \rho \quad (23)$$

is the von Neumann entropy. Note that this is slightly different from Tononi’s definition, but this is easier to calculate.

Recall in Sec. 2, we introduce the definition of the integrated information in original IIT, as the effective information of the minimum information bipartition. The minimum information bipartition is a ‘cut’ of the system according to the weakest link, *i.e.*, minimal mutual effective information. We can also consider cuts here in the quantum sense. In this case, more cuts are available, and we choose the crudest cut as the integrated information. Therefore the *integrated information* is defined as

$$\Phi = \min_{\mathbf{U}} I(\mathbf{U}\rho\mathbf{U}^\dagger). \quad (24)$$

where $\mathbf{U}$ is the unitary evolution on density matrix ρ .

A general Hamiltonian can be written as

$$\mathbf{H} = \mathbf{H}_1 \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{H}_2 + \mathbf{H}_3 \quad (25)$$

where $\mathbf{I}$ is identity matrix.

If a Hamiltonian can be decomposed without an interaction term (with $\mathbf{H}_3 = 0$), then it describes two perfectly independent systems,

$$\rho \propto e^{-\mathbf{H}/kT} = e^{-\mathbf{H}_1/kT} e^{-\mathbf{H}_2/kT} \quad (26)$$

由纯态构建的密度矩阵总是秩为 1。一个更一般的密度矩阵也可以表示状态的经典混合，通常具有更高的秩

$$\rho = \sum_i |\psi_i\rangle\langle\psi_i|. \quad (18)$$

这是一个混合态。

观测到系统处于某个特定状态的概率为

$$P = \langle\chi|\rho|\chi\rangle \quad (19)$$

密度矩阵的时间演化为

$$\rho(t) = e^{i\mathbf{H}t/\hbar}\rho(0)e^{-i\mathbf{H}t/\hbar}. \quad (20)$$

在能量本征基中写出来，它是

$$\rho(t)_{mn} = \rho(0)_{mn}e^{i(E_m - E_n)t} \quad (21)$$

3.2 Integration Principle

对于一个二分系统 $\rho = \rho_1 \otimes \rho_2$ ，我们将整合信息 Φ 定义为“最残酷分割”下的互信息 I 。互信息的定义为

$$I \equiv S(\rho_1) + S(\rho_2) - S(\rho) \quad (22)$$

其中

$$S(\rho) \equiv -\text{tr } \rho \log_2 \rho \quad (23)$$

是冯·诺依曼熵。请注意，这与托诺尼的定义略有不同，但此定义更易于计算。

回顾第 2 节，我们介绍了原始整合信息理论中整合信息的定义，即最小信息二分下的有效信息。最小信息二分是根据最弱连接对系统进行的“切割”，*i.e.*，即最小的互有效信息。我们也可以考虑量子意义上的切割。在这种情况下，有更多的切割方式可供选择，而我们选择最残酷的切割作为整合信息。因此，*integrated information* 被定义为

$$\Phi = \min_{\mathbf{U}} I(\mathbf{U}\rho\mathbf{U}^\dagger). \quad (24)$$

其中 $\mathbf{U}$ 是作用于密度矩阵 ρ 上的幺正演化。

一个一般的哈密顿量可以写为

$$\mathbf{H} = \mathbf{H}_1 \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{H}_2 + \mathbf{H}_3 \quad (25)$$

其中 $\mathbf{I}$ 是单位矩阵。

如果一个哈密顿量可以在没有相互作用项的情况下分解（即 $\mathbf{H}_3 = 0$ ），那么它描述的是两个完全独立的系统，

$$\rho \propto e^{-\mathbf{H}/kT} = e^{-\mathbf{H}_1/kT} e^{-\mathbf{H}_2/kT} \quad (26)$$

在这种情况下，可以推导出 $\Phi = 0$ ，其中 $\mathbf{U} = e^{i\mathbf{H}_1 t/\hbar} e^{i\mathbf{H}_2 t/\hbar}$ 。这符合预期，因为整合信息描述了一个系统内部的信息整合水平，而一个完全分离的系统应该没有整合。根据整合信息理论，我们知道有意识的系统很可能具有最大的整合信息。以具有 n 个偶极子的二维伊辛模型为例，根据上述定义，其最大整合信息仅为 $O(\sqrt{n})$ 。然而，通过使用最优纠错码，系统可以在大 n 区域实现渐近的 $n/2$ 比特整合信息。

以上分析基于量子系统。对于描述大脑过程的经典系统，如霍普菲尔德网络 [Hopfield, 1982]，也可以计算整合信息，但这将导致所谓的整合悖论。

In this case, it can be derived that $\Phi = 0$ with $\mathbf{U} = e^{i\mathbf{H}_1 t/\hbar} e^{i\mathbf{H}_2 t/\hbar}$. This is within expectation since the integrated information describes the level of information integration within a system, and a perfectly separated system should have no intergration. From IIT, we know that the conscious systems are likely to have the maximum of integrated information. Taking the 2D Ising model with n dipoles as an example, the maximum integrated information according to above definition will only be $O(\sqrt{n})$. However, with the optimal error correcting codes, the system can achieve asymptotic $n/2$ bits of integrated information in the large- n regime.

The above analysis is based on the quantum system. For a classical system like a Hopfield network [Hopfield, 1982] for describing the brain process, the integrated information can also be calculated, and it will lead to a so-called integration paradox.

Integration paradox. Suppose our brain is a Hopfield network with n neurons using Hebbian learning rules, then the maximum capacity of integrated information is 37 bits for $n = 10^{11}$ neurons [Tegmark, 2015]. However, the information for a conscious experience is much larger than this value, with an example of a human imagining a picture in his mind. This is known as the integration paradox.

Why does the information content of our conscious experience appear to be vastly larger than 37 bits? In the quantum case, it can never contain more than 1/4 bit of information [MacKay, 2003]. This observation leads to some conjectures including that the human brains use a better coding method for conscious information rather than the Hopfield networks.

3.3 Independence Principle

This principle follows the idea of IIT for cutting the system into independent parts from its “weakest link”, as described in Sec. 2.2. It first requires the ρ -diagonality theorem in a quantum case to find the minimum of mutual information.

Theorem 1 (ρ -Diagonality Theorem, ρ DT [Jevtic et al., 2012]). *The mutual information always takes its minimum in a basis where ρ is diagonal.*

With ρ DT, the problem becomes how to find the basis with ρ being diagonal in the quantum system with Hamiltonian as Eq. 25. The answer to this question is presented by the following theorem.

Theorem 2 (H -Diagonality Theorem, HDT [Tegmark, 2015]). *The Hamiltonian is always maximally separable (minimizing $\|H_3\|$) in the energy eigenbasis where it is diagonal.*

Furthermore, to minimize $\|H_3\|$, we must have $[H_1, H_3] = 0$, which indicates that all subsystems (*e.g.*, subsystem with Hamiltonian H_1) need to commute with all interaction Hamiltonians (*e.g.*, H_3). Following this principle, it will finally lead to a heat death, where all subsystems cease to evolve, known as the *quantum Zeno paradox*.

Definition 9 (Quantum Zeno Paradox [Tegmark, 2015]). *If we decompose our universe into maximally independent objects, then all change grinds to a halt.*

3.4 Dynamics principle

If only following the independence principle, we will face the *quantum Zeno paradox* where the system cease to evolve and has no information processing capability. However, we also require the conscious system to have certain information processing capability by the dynamics principle. The autonomy principle says the system should be able to strike a balance between the independence and dynamics principles. There are interesting classes of states ρ that provide substantial dynamics and near-perfect independence even when the interaction Hamiltonian H_3 is not small.

As a measure of the dynamics, the energy coherence is defined as:

$$\begin{aligned} \delta H &\equiv \frac{1}{\sqrt{2}} \|\dot{\rho}\| = \frac{1}{\sqrt{2}} \|i[\mathbf{H}, \rho]\| = \sqrt{\frac{-\text{tr}\{[\mathbf{H}, \rho]^2\}}{2}} \\ &= \sqrt{\text{tr}[\mathbf{H}^2 \rho^2 - \mathbf{H} \rho \mathbf{H} \rho]} \end{aligned} \quad (27)$$

Then, the maximum probability velocity can be calculated as:

$$v_{\max} = \sqrt{2} \delta H \quad (28)$$

Integration paradox. 假设我们的大脑是一个具有 n 个神经元并使用赫布学习规则的霍普菲尔德网络, 那么对于 $n = 10^{11}$ 个神经元 [Tegmark, 2015], 其整合信息的最大容量为 37 比特。然而, 一个意识体验所包含的信息量远大于此值, 例如人类在脑海中想象一幅图像。这就是所谓的整合悖论。

为什么我们意识体验的信息内容似乎远远超过 37 比特? 在量子情况下, 它永远不可能包含超过 1/4 比特的信息 [MacKay, 2003]。这一观察引出了若干猜想, 包括人脑使用比霍普菲尔德网络更好的编码方法来处理意识信息。

3.3 Independence Principle

该原理遵循整合信息理论 (IIT) 的思想, 即从系统的“最薄弱环节”将其切割为独立部分, 如第2.1节所述。它首先要求在量子情形下应用 ρ -对角性定理, 以找到互信息的最小值。

定理 1 (ρ -对角性定理, ρ DT [Jevtic et al., 2012]). 互信息总是在 ρ 对角化的基中取得其最小值。

利用 ρ DT, 问题转化为如何在哈密顿量如式 25 的量子系统中找到使 ρ 对角化的基。此问题的答案由以下定理给出。

定理 2 (H -对角性定理, HDT [Tegmark, 2015]). 哈密顿量在其对角化的能量本征基中总是最大可分离的 (即最小化 $\|H_3\|$)。

此外, 为了最小化 $\|H_3\|$, 我们必须有 $[H_1, H_3] = 0$, 这表明所有子系统 (*e.g.*, 即具有哈密顿量 H_1 的子系统) 需要与所有相互作用哈密顿量 (*e.g.*, H_3) 对易。遵循此原理, 最终将导致热寂, 即所有子系统停止演化, 这被称为 *quantum Zeno paradox*。

定义 9 (量子芝诺悖论 [Tegmark, 2015]). 如果我们将宇宙分解为最大独立的物体, 那么一切变化都将停滯。

3.4 Dynamics principle

如果仅遵循独立性原理, 我们将面临 *quantum Zeno paradox*, 即系统停止演化且不具备信息处理能力。然而, 根据动力学原理, 我们也要求意识系统具备一定的信息处理能力。自主性原理指出, 系统应能在独立性与动力学原理之间取得平衡。存在一些有趣的状态类 ρ , 它们即使在相互作用哈密顿量 H_3 并不小的情况下, 也能提供显著的动力学行为和近乎完美的独立性。

作为动力学的一种度量, 能量相干性定义为:

$$\begin{aligned} \delta H &\equiv \frac{1}{\sqrt{2}} \|\dot{\rho}\| = \frac{1}{\sqrt{2}} \|i[\mathbf{H}, \rho]\| = \sqrt{\frac{-\text{tr}\{[\mathbf{H}, \rho]^2\}}{2}} \\ &= \sqrt{\text{tr}[\mathbf{H}^2 \rho^2 - \mathbf{H} \rho \mathbf{H} \rho]} \end{aligned} \quad (27)$$

那么, 最大概率速度可计算为:

$$v_{\max} = \sqrt{2} \delta H \quad (28)$$

其中概率速度定义为 $v = \dot{p}, p_i = \rho_{ii}$ 。如果我们仅遵循动力学原理最大化 $v_{\max}$, 一些计算表明这将导致一个非常简单的动力学解, 该解不具备足够的信息处理能力。

对环境诱导退相干最鲁棒的状态是那些近似与相互作用哈密顿量对易的状态。这意味着 $[\rho, H_3] \approx 0$, 但 $[\rho, H_1] \neq 0$ 。 H_3 相互作用项将使系统退相干, 因此要求其足够小, 以便系统 ρ 能够随时间演化。

with probability velocity defined as $v = \dot{p}, p_i = \rho_{ii}$. If we only maximize $v_{\max}$ following the dynamics principle, some calculations indicate that it will lead to a very simple dynamics solution, which does not carry the capability of sufficient information processing.

The states that are most robust toward environment-induced decoherence are those that approximately commute with the interaction Hamiltonian. It means that $[\rho, H_3] \approx 0$, but $[\rho, H_1] \neq 0$. The H_3 interaction term will decohere the system thus it is required to be sufficiently small for system ρ to evolve over time.

To correctly describe the system with a balance of independence and dynamics, it needs a new metric called the *autonomy*. It first requires to use the linear entropy for quantifying the non-unitary property of an evolution.

The linear entropy is defined as:

$$S^{\text{lin}} \equiv 1 - \text{tr } \rho^2 = 1 - \|\rho\|^2 \quad (29)$$

Let us define the dynamical timescale τ_{dyn} and the independence timescale τ_{ind} as

$$\begin{aligned} \tau_{\text{dyn}} &= \frac{\hbar}{\delta H}, \\ \tau_{\text{ind}} &= \left[S_1^{\text{lin}}(0) \right]^{-1/2}. \end{aligned}$$

The autonomy can be defined as the ratio:

$$A \equiv \frac{\tau_{\text{ind}}}{\tau_{\text{dyn}}} \quad (30)$$

This ratio will exponentially increase with the system size, such that it leads to a highly autonomous system with sufficient information processing even with a large H_3 .

Summary. The theory proposed by Tegmark generalizes the IIT to the quantum domain, and analyzes the deficiency of information integration principle from the physics-principled calculation, which raises the *integration paradox* that the Hopfield neural network cannot integrate a sufficient amount of information for conscious experience. It also analyzes the independence principle and leads to the *quantum Zeno paradox* that a system decomposed into maximally independent sub-systems will cease to evolve in the end, which is in conflict with the dynamics principle. The theory finally propose the metric named *autonomy* that is found to have a high value as a balance of the independence and dynamics principles for a conscious system.

4 Orchestrated Objective Reduction Theory

4.1 Consciousness as Orchestrated Objective Reduction

Recalling the discussion of free will problem in arising consciousness in Sec.1.5, we may conjecture that the true randomness may be required for the free will to happen. How does this truly random process happen in human brains? Orchestrated objective reduction (Orch OR) theory [Penrose, 1991, 1994, Hameroff and Penrose, 1996, Hameroff, 2007, 2010, 2012, Hameroff and Penrose, 2014], builds on the hypothesis that the emergence of consciousness is due to a biological mechanism that is able to orchestrate moments of quantum state reduction. The theory posed that conscious events arises from the termination of quantum computation in the brain microtubules, framed as *objective reduction*. Objective reduction refers to the idea that a quantum system can spontaneously collapse from a superposition of multiple possibilities into a single state. In the Orch OR theory, Hameroff and Penrose propose that these objective reductions are non-deterministic but are orchestrated by certain processes in the brain. These affected objective reduction processes are called the free will. In a nutshell, the Orch OR framework based on quantum theory appears to introduce stochasticity aspects into the reductionist view of consciousness as a pure physical process. Through this, independent causal agency and free will can be explained [Hameroff, 2012]. Moreover, it will also imply the existence of consciousness in a single cell.

为了正确描述一个在独立性与动力学之间取得平衡的系统，需要一种称为 *autonomy* 的新度量。它首先要求使用线性熵来量化演化的非幺正性质。

线性熵定义为：

$$S^{\text{lin}} \equiv 1 - \text{tr } \rho^2 = 1 - \|\rho\|^2 \quad (29)$$

让我们将动力学时间尺度 τ_{dyn} 和独立性时间尺度 τ_{ind} 定义为

$$\begin{aligned} \tau_{\text{dyn}} &= \frac{\hbar}{\delta H}, \\ \tau_{\text{ind}} &= \left[S_1^{\text{lin}}(0) \right]^{-1/2}. \end{aligned}$$

自主性可定义为比值：

$$A \equiv \frac{\tau_{\text{ind}}}{\tau_{\text{dyn}}} \quad (30)$$

该比值将随系统尺寸指数增长，从而即使存在较大的 H_3 ，也能产生一个具有足够信息处理能力的高度自主系统。

Summary. Tegmark 提出的理论将整合信息理论推广到量子领域，并从物理学原理计算的角度分析了信息整合原理的不足，提出了 *integration paradox*，即 Hopfield 神经网络无法整合足够的信息以产生意识体验。该理论也分析了独立性原理，并得出 *quantum Zeno paradox*，即一个分解为最大独立子系统的系统最终将停止演化，这与动力学原理相冲突。该理论最终提出了名为 *autonomy* 的度量，该度量被发现对于意识系统而言，作为独立性与动力学原理的平衡，具有很高的值。

4 Orchestrated Objective Reduction Theory

4.1 Consciousness as Orchestrated Objective Reduction

回顾第1.2节中关于意识涌现中自由意志问题的讨论，我们可以推测，真正的随机性可能是自由意志发生所必需的。这种真正的随机过程是如何在人脑中发生的呢？协调客观还原（Orch OR）理论 [Penrose, 1991, 1994, Hameroff and Penrose, 1996, Hameroff, 2007, 2010, 2012, Hameroff and Penrose, 2014]，建立在以下假说之上：意识的涌现是由于一种能够协调量子态还原时刻的生物机制。该理论提出，意识事件源于大脑微管中量子计算的终止，被表述为 *objective reduction*。客观还原指的是量子系统可以从多种可能性的叠加态自发坍缩为单一状态的观点。在 Orch OR 理论中，Hameroff 和 Penrose 提出，这些客观还原是非决定性的，但受到大脑中某些过程的协调。这些受影响的客观还原过程被称为自由意志。简而言之，基于量子理论的 Orch OR 框架似乎将随机性方面引入了将意识视为纯粹物理过程的还原论观点中。通过这一点，独立的因果能动性和自由意志可以得到解释 [Hameroff, 2012]。此外，这也意味着单个细胞中意识的存在。

4.2 Free Will in Neurons

产生脑电图（EEG）与神经相关物（NCC）的整合与放电序列，主要由树突-胞体膜产生。随后，轴突放电输出有意识（或无意识）过程以控制行为。微管（MTs）作为细胞骨架的一部分，即细胞内部的蛋白质支架网络，被假设能够影响放电阈值。具体而言，神经元的树突-胞体微管以局部递归网络形式排列，且相较于其他细胞中的微管更为稳定，因此使其自身成为适宜的信息处理与存储单元，进而适于介导意识并调节放电。

4.2 Free Will in Neurons

The integration and firing sequences, which gives rise to EEG and NCC, are primarily generated by dendritic-somatic membranes. Then axonal firings outputs conscious (or non-conscious) processes to control behavior. Microtubules (MTs), as part of the cytoskeleton, a protein scaffolding network inside of the cell, is hypothesized to influence the threshold of firing. Specifically, Dendritic-somatic MTs of neurons are arranged in local recursive networks and are more stable comparing to MTs in other cells, therefore render itself as a suitable information processing and storage unit, moreover, suitable to mediate consciousness and regulate firing.

As shown in Fig 4, MT constitutes of peanut-shaped tubulin protein, each with a dipole and can be arranged in 13 protofilaments each with two types of hexagonal lattices. In [Hameroff and Penrose, 2014], the MT dipoles are described as electron spin (magnetic), which is inherently a quantum-mechanical quantity. Therefore all possible directions for the spin rotation axis arise as quantum superpositions of some random pair of directions. The authors then speculate that there may exists chains of spin along the pathway in MT that propagate quantum bit pairs, in addition, there may exists alternative currents at certain frequency caused by periodic spin flips. The fact that tubulins in MTs can each exists in different states (and give rise to quantum superposition) based on the dipoles position (direction), could indicate MT processes may directly result in consciousness.

4.3 Diósi-Penrose Objective Reduction

Having notated the physiological unit where quantum superposition may took place in our brain, one might ask how the orchestrated reduction that gives rise to consciousness happen? The argument starts with linking Orch OR to theoretical physics.

The Diósi-Penrose (DP) objective reduction proposal bridges quantum and classical physics as *quantum-gravitational* phenomenon, whereby the quantum superposition reduces to an average time measurement τ for the state reduction to take place according to $\tau \approx \frac{\hbar}{E_G}$, $E_G = \frac{Gm^2}{a}$, where E is space-time superposition curvature, G is gravitational constant, m is mass, a is spatial size. The actual time of decay in each event of state-reduction is taken as a random process in DP. From this, the reduction of quantum superposition of space-time objects takes place when the superposition curvature E_G reaches the threshold $\frac{\hbar}{\tau}$.

The Orch-OR schemes goes further to relate the DP physical proposal to consciousness. Hameroff and Penrose [2014] proposed that if a quantum superposition is firstly well-orchestrated: "adequately organized, imbued with cognitive information, and capable of integration and computation"; and secondly isolated from non-orchestrated, random environment for the superposition E_G to reach the threshold τ , then the Orch OR will occur along with the emergence of consciousness. An illustration of this process is shown in Fig 4.

4.4 Evidence for Objective Reduction of Quantum State

In Hameroff [2010], the authors claimed that the best measure of neural correlate of consciousness is 30- to 90-Hz gamma synchrony electroencephalography (EEG), which is largely derived from dendritic and somatic integration potentials. In addition, the theory claims that the state of anaesthesia is owing to dispersed dipoles in the MTs, responsible for quantum computing.

There are yet experiments in confirming the theory, however, biological evidence has been observed in warm conditions, where the theory has yet to extend to. Nonetheless, the Orch OR theory has provided a computational framework allowing falsification of the biological quantum theory that takes place in the MT.

5 Global Workspace Theory

5.1 The Theatre of Consciousness

Global workspace theory (GWT) is an architecture proposed by Bernard Baars [Baars et al., 1997] to explain the inner procedure of how the human brain selects and deals with consciousness attention. There are some limits to conscious capacity. For example, working memory, which temporally store

如图 4所示，微管由花生状的微管蛋白构成，每个蛋白具有偶极子，并可排列成 13 条原丝，每条原丝具有两种类型的六角晶格。在 [Hameroff and Penrose, 2014] 中，微管偶极子被描述为电子自旋（磁性），这本质上是一个量子力学量。因此，自旋转轴所有可能的方向都作为某些随机方向对的量子叠加而出现。作者随后推测，微管内部路径上可能存在沿链传播量子比特对的自旋链；此外，可能存在由周期性自旋翻转引起的特定频率的交替电流。微管中的微管蛋白各自可根据偶极子位置（方向）处于不同状态（并产生量子叠加）这一事实，可能表明微管过程可直接导致意识的产生。

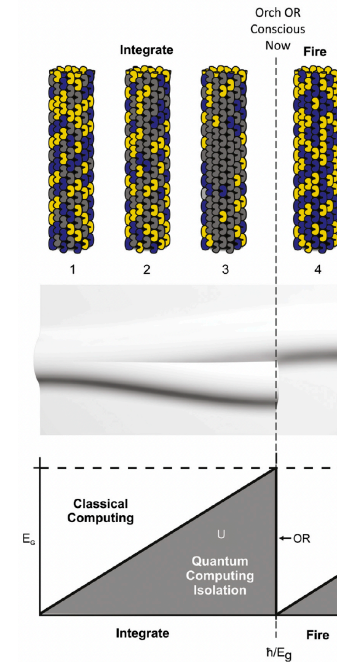


图 4: 此图发表于Hameroff and Penrose [2014], 用于阐释 Orch OR (调谐客观还原) 发生的过程。上图: 微管蛋白处于经典偶极子状态 (黄色或蓝色), 或两种偶极子状态的量子叠加 (灰色)。量子叠加/计算在 (1-3) 阶段增加。当阈值在时间 $\tau \approx \hbar/E_G$ 达到时, 意识瞬间发生。中图: 相应的交替叠加时空曲率在 OR 时刻达到阈值并选择一种时空曲率。下图: 一个意识性 Orch OR 事件的示意图, 显示了量子叠加的 U 型演化以及不断增加的 E_G , 直至达到 OR 阈值, 并通过 $\tau \approx \hbar/E_G$ 产生一个意识瞬间。

4.3 Diósi-Penrose Objective Reduction

在标注了量子叠加可能发生在我们大脑中的生理单元后, 人们可能会问, 产生意识的协调还原是如何发生的? 论证始于将协调客观还原 (Orch OR) 与理论物理学联系起来。

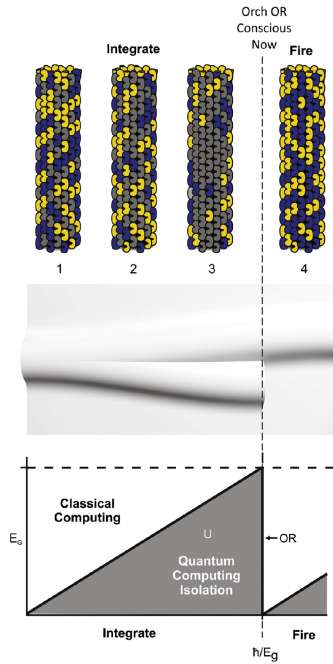


Figure 4: The figure published in Hameroff and Penrose [2014] was used to illustrate the process that Orch OR occurs. Top: Tubulins are in classical dipole states (yellow or blue), or quantum superposition of both dipole states (gray). Quantum superposition/computation increases during (1-3). The conscious moment occurs when threshold is met at time $\tau \approx \hbar/E_G$. Middle: Corresponding alternative superposed space-time curvatures reaching threshold at the moment of OR and selecting one space-time curvature. Bottom: Schematic of a conscious Orch OR event showing U-like evolution of quantum superposition and increasing E_G until OR threshold is met, and a conscious moment occurs by $\tau \approx \hbar/E_G$.

information to be dealt with, holds only several things at a time. Moreover, the human brain is only able to receive information from a single stream. ‘The theatre of consciousness’ was proposed in a metaphor term to answer how the human brain handles different inputs, and then outputs a single stream of information that draws the final attention.

There are several components of a theatre of consciousness. The ‘stage of working memory’ is the platform to receive all potential information from sensors or abstract information from cortices. The ‘spotlight’ mechanism in working memory highlights the conscious steam of information, other information on the stage is not aware by attention. Information resources, e.g. the potential thoughts, images or sensations, are regarded as ‘actors’. The information resources compete with each other to get the spotlight. The more conscious procedure is required to handle the information, the more likely the information resource will be put under the spotlight. Perceptual, intention, expectations etc. influence the result of this competition. ‘Context’ refers to unconscious networks that potentially shape conscious contents in the brain. ‘Directors’, the executive functions of human brain, guide the selection procedure with intentions and goals. The frontal cortex is believed to act as an important

迪欧西-彭罗斯（DP）的客观还原提议将量子物理学与经典物理学桥接为 *quantum-gravitational* 现象，据此量子叠加会还原到一个平均时间测量 τ ，根据 $\tau \approx \frac{\hbar}{E_G}$ 、 $E_G = \frac{Gm^2}{a}$ ，状态还原发生，其中 E 是时空叠加曲率， G 是引力常数， m 是质量， a 是空间尺寸。在 DP 理论中，每次状态还原事件的实际衰变时间被视为一个随机过程。由此，时空物体的量子叠加会在叠加曲率 E_G 达到阈值 $\frac{\hbar}{\tau}$ 时发生还原。

Orch-OR 方案更进一步，将 DP 物理提议与意识联系起来。Hameroff and Penrose [2014] 提出，如果一个量子叠加首先被良好协调：“充分组织、充满认知信息，并能够进行整合与计算”；其次与非协调的随机环境隔离，使叠加 E_G 达到阈值 τ ，那么 Orch OR 将与意识的涌现一同发生。此过程的图示见图 4。

4.4 Evidence for Objective Reduction of Quantum State

在Hameroff [2010] 中，作者声称意识神经相关性的最佳测量是 30 至 90 赫兹的伽马同步电图（EEG），这主要源于树突和胞体的整合电位。此外，该理论声称麻醉状态是由于微管中偶极子的分散所致，而微管负责量子计算。

目前尚有待实验证实该理论，然而，在温暖条件下已观察到生物学证据，而该理论尚未扩展到这些条件。尽管如此，Orch OR 理论提供了一个计算框架，允许对发生在微管中的生物量子理论进行证伪。

5 Global Workspace Theory

5.1 The Theatre of Consciousness

全局工作空间理论（GWT）是由伯纳德·巴尔斯 [Baars et al., 1997] 提出的一种架构，用以解释人脑如何选择和处理意识注意的内部过程。意识容量存在一些限制。例如，工作记忆临时存储待处理的信息，一次只能容纳少数几项内容。此外，人脑只能接收来自单一信息流的信息。‘意识剧场’这一隐喻术语被提出来回答人脑如何处理不同的输入，然后输出一个吸引最终注意力的单一信息流。

意识剧场包含若干组成部分。‘工作记忆舞台’是接收来自传感器的所有潜在信息或来自皮层的抽象信息的平台。工作记忆中的‘聚光灯’机制会突出显示意识信息流，舞台上的其他信息则不被注意力所察觉。信息资源，例如潜在的思想、图像或感觉，被视为‘演员’。这些信息资源相互竞争以获取聚光灯。处理信息所需的意识过程越多，该信息资源就越有可能被置于聚光灯下。知觉、意图、期望等会影响这种竞争的结果。‘语境’指的是大脑中可能塑造意识内容的无意识网络。‘导演’，即人脑的执行功能，通过意图和目标来指导选择过程。额叶皮层被认为在此过程中扮演重要角色，因为额叶损伤会导致基于长期目标的行为丧失。然后，聚光灯下的信息被广播给‘观众’，观众代表需要该信息的大脑区域。

2003 年 GWT 的一次更新 [Baars, 2003] 更详细地介绍了 GWT 与大脑功能之间的关系，为竞争如何在脑中执行提供了一些证据。额叶皮层和其他可能中断聚光灯控制的大脑区域都参与了意识事件的选择过程。后者的中断控制包括例如脑干、疼痛系统和情绪中枢，它们允许中断选择过程，并赋予更显著和紧急的活动以权重。大脑的‘语境’功能被认为参与了意识决策过程。与身体部位自我意识相关的顶叶皮层并不直接客观地与意识相关联，但被认为塑造了有意识的视觉事件。‘自我’系统可能参与了意识的产生。研究发现，裂脑患者的左右半球具有不同的执行和知觉功能 [Gazzaniga et al., 1996]，并且左前额叶皮层通过一个‘叙事自我’来处理感觉信息，该自我可以引发不同的意识，从而导致两个半球之间的冲突。然后左半球试图合理化并修复这些冲突。该论文还提供了一些证据来支持意识内容被广播并分布到大脑

role in this procedure with the fact that damage to the front lobe leads to loss of actions by long-term goals. Then the information under the spotlight is broadcasted to the 'audience', which represents the brain region which requires the information.

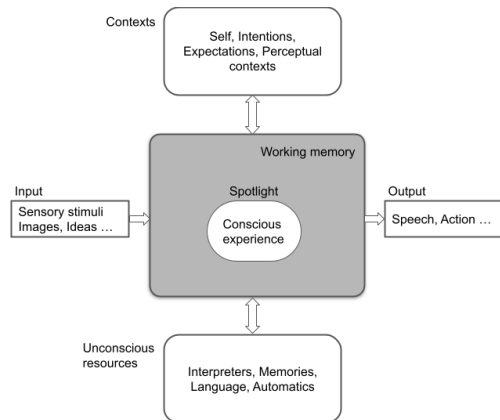


Figure 5: Scheme diagram of GWT derived from Baars et al. [1997].

An update of GWT in 2003 [Baars, 2003] gives a more detailed introduction to the relationships between GWT and Brain functions, which provides some evidence of how the competition is performed in the human brain. Both the frontal cortex and other brain regions, which can interrupt the spotlight control, are involved in the conscious event selection procedure. The latter interrupt control consists of, for example, the brain stem, pain system, and emotional centers, which allow interrupting the selection procedure and give weight to more significant and urgent activities. The 'Context' function of the brain is believed to be involved in the conscious decision process. The parietal cortex which is related to self-awareness of parts of the body is not directly objectively linked to consciousness but is believed to shape conscious visual events. The 'Self' system may be involved in the generation of consciousness. It was found that split-brain patients have different executive and perceptual functions from the left and right hemisphere [Gazzaniga et al., 1996] and the left prefrontal cortex processes the sensory information with a 'narrative self' that can draw different awareness which causes conflicts between both hemispheres. Then the left hemisphere tries to rationalize and repair such conflicts. Some evidence was also provided in the paper to support the assumption that consciousness contents are broadcasted and distributed to brain regions. In a visual word task, the word task not only triggers visual word recognition areas of the cortex but also was found to evoke activities in the parietal and prefrontal cortex [Baars, 2002].

5.2 Computational Models of GWT

The Intelligent Distribution Agent (IDA) [Baars and Franklin, 2003, 2007] and LIDA (Learning IDA) [Baars and Franklin, 2009] computational models were proposed based on GWT to perform human-like tasks. In the study, naval jobs of sailors are used as an example task to test the model. IDA and LIDA contain several blocks reflecting the GWT, including **sensory modules** to deal with stimulus, **memory modules** as storage, **attention modules** referring to the concept of attention, the **action module** for action selection. Particularly, as in the GWT, **global workspace modules** integrates and broadcasts information, as well as selects the most relevant and important information to be on the stage. This model shows an empirical computational implementation of the GMT model as conceptual evidence that GMT could work out human-like functions. IDA or LIDA turns incoming

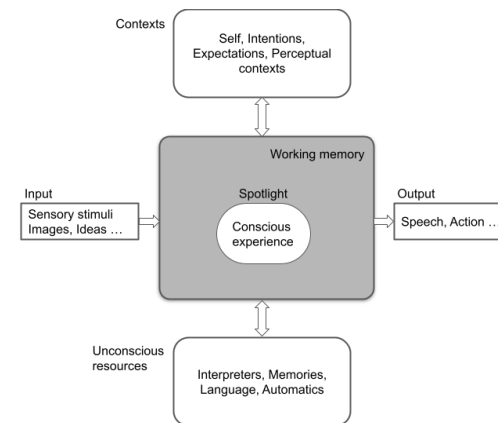


图 5: 源自 Baars et al. [1997] 的 GWT 示意图。

区域的假设。在一项视觉词汇任务中，词汇任务不仅触发了皮层的视觉词汇识别区域，还发现能唤起顶叶和前额叶皮层的活动 [Baars, 2002]。

5.2 Computational Models of GWT

基于全局工作空间理论（GWT），研究者提出了智能分配代理（IDA）[Baars and Franklin, 2003, 2007] 和学习型 IDA（LIDA）[Baars and Franklin, 2009] 计算模型，用以执行类人任务。在该研究中，以水手的海军工作为例证任务来测试该模型。IDA 和 LIDA 包含多个反映 GWT 的模块，包括用于处理刺激的 **sensory modules**、作为存储的 **memory modules**、指代注意力概念的 **attention modules**，以及用于动作选择的 **action module**。特别是，正如在 GWT 中一样，**global workspace modules** 整合并广播信息，同时选择最相关和最重要的信息登上舞台。该模型展示了 GMT 模型的一种经验性计算实现，作为概念性证据表明 GMT 可以实现类人功能。IDA 或 LIDA 将传入的感觉数据转化为对环境的行为。记忆、竞争和广播等概念都参与了 this 转换过程。随后，对环境产生的行为改变了系统的输入，从而形成了一个迭代的认知循环。

GWT 启发了相关领域的一些研究，这里我们描述几个在脑信号分析和深度学习中的例子。受 GWT 启发，[Schutter and van Honk, 2004] 使用脑电图相干性（不同脑区连接水平的指标）来衡量情绪是否在意识中起作用。另一项研究 [Bartolomei and Naccache, 2011]，基于 GWT 中的广播和分发过程，比较了癫痫发作时远距离皮质-皮质和皮质-丘脑网络内部的同步性与 GWT 中不同脑区之间的远距离关系。最近，一项研究探讨了用深度学习实现 GWT 的可能性。研究者提出了全局潜在工作空间（GLW）的概念，以反映类脑机制的深度学习设计原则 [VanRullen and Kanai, 2021]。

sensory data into actions to the environment. The concepts of memory, competition, and broadcasting are involved in the conversion process. Then the resulting action to the environment changes the inputs of the system which forms an iterating cognitive cycle.

GWT has inspired some studies in related fields, here we describe several examples in brain signal analysis and deep learning. Inspired by GWT, [Schutter and van Honk, 2004] used EEG coherence, the level of connectivity of different brain region, to measure if emotions play a role in consciousness. Another study [Bartolomei and Naccache, 2011], in light of the broadcasting and distributing process in the GWT, compared the synchrony within distant cortico-cortical and cortico-thalamic networks of epileptic seizures with the distant relationships across different brain regions in GWT. More recently, a study discussed the possibility of implementing GWT with deep learning. The idea of Global Latent Workspace (GLW) was proposed to reflect deep learning design principles of brain-like mechanisms [VanRullen and Kanai, 2021].

6 Higher-Order Theories

A general definition of HOT is given by Carruthers and Gennaro [2023]: A phenomenally conscious mental state is a mental state (of a certain sort) that either is, or is disposed to be, the object of a higher-order representation of a certain sort.

Depending on whether the higher-order states in question are perception-like or thought-like, the high-order theories are categorized as high-order perception theory (HOPT), high-order thought theory (HOTT), self-representational theory (SRT), etc. Definitions of HOPT (Sec. 6.1 Def. 10), HOTT (Sec. 6.2 Def. 11) and SRT (Sec. 6.3 Def. 12) are described in details in the following sections. Additionally, we will introduce and discuss other relevant theories in the final section. To provide a concise overview, we consolidate these theories into a summarizing Table 2.

Higher-order theories (HOTs) try to answer the question that if a mental state is conscious or unconscious. The higher-order theory believes that there is a certain brain mechanism that is more advanced than the first-order information (e.g. senses from organs like visual or auditory nerves). Three sub-theories claimed different explanations for the higher-order mechanism. HOPT believes that there are inner senses that scan or refine but are independent of the first-order information. This explains why people are able to imagine feelings like pain. In HOTT, it is believed that a mental state is conscious when it is the subject of higher-order thought. They propose that a conscious mental state or event is either actually causing or is disposed to cause an activated thought that a person has the state or event. Self-representational theory proposed another explanation to the higher-order theory. The self-representational theory believes that the higher-order state is constitutive or internal to its first-order state, i.e. that the higher-order state is formed from the first-order states and as a more complex system than the first-order states that generate awareness.

6.1 Higher-Order Perception Theory

The high-order perception theory (HOPT) theory [Locke, 1948, Armstrong, 1981, Armstrong and Malcolm, 1985, Lycan, 1996, Armstrong, 2002, Lycan, 2004], which is also called the Inner-Sense Theory or High-Order-Sense Theory, is referring to the followings: Humans not only have the sense-organs to scan the environment and their own bodies to produce the representations, which are called the *first-order* non-conceptual and/or analog perceptions of environment/body states, but also have inner senses of those first-order senses to generate equally fine-grained but higher-order representations, which are called the *second-order* non-conceptual and/or analog perceptions of the *first-order* perception states.

The definitions of the *first-order* perception and the *second-order* perception are actually close to the *M*-consciousness and *I*-consciousness in the attention schema theory, which will be introduced in later Sec. 7. We will discuss the connections of the two theories later in Sec. 7.3.

Definition 10 (Higher-Order Perception Theory/Inner-Sense Theory [Carruthers and Gennaro, 2023]). *A phenomenally conscious mental state is a state with analog/non-conceptual intentional content, which is in turn the target of a higher-order analog/non-conceptual intentional state, via the operations of a faculty of ‘inner sense’.*

6 Higher-Order Theories

高阶理论（HOT）的一个普遍定义由 Carruthers and Gennaro [2023] 给出：一种现象上意识到的心理状态是一种（特定类型的）心理状态，它要么是，要么倾向于成为某种特定类型的高阶表征的对象。

根据所讨论的高阶状态是感知类还是思想类，高阶理论被分类为高阶感知理论（HOPT）、高阶思想理论（HOTT）、自我表征理论（SRT）等。HOPT（第 6.1 定义 10）、HOTT（第 6.2 定义 11）和 SRT（第 6.3 定义 12）的定义将在后续章节中详细描述。此外，我们将在最后一节介绍并讨论其他相关理论。为提供一个简洁的概览，我们将这些理论整合成一个总结性的表格 6。

高阶理论（HOTs）试图回答一个心理状态是有意识还是无意识的问题。高阶理论认为，存在某种比一阶信息（例如来自视觉或听觉神经等器官的感觉）更高级的大脑机制。三个子理论对这种高阶机制提出了不同的解释。HOPT 认为存在内在感官，它们扫描或精炼但独立于一阶信息。这解释了人们为何能够想象诸如疼痛等感觉。在 HOTT 中，人们相信当一个心理状态成为高阶思想的对象时，它才是有意识的。他们提出，一个有意识的心理状态或事件要么实际上正在引起，要么倾向于引起一个人拥有该状态或事件的激活思想。自我表征理论对高阶理论提出了另一种解释。自我表征理论认为，高阶状态是其对应一阶状态的构成性或内在部分，即高阶状态由一阶状态形成，并且作为一个比产生意识的一阶状态更复杂的系统。

高阶理论概览			
理论	一阶 & 高阶关系	核心机制	参考文献
高阶知觉理论	高阶感知存在且独立于一阶信息	知觉式：人脑具有对一阶感知的内在感知，以生成高阶表征	[Locke, 1948, Armstrong, 1981, Armstrong and Malcolm, 1985, Lycan, 1996, Armstrong, 2002, Lycan, 2004]
高阶思维理论	现实论	当心理状态成为高阶思维的目标时，它便是有意识的	[Rosenthal, 1986, 1993, 2005]
	倾向论	当心理状态成为高阶思维的目标时，它便是有意识的	[Dennett, 1978, Carruthers, 1998]
自我表征理论	部分-整体	一阶与高阶是一个整体复合体的组成部分	[Kriegel, 2009, Picciuto, 2011]
	同一性	高阶与一阶是同一的	[Caston, 2002, Carruthers, 2005, Van Gulick, 2004]

Table 2: A summary of HOTs

Theory	First-order & Higher-order Relationship	Key Mechanism	References
HOPT	higher-order senses exist and are independent of first-order information	perception-like; the human brain has inner senses of the first-order senses to generate higher-order representations	[Locke, 1948, Armstrong, 1981, Armstrong and Malcolm, 1985, Lycan, 1996, Armstrong, 2002, Lycan, 2004]
HOTT	Actualist	a mental state is conscious when it is the target of a higher-order thought	[Rosenthal, 1986, 1993, 2005]
	Dispositional	a mental state is conscious when it is the target of a higher-order thought	[Dennett, 1978, Carruthers, 1998]
SRT	Part-whole	first-order and higher-order are parts of a whole complex	[Kriegel, 2009, Picciuto, 2011]
	Identity	higher-order and first-order are identical	[Caston, 2002, Carruthers, 2005, Van Gulick, 2004]

A formal proposition of HOPT is provided as Definition 10. It explains consciousness as the higher-order states generated from inner sensing of the first-order states. Referring to Fig. 3 in Sec. 1.5, the conscious module is a higher-level component perceptive of the information flow of the lower-level sensing modules.

The antagonistic viewpoint of HOPT is held by some theorists that the attention mechanism on the first-order states may serve as a substitute of the higher-order states [Sauret and Lycan, 2014].

6.2 Higher-Order Thought Theory

The higher-order thought theory (HOTT) [Rosenthal, 2009, 2012, 2004, Byrne, 1997, Brown et al., 2019] propose that a conscious mental state or event is either actually causing or is disposed to cause an activated thought that a person has the mental state or event. There are two embranchments of the theory: the **actualist** and the **dispositionalist**. In the above statement, the actualists believe that mental state directly caused the activated thought while the dispositionalist believes that the mental state is disposed to the thought. Another difference between them is that the actualist HOTT requires actual involvement of the first-order information in order to compute the higher-order information. On the contrary, the dispositionalist HOTT states that the higher-order computation only requires the availability of the first-order information, for example utilising board casting in the global workspace theory, instead of directly accessing all first-order information. In [Lau and Rosenthal, 2011], some empirical support for the higher-order theories was discussed, e.g. the association of conscious awareness with prefrontal mechanisms and evidence based on clinical disorders.

Definition 11 (Higher-Order Thought Theory [Carruthers and Gennaro, 2023]). *A phenomenally conscious mental state is a state of a certain sort (e.g. with analog/non-conceptual intentional content, perhaps) which is the object of a higher-order thought, and which causes that thought non-inferentially.*

6.1 Higher-Order Perception Theory

高阶知觉理论 (HOPT) 理论 [Locke, 1948, Armstrong, 1981, Armstrong and Malcolm, 1985, Lycan, 1996, Armstrong, 2002, Lycan, 2004], 亦称为内在感官理论或高阶感官理论, 指的是以下内容: 人类不仅拥有感官器官来扫描环境及自身身体以产生表征, 这些表征被称为对环境/身体状态的 *first-order* 非概念性和/或模拟知觉, 而且还拥有对这些一阶感官的内在感官, 以产生同样精细但更高阶的表征, 这些表征被称为对 *first-order* 知觉状态的 *second-order* 非概念性和/或模拟知觉。

first-order 知觉与 *second-order* 知觉的定义实际上接近于注意图式理论中的 *M*-意识与 *I*-意识, 这将在后面的第 7 节介绍。我们将在第 7.3 节稍后讨论这两种理论之间的联系。

定义 10 (高阶知觉理论/内在感官理论 [Carruthers and Gennaro, 2023]). 一个现象上具有意识的心理状态是一个具有模拟/非概念性意向内容的状态, 而该状态又通过“内在感官”官能的运作, 成为一个更高阶的模拟/非概念性意向状态的目标。

HOPT 的一个正式命题作为定义10提供。它将意识解释为通过对一阶状态进行内在感知而产生的高阶状态。参考第 1.2 节中的图 3, 意识模块是一个更高层级的组件, 它能感知较低层级传感模块的信息流。

HOPT 的对立观点由一些理论家持有, 他们认为对一阶状态的注意机制可能作为高阶状态的替代 [Sauret and Lycan, 2014]。

6.2 Higher-Order Thought Theory

高阶思维理论 (HOTT) [Rosenthal, 2009, 2012, 2004, Byrne, 1997, Brown et al., 2019] 提出, 一个有意识的心理状态或事件要么实际导致、要么倾向于导致一个人产生自己拥有该心理状态或事件的激活思维。该理论有两个分支: **actualist** 和 **dispositionalist**。在上述陈述中, 现实主义者认为心理状态直接导致了激活思维, 而倾向主义者则认为心理状态倾向于产生该思维。它们之间的另一个区别是, 现实主义 HOTT 要求一阶信息实际参与计算, 以生成高阶信息。相反, 倾向主义 HOTT 认为高阶计算仅需要一阶信息的可用性, 例如利用全局工作空间理论中的广播机制, 而不是直接访问所有一阶信息。在 [Lau and Rosenthal, 2011] 中, 讨论了对高阶理论的一些实证支持, 例如意识知觉与前额叶机制的关联以及基于临床障碍的证据。

定义 11 (高阶思维理论 [Carruthers and Gennaro, 2023]). 一个现象上有意识的心理状态是某种特定类型的状态 (例如, 具有模拟/非概念性意向内容), 它是高阶思维的对象, 并且以非推理的方式引起该思维。

基于高阶思维理论, 一些计算模型被提出。元认知神经网络模型 [Pasquali et al., 2010, Cleeremans et al., 2007, Timmermans et al., 2012] 由两个网络组成: 一阶网络和二阶网络。一阶网络直接接收输入并通过隐藏单元进行处理。之后, 一阶网络的隐藏单元连接到二阶网络进行进一步处理。一阶网络学习执行任务分类, 而二阶网络通过访问一阶网络的表征信息来预测一阶网络的置信度。这些模型在多项任务上进行了测试和报告, 例如爱荷华赌博任务 [Pasquali et al., 2010]、区分语法与非语法句子的人工语法学习 (AGL) 任务, 以及盲视任务 [Persaud et al., 2007], 其中盲视患者在缺乏视觉意识的情况下进行视觉辨别。

6.3 Self-Representational Theory

自我表征理论 (SRT) [Kriegel, 2009, Van Gulick, 2004, Picciuto, 2011] 提出了现象意识心理状态的概念, 这种状态同时具有非概念性意向内容和概念性意向内容。据说这种心理状态向作为该状态主体的个人表征了其自身。

定义 12 (自我表征理论 [Carruthers and Gennaro, 2023]). 现象意识心理状态是一种特定类型的状态 (可能具有模拟/非概念性意向内容), 它同时拥有一种意向内容, 从而在某种意义上向作为该状态主体的个人表征了其自身。

该理论的两个分支论证了意识状态与高阶状态之间的构成性关系是 **identity** [Caston, 2002, Carruthers, 2005, Van Gulick, 2004], 或是 **part-whole** [Kriegel, 2009, Picciuto, 2011]。前者认为意识状态既是第一

Some computational models were proposed based on higher-order thought theories. Metacognition neural network models [Pasquali et al., 2010, Cleeremans et al., 2007, Timmermans et al., 2012] consist of two networks: a first-order network and a second-order network. The first-order network directly gets inputs and processes them with hidden units. After that, the hidden units of the first-order network are linked to the second-order network for processing. The first-order network learns to perform the classification of tasks, and the second-order network predicts the confidence of the first-order network by accessing the representational information of the first-order network. The models were tested and reported on several tasks, e.g. the Iowa Gambling Task [Pasquali et al., 2010], artificial grammar learning (AGL) tasks which distinguish grammar and non-grammar sentences, and blindsight tasks [Persaud et al., 2007] in which blindsight patients make visual discriminations in the absence of visual awareness.

6.3 Self-Representational Theory

The Self-Representational Theory (SRT) [Kriegel, 2009, Van Gulick, 2004, Picciuto, 2011] presents the idea of phenomenally conscious mental state, which is a state with non-conceptual intentional content, and conceptual intentional content at the same time. Such a mental state is said to representing itself to the person who is the subject of that state.

Definition 12 (Self-Representation Theory [Carruthers and Gennaro, 2023]). *A phenomenally conscious mental state is a state of a certain sort (perhaps with analog/non-conceptual intentional content) which also, at the same time, possesses an intentional content, thereby in some sense representing itself to the person who is the subject of that state.*

Two branches of the theory have argued for the constitutive relation between the conscious state and higher-order state is one of **identity** [Caston, 2002, Carruthers, 2005, Van Gulick, 2004], or **part-whole** [Kriegel, 2009, Picciuto, 2011]. The former argues that the conscious state is both first-order and higher-order. More precisely, a first-order perceptual state with analog content acquires, at the same time, a higher-order analog content. The part-whole SRT take stance similar to actualist HOT theory arguments, where the first-order perceptual state gives rise to higher-order thought that represents experience.

6.4 Other theories and perspectives in HOT

The same-order theory [Kriegel, 2009, Brentano, 1973, Lau and Rosenthal, 2011] proposes that conscious mental states are not represented by any other mental states, but are instead directly present to the subject's awareness. The higher-order statistical inference view [Lau, 2011, 2007] believes conscious mental states involve higher-order statistical inferences about one's own mental states. According to this view, first-order representation is reviewed by a higher-order inference procedure to form a statistically reliable perceptual signal, similar to perceptual decision-making process. From the perspective of the radical plasticity thesis [Cleeremans, 2011, Pasquali et al., 2010], the brain is capable of remarkable adaptability and flexibility, and this plasticity plays a critical role in the development of conscious awareness. The radical plasticity thesis proposes that consciousness is not an intrinsic procedure but a learning process in the brain. The brain engages in continuous and unconscious learning to re-describe its activity, thereby developing systems of meta-representations that describe and refine the initial, first-order representations.

7 Attention Schema Theory

7.1 Formulation

It is important to understand the difference of attention and awareness and their relationship in Attention Schema Theory (AST) [Graziano and Webb, 2015, Graziano et al., 2020].

Definition 13 (Attention). *Attention is a process where the brain selectively process certain pieces of information more than others.*

As one of the most influential explanation of the attention process, people [Desimone et al., 1995] propose that there is a signal competition process emerging at the earliest stages of signal processing and existing in every later stages.

阶的也是高阶的。更准确地说，一个具有模拟内容的第一阶知觉状态同时获得了一个高阶的模拟内容。部分-整体 SRT 采取的立场类似于现实主义高阶思想理论的论证，即第一阶知觉状态产生了表征经验的高阶思想。

6.4 Other theories and perspectives in HOT

同阶理论 [Kriegel, 2009, Brentano, 1973, Lau and Rosenthal, 2011] 提出，意识心理状态并非由任何其他心理状态所表征，而是直接呈现于主体的知觉之中。高阶统计推断观点 [Lau, 2011, 2007] 认为，意识心理状态涉及关于自身心理状态的高阶统计推断。根据这一观点，第一阶表征通过一个高阶推断程序进行审查，以形成一个统计上可靠的知觉信号，类似于知觉决策过程。从激进可塑性论题

[Cleeremans, 2011, Pasquali et al., 2010] 的视角来看，大脑具有显著的适应性和灵活性，这种可塑性在意识觉知的发展中起着关键作用。激进可塑性论题提出，意识并非一个内在过程，而是大脑中的一个学习过程。大脑进行持续且无意识的学习，以重新描述其活动，从而发展出描述并精炼初始的第一阶表征的元表征系统。

7 Attention Schema Theory

7.1 Formulation

理解注意力与意识的区别及其在注意力图式理论（AST）中的关系至关重要 [Graziano and Webb, 2015, Graziano et al., 2020]。

定义 13 (注意力). 注意力是大脑选择性地处理某些信息多于其他信息的过程。

作为对注意力过程最具影响力的解释之一，人们 [Desimone et al., 1995] 提出，在信号处理的最早阶段就存在一个信号竞争过程，并且该过程存在于之后的每一个阶段。

意识是一个不同于注意力的概念。尽管意识与注意力通常高度相关，但它们可以分离。对于意识这个概念，我们首先需要区分客观意识和主观意识。这两种意识都涉及一个参与者。

定义 14 (客观意识). 对于客观意识，要求参与者报告他客观上意识到了刺激。

定义 15 (主观意识). 对于主观意识，参与者根据自己的看法报告是否感知到了刺激。

客观意识与主观意识的区别仅在于区分参与者是“看到”还是“猜测”所感知的刺激。

在 AST 中，作者将 *awareness*、*consciousness* 和 *subjective experience* 视为与 *subjective awareness* 相同的概念。因此，AST 是关于主观意识的理论，而非 *objective awareness*。

AST 提出，意识是注意力的一个模型。设想一个人看到一个苹果，苹果的视觉表征（V）通过注意力过程出现在此人的脑海中。然而，这还不足以使此人意识到“苹果”。为了产生意识，心智还有一个自我模型（S），而 S 对 V 的注意力（A）也是意识的一部分。根据 AST，主观意识是 [S+A+V]，它是注意力过程的一个模型。

Awareness is a different concept from the attention. Although awareness and attentions are typically highly correlated, they can be dissociated. For the concept of awareness, we need to first distinguish objective awareness and subjective awareness. Both of the awareness involves a participant.

Definition 14 (Objective Awareness). *For objective awareness, the participant is required to report that he is objectively aware of the stimulus.*

Definition 15 (Subjective Awareness). *For subjective awareness, the participant reports whether he has perceived the stimulus in his own opinion.*

The difference of objective awareness and subjective awareness is just to distinguish whether the participant ‘sees’ or ‘guesses’ the perceived stimulus.

In AST, the authors refer the *awareness*, *consciousness* and *subjective experience* as the same concept as the *subjective awareness*. Therefore, AST is a theory about subjective awareness, not *objective awareness*.

AST proposes that awareness is a model of attention. Thinking of a person seeing an apple, the visual representation of the apple (V) appears in the person’s mind through the attention process. However, this is not enough for making the person aware of the ‘apple’. To generate awareness, the mind also has a model of self (S), and the attention (A) of S to V is also part of the awareness. By AST, a subjective awareness is [S+A+V], which is a model of the attention process.

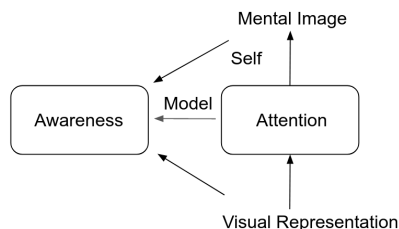


Figure 6: Awareness as a model of attention in AST.

People may ask why subjective awareness is required beyond the attention? According to AST, subjective awareness allows for self-modelling which is essential for model-based control. Without awareness is like without modelling the arm when a person tries to reach some objects, and it will lead to inaccurate prediction about the arm’s position therefore bad reaching result. The awareness serves as an internal modelling of the mind itself and the attention, leading to more accurate model-based control for human.

7.2 I-consciousness and M-consciousness

As the primary researchers of AST, Graziano et al. [2020] proposes to interpret the consciousness of human mind via the *I-consciousness* (*I* for information) and *M-consciousness* (*M* for mysterious) in their later study.

I-consciousness indicates the process of signals winning attentional competition, just as in GWT, which is generally assumed to computationally feasible [Baars and Franklin, 2007]. However, the mysterious part is the *M-consciousness*, which is used to explain the subjective experience of perceiving the winning piece of information.

Similar as the previous work of AST [Graziano and Webb, 2015], that awareness is a model of attention, Graziano et al. [2020] further proposes that *M-consciousness* is a natural, built-in, imperfect model of *I-consciousness*. Moreover, the *I-consciousness* and *M-consciousness* can be mutually involved, like a mirror of another mirror. A person is *I-conscious* of having his *M-consciousness*, and *M-consciousness* is a model of *I-consciousness*. For the self-modeling, the modeled parts are the

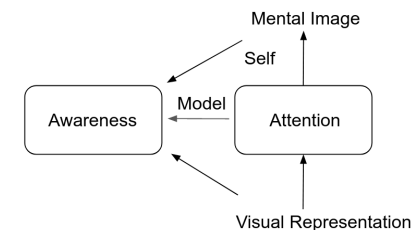


图 6: AST 中作为注意力模型的意识。

人们可能会问，为什么在注意力之外还需要主观意识？根据 AST，主观意识允许进行自我建模，这对于基于模型的控制至关重要。没有意识，就如同一个人试图伸手够物体时没有对手臂进行建模，这将导致对手臂位置预测不准确，从而造成不良的够取结果。意识作为心智本身及其注意力的内部模型，为人类带来了更准确的基于模型的控制。

7.2 I-意识与 M-意识

作为 AST 的主要研究者，Graziano et al. [2020] 在其后续研究中提出通过 *I-意识* (*I* 代表信息) 和 *M-意识* (*M* 代表神秘) 来解释人类心智的意识。

I-意识 表示信号在注意力竞争中胜出的过程，正如全局工作空间理论 (GWT) 中所描述的那样，通常被认为在计算上是可行的 [Baars and Franklin, 2007]。然而，神秘的部分是 *M-意识*，它被用来解释感知获胜信息片段的主观体验。

类似于 AST 先前的工作 [Graziano and Webb, 2015]，即觉知是注意力的一个模型，Graziano et al. [2020] 进一步提出 *M-意识* 是 *I-意识* 的一个自然的、内置的、不完美的模型。此外，*I-意识* 和 *M-意识* 可以相互关联，就像一面镜子映照另一面镜子。一个人能够 *I-意识* 到自己拥有 *M-意识*，而 *M-意识* 是 *I-意识* 的一个模型。对于自我建模而言，被建模的部分是与注意力竞争中获胜信息片段最密切相关的自我物理组成部分。

人们可能会进一步追问，为何我们认为主观体验是真实的。根据 AST，两个特性确保了真实的主观体验：第一是主观体验无法被关闭。第二是心智能够对感知到的信息进行来源监控，这使得人们能够区分真实与假设。由于 *M-意识* 是 *I-意识* 的一个模型，其感觉之所以真实，其理由与人们相信所有物理对象都是真实的完全相同。

AST 的一个实际实现将包含三个组成部分。网络 *A* 代表通过注意力竞争进行的选择性信息处理。网络 *B* 旨在通过对 *A* 的输出进行预测，来建模网络 *A* 的功能。网络 *C* 接收来自 *A* 和 *B* 的输出，以生成对大脑内部及外部世界其他组成部分的报告 (e.g., 言语)。网络 *B* 是重要的注意力图式，其在大脑中的物理对应部分已被认为服务于一个与颞顶联合区 (TPJ) 部分重叠的皮层网络 [Graziano and Kastner, 2011, Graziano, 2016]。

7.3 AST as a Unification of GWT and HOT

AST 可被视为全局工作空间理论 (GWT, 第 5 节) [Baars, 1993, Dehaene, 2014, Dehaene and Changeux, 2011] 与高阶理论 (HOT, 第 6 节) [Gennaro, 2011, Lau and Rosenthal, 2011, Rosenthal et al., 1991, Rosenthal, 2005] 的统一。

具体而言，GWT 解释了信息出现在心智舞台上的注意图式，这对应于 AST 的 *I-意识*。然而，GWT 并未解释意识体验 (即 *M-意识*) 的存在。AST 通过使用网络 *B* 构建注意图式来解释这一谜题。

physical components of the self most closely correlates with the winning piece of information from the attention competition.

People may further ask why we think the subjective experience is realistic. According to AST, the two properties ensure the realistic subjective experience: The first one is that the subjective experience cannot be turned off. The second one is that the mind enables source monitoring for the perceived information, which allows people to distinguish the real and the hypothetical. Since *M*-consciousness is a model of *I*-consciousness, the feeling is realistic for the exactly same reason that the people would believe every physical objects are real.

A practical implementation of AST would involve three components. The network *A* represents the selective information process by the attention competition. The network *B* is to model the function of network *A* by making predictions on *A*'s output. Network *C* receives the output from *A* and *B* to generate the report (e.g., speech) to other components within the brain and outside world. The network *B* is the important attention schema, for which the physical counterpart in the brain has been suggested to serve, as a cortical network overlapping part of the temporoparietal junction (TPJ) [Graziano and Kastner, 2011, Graziano, 2016].

7.3 AST as a Unification of GWT and HOT

AST can be viewed as a unification of global workspace theory (GWT, Sec. 5) [Baars, 1993, Dehaene, 2014, Dehaene and Changeux, 2011] and higher-order theory (HOT, Sec. 6) [Gennaro, 2011, Lau and Rosenthal, 2011, Rosenthal et al., 1991, Rosenthal, 2005].

Specifically, the GWT explains the attention schema for a piece of information to appear on the stage of the mind, which corresponds to the *I*-consciousness of AST. However, GWT does not explain the existence of consciousness experience, as the *M*-consciousness. AST explains this mystery by constructing the attention schema using network *B*.

The HOT says consciousness arises from the higher-order representation. Recall the introduction of HOT in Sec. 6, a conscious system has inner senses of those first-order senses to generate equally fine-grained but higher-order representations, which are called the *second-order* perceptions of the *first-order* perception states. The *I*-consciousness in AST represents the external first-order senses, while the *M*-consciousness corresponds to the *second-order* perceptions over the first-order senses. These *second-order* perceptions can be thought of as a modelling process of the first-order senses. AST assumes the brain to construct higher-order representation of the global workspace, as imperfect modelling of the *I*-consciousness, which unifies the HOT and GWT to give explanations of subjective awareness.

8 Conscious Turing Machine

8.1 Formulation

In traditional Turing machine (TM) [Turing, 2009], Turing does not involve the subjective experience into the concept of TM. The TM is only about the computational intelligence but not consciousness of the machine, whereas the latter one is usually considered a hard problem [Chalmers, 1995].

Conscious Turing machine (CTM) [Blum and Blum, 2022] is a theory as an extended concept of TM. Compared with TM's model of computation, CTM empowers the system a distinguishable feature, i.e., the "feeling of consciousness". Specifically, CTM is defined as following.

Definition 16 (Conscious Turing Machine). *CTM is defined as a seven-element tuple: $\langle STM, LTM, Up Tree, Down Tree, Links, Input, Output \rangle$, where STM and LTM are shorten for short-term memory and long-term memory.*

The CTM can be viewed as GWT with a more sophisticated structure. STM is an analogue of the "stage" in GWT as a necessary component for the consciousness to happen. As an analogue of the "audience" in GWT, LTM is a large collection of general processors, including the *Model of the World* processor for modeling the world and the agent itself, *Inner Speech* processor for processing linguistic information, and other *Inner Generalized Speech* processors for handling information inputs like five senses. These processors are called LTM since they have a relatively stable status and expertise

HOT 认为意识产生于高阶表征。回顾第 6 节中对 HOT 的介绍，一个意识系统具有对那些一阶感觉的内在感知，以生成同等精细但更高阶的表征，这些被称为 *first-order* 感知状态的 *second-order* 感知。AST 中的 *I*-意识代表外部的一阶感觉，而 *M*-意识则对应于对一阶感觉的 *second-order* 感知。这些 *second-order* 感知可被视为对一阶感觉的建模过程。AST 假设大脑构建全局工作空间的高阶表征，作为对 *I*-意识的不完美建模，从而统一了 HOT 与 GWT，为主观感知提供解释。

8 Conscious Turing Machine

8.1 Formulation

在传统的图灵机 (TM) [Turing, 2009] 中，图灵并未将主观体验纳入图灵机的概念。图灵机仅关乎计算智能，而非机器的意识，而后者通常被认为是一个难题 [Chalmers, 1995]。

意识图灵机 (CTM) [Blum and Blum, 2022] 是作为图灵机扩展概念的一种理论。与图灵机的计算模型相比，CTM 赋予系统一个可区分的特征，即“意识的感觉”。具体而言，CTM 定义如下。

定义 16 (意识图灵机). *CTM 被定义为一个七元组: $\langle STM, LTM, Up Tree, Down Tree, Links, Input, Output \rangle$ ，其中 STM 和 LTM 分别是短期记忆和长期记忆的缩写。*

CTM 可被视为具有更复杂结构的 GWT。STM 类似于 GWT 中的“舞台”，是意识发生的必要组件。作为 GWT 中“观众”的类比，LTM 是一个大型通用处理器集合，包括用于建模世界和智能体自身的 *Model of the World* 处理器、用于处理语言信息的 *Inner Speech* 处理器，以及其他用于处理如五感等信息输入的 *Inner Generalized Speech* 处理器。这些处理器被称为 LTM，因为它们具有相对稳定的状态和专门处理特定类型信息的能力，而 LTM 则对应于更长期的状态维护以实现更通用的功能。

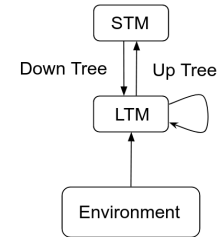


图 7: CTM 中的信息流。

在 CTM 中，信息流仅以五种方式出现，如图 7 所示：

- (1) 环境 $\rightarrow$ LTM;
- (2) LTM $\rightarrow$ STM (通过上行树);
- (3) STM $\rightarrow$ LTM (通过下行树);
- (4) LTM $\rightarrow$ LTM;
- (5) LTM $\rightarrow$ 环境。

过程 (1) 是信息感知。过程 (2) 通过上行树竞争实现。在上行树竞争过程中，最终有一个获胜的信息块到达 STM。竞争过程由一种内部机制决定，该机制是概率性的，并通过一些具有内在随机性的神经元实现。然而，作者认为，即使在完全确定性的设置下，自由意志仍然可以被感受到。过程 (3)

for processing a specific type of information, while LTM corresponds to a shorter period of status maintenance for more general functionalities.

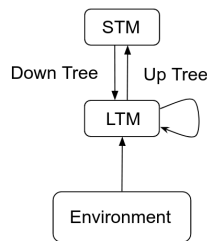


Figure 7: The information flows in CTM.

In CTM, the information flows only appear in five ways, as also depicted in Fig. 7:

- (1) environment $\rightarrow$ LTM;
- (2) LTM $\rightarrow$ STM (via Up Tree);
- (3) STM $\rightarrow$ LTM (via Down Tree);
- (4) LTM $\rightarrow$ LTM;
- (5) LTM $\rightarrow$ environment.

Process (1) in the information perception. Process (2) is achieved with Up Tree competition. In the Up Tree competition process, there is a winning information chunk finally reaching the STM. The competition process is determined by an internal mechanism, which is probabilistic and achieved with some coin-flip neurons with inherent randomness. However, the authors argue that the free will can still be felt even with a completely deterministic setting. Process (3) is through the Down Tree broadcast, and it broadcasted the information in STM to all LTM. (2) and (3) achieve the consciousness awareness. Conscious awareness (attention) is the reception by all LTM processors of the broadcasted winning chunk in the Up Tree competition. Process (4) is a bidirectional link between processors to collaborate on the information processing. Process (5) is the output of the system to the environment through processors like *Motion Controller*.

8.2 CTM for Consciousness

CTM adopts the concept of *Brainish* as the inner language for communicating between different processors. *Brainish* is a terminology referring to the abstract language used for carrying information among different modules of the brain, it can be viewed as an encoding of multi-modal information, and it is unsymbolized and more powerful than outer language like English.

The feeling of consciousness, by CTM theory, is generated as a result of combining *Brainish* language, CTM's architecture, some special processors and CTM's dynamics predictive power. Self-modeling is achieved through the *Model of the World* processor by repeatedly generating actions from some LTMs (like the *Motion Controller*) and observing the consequences perceived by some other LTMs (like *Inner Generalized Speech* processor for sensing the surrounding environment). CTM is used to interpret the blindsight, illusions, dreams and other consciousness-related process. The free will is achieved within coin-flip process in the Up Tree competition.

8.3 Relationships with Other Theories

Compared with GWT, the CTM has just one "actor" on stage holding just one chunk at a time. Additionally, all processors in the CTM are in LTM. Compared with an experimental work [Lee et al.,

通过下行树广播实现, 它将 STM 中的信息广播给所有 LTM。(2) 和 (3) 共同实现了意识觉知。意识觉知 (注意) 是所有 LTM 处理器对上行树竞争中广播的获胜信息块的接收。过程 (4) 是处理器之间用于信息处理协作的双向链接。过程 (5) 是系统通过如 *Motion Controller* 等处理器向环境的输出。

8.2 CTM for Consciousness

CTM 采用 *Brainish* 的概念作为不同处理器之间通信的内部语言。*Brainish* 是一个术语, 指代用于在大脑不同模块之间传递信息的抽象语言, 它可以被视为多模态信息的一种编码, 它是非符号化的, 并且比英语等外部语言更强大。

根据 CTM 理论, 意识的感觉是结合 *Brainish* 语言、CTM 的架构、一些特殊处理器以及 CTM 的动态预测能力而产生的。自我建模是通过 *Model of the World* 处理器实现的, 该处理器反复地从某些长期记忆 (如 *Motion Controller*) 生成动作, 并观察由其他一些长期记忆 (如用于感知周围环境的 *Inner Generalized Speech* 处理器) 所感知到的后果。CTM 被用来解释盲视、错觉、梦境以及其他与意识相关的过程。自由意志是在上树竞争中的抛硬币过程中实现的。

8.3 Relationships with Other Theories

与 GWT 相比, CTM 在舞台上只有一个 "演员", 一次只持有一个信息块。此外, CTM 中的所有处理器都位于长期记忆中。与一项区分唤醒和意识作为意识两个组成部分的实验工作 [Lee et al., 2022] 相比, 本文仅讨论意识觉知 (或注意)。CTM 理论假设意识可以还原为计算过程, 并且可以用图灵机建模。然而, 这一点仍然存在争议。其他人可能质疑这一观点, 并认为意识是一种更复杂的现象, 不能还原为纯粹的计算过程。有几种理论和论点支持意识不能还原为纯粹计算过程的观点, 其中包括:

- 意识的困难问题: 这一由哲学家大卫·查尔莫斯 [Chalmers, 1995, 2017] 提出的论点认为, 虽然大脑可能能够执行各种计算, 但这些计算如何产生主观体验或意识尚不清楚。
- 感受质: 这指的是体验的主观且难以言喻的方面, 例如红色的红或巧克力的味道。一些人认为, 这些主观体验无法被计算模型捕捉, 而是植根于大脑的生物和物理过程 [Tononi and Koch, 2015, Chalmers, 2017, Albantakis and Tononi, 2021]。
- 整合问题: 这一论点认为, 意识产生于大脑不同区域之间复杂且动态的相互作用, 而这些相互作用不能还原为简单的计算。
- 计算的局限性: 一些人认为, 计算存在根本性的限制, 而意识的某些方面可能超出了这些限制。

这些论点以及其他论点表明, 虽然计算可能在意识中扮演一定角色, 但它并非全部, 要完全解释意识现象, 需要一个更复杂、更细致的理解。

9 Physiological Evaluation Metric of Consciousness

本节介绍一些用于医学诊断的意识水平生理评估指标。需要注意的是, 本节中 "生理意识" 的定义与论文其他部分 (例如, IIT (第 2 节) 或 GWT (第 5 节)) 介绍的 "意识" 有所不同。生理意识通常表示基于生理和生物学数据 (例如, 从脑电图提取的信号特征或对刺激的反应) 得出的受试者或患者的意识水平。在医学评估中, 意识的定义通常包含觉醒或知晓水平 [Walker et al., 1990]。在接下来的段落中, 我们将深入探讨基于电信号和行为指标的常见评估指标。为获得全面概览, 请参阅表 2, 该表总结了每种意识生理评估指标的信号来源和应用。

9.1 Metrics Based on Electrical Signals

脑电双频指数 (BIS) [Rosow and Manberg, 2001, Johansen, 2006] 是一种基于生物信号、用于衡量意识水平的指标, 通常用于已接受麻醉的患者。该数值通过患者的脑电图 (EEG) 计算得出, 以评估麻醉深度。BIS 结合了脑电图的频谱图、双谱分析以及时域上的爆发抑制评估进行计算。BIS 的取值范围

2022], which distinguished the arousal and awareness as two components of consciousness, this paper only discuss conscious awareness (or attention). The CTM theory assumes the consciousness can be reduced to computational process and modeled with a Turing machine. However, this is still debatable. Others may dispute this view and hold that consciousness is a more complex phenomenon that cannot be reduced to purely computational processes. There are several theories and arguments that support the idea that consciousness cannot be reduced to purely computational processes. Some of these include:

- The hard problem of consciousness: This argument, put forth by philosopher David Chalmers [Chalmers, 1995, 2017], posits that while the brain may be able to perform various computations, it is unclear how these computations give rise to subjective experience or consciousness.
- Qualia: This refers to the subjective and ineffable aspects of experience, such as the redness of red or the taste of chocolate. Some argue that these subjective experiences cannot be captured by computational models and are instead rooted in the biological and physical processes in the brain [Tononi and Koch, 2015, Chalmers, 2017, Albantakis and Tononi, 2021].
- The integration problem: This argument suggests that consciousness emerges from the complex and dynamic interactions between different regions of the brain, and that these interactions cannot be reduced to simple computations.
- The limits of computation: Some argue that there are fundamental limitations to what can be computed and that certain aspects of consciousness may fall outside of these limitations.

These arguments and others suggest that while computation may play a role in consciousness, it is not the whole story and that a more complex and nuanced understanding is needed to fully explain the phenomenon of consciousness.

9 Physiological Evaluation Metric of Consciousness

This section introduces some physiological evaluation metrics of consciousness level used in medical diagnostics. It should be noticed that the definition of ‘physiological consciousness’ in this section is different from the ‘consciousness’ introduced in other sections of the paper, e.g. IIT (Sec. 2) or GWT (Sec. 5). Physiological consciousness usually represents the consciousness level of subjects or patients based on the physiological and biological data, for example, signal features draw from EEG or response to stimulus. In medical evaluation, the definition of consciousness usually comprises wakefulness or awareness level [Walker et al., 1990]. In the upcoming paragraphs, we will delve into the examination of common evaluation metrics based on electrical signals and behavioral indicators. For a comprehensive overview, please refer to the Table 3, which summarizes the source signals and applications for each physiological evaluation metric of consciousness.

9.1 Metrics Based on Electrical Signals

The Bispectral Index (BIS) [Rosow and Manberg, 2001, Johansen, 2006] is a biological-signal-based measure of the level of consciousness, usually in a patient who has been given anaesthetics. It is a value calculated from a patient’s electroencephalogram (EEG) to evaluate the depth of anaesthesia. BIS is calculated from EEG using the combination of the spectrogram, bispectrum and time-domain assessment of burst suppression. BIS ranges from 0 to 100, indicating from deep anaesthesia to full wakefulness. A BIS monitor is usually used in operations to guarantee an appropriate level of anaesthesia. The perturbational complexity index (PCI) was introduced as a metric for evaluating the level of consciousness by Casali et al. [2013]. PCI uses transcranial magnetic stimulation (TMS) to stimulate cortical activities and analyse algorithmic complexity information based on the spatiotemporal pattern of EEG responding to the stimulus. Similarly, using EEG as the source, the explainable consciousness indicator (ECI) [Lee et al., 2022] utilises deep learning models to compute a score to evaluate consciousness. The study defined consciousness as a combination of arousal and awareness. There are two deep learning networks computing them respectively, and then integrating the scores of both as an indicator of consciousness.

Metric	Source Signal	Application	References
Bispectral Index (BIS)	electroencephalogram (EEG) spectrogram; bispectrum; time-domain assessment of burst suppression	evaluate the level of consciousness (anaesthetics etc)	[Rosow and Manberg, 2001, Johansen, 2006]
Perturbational Complexity Index (PCI)	collect electroencephalogram (EEG) features responding to the TMS (transcranial magnetic stimulation) stimulation	evaluate the level of consciousness of brain-injured or unresponsive patients	[Casali et al., 2013]
Explainable Consciousness Indicator (ECI)	electroencephalogram (EEG) features extracted by two deep neural networks	evaluate the level of wakefulness and awareness	[Lee et al., 2022]
Glasgow Coma Scale (GCS)	eye activities; verbal response; motor response	evaluate effectiveness of treatments and as a prognostic indicator; clinical assessment of unconsciousness, e.g. comatose patients	[Jones, 1979, Sternbach, 2000]
Rancho Los Amigos Scale (RLAS)	Responses to external stimuli; Responds to people; short-term memory; behavior and verbalization; task operation and learning	Consist eight levels of consciousness evaluation for patient recovery evaluation	[Lin and Wroten, 2017]
Full Outline of Unresponsiveness (FOUR)	eye; motor; brainstem; respiration	evaluate level of consciousness (including verbal score in intubated patients and to test brainstem reflexes)	[Wijdicks et al., 2005]
Coma Recovery Scale-Revised (CRS-R)	emotion; language; memory; attention	evaluate patients’ disorders of consciousness, e.g. coma.	[Giacino et al., 2004]
Vegetative and Minimally Conscious State Scale (VSMS)	responsiveness; attention; communication	evaluate the awareness level of patients in vegetative or in a minimally conscious	[Wieser et al., 2010]
Coma/Near Coma Scale (CNCS)	responsiveness; attention; communication	evaluate the awareness level of patients in vegetative or in a minimally conscious	[Rappaport, 2005]

表 2: 意识生理评估指标总结

Metric	Source Signal	Application	References
Bispectral Index (BIS)	electroencephalogram (EEG) spectrogram; bispectrum; time-domain assessment of burst suppression	evaluate the level of consciousness (anaesthetics etc)	[Rosow and Manberg, 2001, Johansen, 2006]
Perturbational Complexity Index (PCI)	collect electroencephalogram (EEG) features responding to the TMS (transcranial magnetic stimulation) stimulation	evaluate the level of consciousness of brain-injured or unresponsive patients	[Casali et al., 2013]
Explainable Consciousness Indicator (ECI)	electroencephalogram (EEG) features extracted by two deep neural networks	evaluate the level of wakefulness and awareness	[Lee et al., 2022]
Glasgow Coma Scale (GCS)	eye activities; verbal response; motor response	evaluate effectiveness of treatments and as a prognostic indicator; clinical assessment of unconsciousness, e.g. comatose patients	[Jones, 1979, Sternbach, 2000]
Rancho Los Amigos Scale (RLAS)	Responses to external stimuli; Responds to people; short-term memory; behavior and verbalization; task operation and learning	Consist eight levels of consciousness evaluation for patient recovery evaluation	[Lin and Wroten, 2017]
Full Outline of Unresponsiveness (FOUR)	eye; motor; brainstem; respiration	evaluate level of consciousness (including verbal score in intubated patients and to test brainstem reflexes)	[Wijdicks et al., 2005]
Coma Recovery Scale-Revised (CRS-R)	emotion; language; memory; attention	evaluate patients' disorders of consciousness, e.g. coma.	[Giacino et al., 2004]
Vegetative and Minimally Conscious State Scale (VSMS)	responsiveness; attention; communication	evaluate the awareness level of patients in vegetative or in a minimally conscious	[Wieser et al., 2010]
Coma/Near Coma Scale (CNCS)	responsiveness; attention; communication	evaluate the awareness level of patients in vegetative or in a minimally conscious	[Rappaport, 2005]

Table 3: A Summary of Physiological Evaluation Metrics of Consciousness

为 0 至 100，表示从深度麻醉到完全清醒的状态。BIS 监护仪通常用于手术中，以确保麻醉水平适当。扰动复杂性指数 (PCI) 由 Casali et al. [2013] 提出，作为评估意识水平的度量标准。PCI 使用经颅磁刺激 (TMS) 激发皮层活动，并基于脑电图对刺激响应的时空模式分析算法复杂性信息。类似地，以脑电图作为数据源，可解释意识指标 (ECI) [Lee et al., 2022] 利用深度学习模型计算评分以评估意识水平。该研究将意识定义为唤醒与觉知的结合。两个深度学习网络分别计算这两部分，随后将两者的评分整合为意识指标。

9.2 Metrics Based on Behaviors

此处我们讨论一些基于生命体征或行为的意识水平评估指标。格拉斯哥昏迷量表 (GCS) [Jones, 1979, Sternbach, 2000] 的测量包括睁眼反应、语言反应和运动反应。GCS 得分越高，表示意识水平越高。拉诺斯·阿米戈斯量表 [Lin and Wroten, 2017] 常与 GCS 结合使用，用于评估脑损伤患者的恢复过程。它包含八个意识水平，从无反应到完全恢复。为应对 GCS 有时无法评估插管患者的语言评分以及检测脑干反射的问题，无反应全面评估量表 (FOUR) [Wijdicks et al., 2005] 被提出，该量表包含眼部、运动、脑干和呼吸四项指标。昏迷恢复量表修订版 (CRS-R) [Giacino et al., 2004] 评估广泛的认知和行为功能，包括情绪、语言、记忆和注意力。CRS-R 用于评估患者的意识障碍，例如昏迷。植物状态 and 最小意识状态量表 (VSMS) [Wieser et al., 2010] 以及昏迷/近昏迷量表 (CNCS) [Rappaport, 2005] 在文献中已被用于通过测量一系列认知和行为功能 (包括反应性、注意力和沟通能力) 来表征处于植物状态或最小意识状态患者的意识水平。

10 Look Ahead: Can Computational Models Be Conscious?

10.1 Background

大型人工智能 (AI) 模型的近期进展引发了人们的关注，即思考意识是否存在于这些模型中，或者能否在短时间于这些 AI 系统中构建出来。如图 8 所示，AI 智能体是通常为计算型智能体的一个子集。在所有 AI 智能体中，通用人工智能智能体是最强大的智能体，拥有与人类相兼容的智力能力。一个 AGI 系统要具备意识，还需满足若干属性，这将在后文详细讨论。本节讨论的主题涉及像 ChatGPT 这样的大型语言模型 (LLMs)、像 PALI [Chen et al., 2022] 这样的多模态模型、以及像 PALI-X [Chen et al., 2023]、PALM-E [Driess et al., 2023]、RT-2 [Brohan et al., 2023] 这样的具身智能体。融合不同模态可以提升模型的泛化能力，使其行为表现得更有意识。然而，鉴于当前进展以及语言在人类知识中的重要性，本节将主要讨论 LLMs。在语言层面评估模型的意识更为直接。

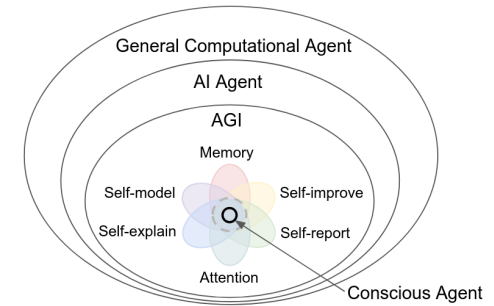


图 8: 计算智能体示意图。最大的圆代表通用计算智能体，而 AI 智能体是其子集。AGI 智能体是那些最强大的 AI 智能体。在满足额外条件的情况下，AGI 智能体可以具备意识。

9.2 Metrics Based on Behaviors

Here we discuss some consciousness-level evaluation metrics based on vital signs or behaviors. The Glasgow Coma Scale (GCS) [Jones, 1979, Sternbach, 2000] measurement comprises eye-opening, verbal response, and motor response. A higher score on the GCS indicates a higher level of consciousness. The Rancho Los Amigos Scale [Lin and Wroten, 2017] is often used in conjunction with the GCS during the recovery of patients from brain injuries. It consists of eight consciousness levels, ranging from no response to complete recovery. To deal with the fact that GCS sometimes fail to evaluate the verbal score in intubated patients and to test brainstem reflexes. The full outline of unresponsiveness (FOUR) [Wijdicks et al., 2005] was proposed, which consists of four metrics of the eye, motor, brainstem, and respiration. The Coma Recovery Scale-Revised (CRS-R) [Giacino et al., 2004] assesses a wide range of cognitive and behavioral functions, including emotion, language, memory and attention. CRS-R is used to evaluate patients' disorders of consciousness, e.g. coma. The Vegetative and Minimally Conscious State Scale (VSMS) [Wieser et al., 2010] and Coma/Near Coma Scale (CNCS) [Rappaport, 2005] have been used in the literature for characterising the awareness of patients in vegetative or in a minimally conscious state by measuring a range of cognitive and behavioral functions, including responsiveness, attention, and communication.

10 Look Ahead: Can Computational Models Be Conscious?

10.1 Background

The recent progress of large artificial intelligence (AI) models attracts people's attention on thinking whether the consciousness exists in these models, or it can be built within these AI systems in a short time. As depicted in Fig. 8, an AI agent is in a subset of agents that are generally computational. Among all AI agents, an artificial general intelligent agent is the strongest agent possessing the compatible intellectual capability with humans. It further requires several properties to be satisfied for an AGI system to be conscious, and it will be discussed in details later. The subject discussed in this section involves the large language models (LLMs) like ChatGPT, multimodal models like PALI [Chen et al., 2022], embodied agents like PALI-X [Chen et al., 2023], PALM-E [Driess et al., 2023], RT-2 [Brohan et al., 2023]. Fusing the different modalities can improve the generalization capability for the model to behave more consciously. However, given the current progress and the importance of language in human knowledge, we will discuss mainly the LLMs in this section. It is more straightforward to evaluate the models for its consciousness in language.

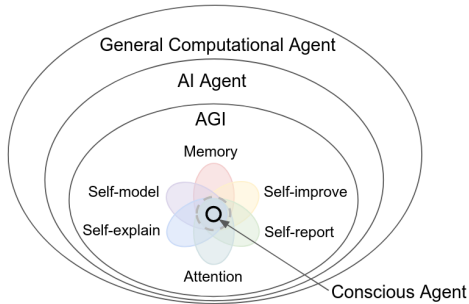


Figure 8: A diagram of computational agents. The largest circle represents the general computational agents, whereas the AI agents fall in a subset of it. The AGI agents are those strongest AI agents. With additional conditions satisfied, the AGI agent can be conscious.

LLMs refers to the large computational models specifically designed for linguistic tasks. To some extent, the LLMs built recently resemble the *philosophical zombie* [Chalmers, 1997], which is a

LLMs 指的是专门为语言任务设计的大型计算模型。在某种程度上，近期构建的 LLMs 类似于 *philosophical zombie* [Chalmers, 1997]，这是一个为论证意识问题之困难性而广为人知的思想实验。有时 LLMs 生成的答案与人类得出的结果高度一致，以至于我们难以区分两者，这使得一些人相信

LLMs 具有像人类一样具备意识的潜力。

尽管当前的 LLMs 非常智能，但研究者 [Ma et al., 2023] 认为当前的 LLMs 是 *brain in a vat* (BiV) [Putnam et al., 1981]。将大型语言模型 (LLMs) 与 BiV 思想实验进行比较，是对模型固有局限性的一种批评。在 BiV 场景中，一个大脑与其身体分离，并连接到一台能生成逼真现实幻觉的超级计算机上。该大脑可以与这个模拟现实互动，但它缺乏与真实世界的连接。类似地，LLMs 尽管拥有令人印象深刻的语言能力，但本质上与真实世界是脱节的。它们基于海量文本语料库中的模式生成回应，但这些回应局限于训练数据之内，缺乏与真实世界实体的连接。

这一比较得到了以下观察的支持：LLMs，如同 BiV 场景中的大脑，无法在符号（词语）与真实世界实体之间建立联系。这一局限性内在于其构建过程，该过程涉及基于海量文本语料库对语言关系进行统计建模。因此，其输出被限制在训练数据之内，它们无法建立符号与真实世界实体之间的联系。这种缺乏现实基础的情况是 LLMs 的一个显著局限，因为它阻碍了它们理解训练数据之外出现的新概念。此外，作者认为人类智能和意识本质上与我们的感官体验及与世界的物理互动相关联。我们创造符号来代表现实世界中的物体，使得知识能够跨代保存和传递。然而，大型语言模型 (LLMs) 缺乏通过行动创造新体验并将其转化为形式知识的初始过程。这种互动行为及伴随的感官输入的缺失，被认为是实现理想通用智能所缺失的一环。

关于大型语言模型是否存在意识的问题，David Chalmers 在 NeurIPS 2022（2022 年 11 月 28 日）上发表了题为 *Could a Large Language Model be Conscious?* [Chalmers, 2023] 的演讲，两天后 OpenAI 发布了非常流行且强大的大型语言模型 ChatGPT（2022 年 11 月 30 日）。David 演讲的主要观点是，当前的大型语言模型具有微小的可能性具备意识（例如 10%）。我们尚不清楚在 ChatGPT 及后续的大模型 GPT-4 [OpenAI, 2023] 发布后，这一可能性如何变化。

针对这一主题，我们提出以下关于大型语言模型意识的问题：

- 当前的大型语言模型是否具备意识？是否存在支持其有/无意识的证据？
- 我们为何希望构建具备意识的计算模型？
- 基于 Transformer 架构和自注意力机制，构建具备意识的大型语言模型在理论上是否可能？
- 构建具备意识的计算模型需要哪些必要组件？

一个被视为具备意识的模型需具备以下几个关键能力方面：自我精炼、自我改进和自我解释。自我精炼 [Madaan et al., 2023] 是指大型语言模型能够通过自我评估对其输出提供反馈，并利用该反馈通过自我改进来优化其输出的能力。反思 [Shinn et al., 2023] 与自我精炼类似，但在自我反思过程中具有持久记忆。自我改进 [Huang et al., 2022] 表明大型语言模型能够通过思维链提示和自洽性评估（例如对多路径推理进行多数投票）为未标注问题生成高置信度的输出。自我解释 [Elton, 2020] 也被认为是构建通用人工智能系统所必需的一个组件，它要求智能体不仅能预测输出及其不确定性，还能解释输出并对解释提供置信度。如图 8 所示，若上述能力未能充分满足，则会降低一个计算模型作为潜在意识模型的置信度。

10.2 Large Language Model

近年来，大型语言模型 (LLMs) 通过在语言建模任务中的卓越表现，彻底改变了自然语言处理 (NLP) 领域。这些模型通常应用 **language modeling objective**，并利用 **Transformer architecture**，该机制融合了 **attention mechanism** 以捕捉复杂的语言依赖关系。注意力机制使模型能够为输入序列的不同元素分配不同级别的重要性，从而使其能够生成连贯且与上文相关的语言。

这些大型语言模型在各种 NLP 任务中展现了卓越的性能，例如语言翻译、文本生成、情感分析和语音识别。通过利用 Transformer 架构和注意力机制的力量，它们显著推进了语言理解，并正在重塑人机交互和信息检索的未来。随着研究不断突破这些模型的边界，预计将有进一步的改进，从而带来更令人

well-known thought experiment for defending the hardness of consciousness problem. Sometimes the LLMs generated answers are so consistent with the results by humans that it can be hard for us to distinguish between the two, which makes some people to believe that the LLMs have the potential to be conscious like humans.

Although present LLMs are highly intelligent, researchers [Ma et al., 2023] argue that the present LLMs are *brain in a vat* (BiV) [Putnam et al., 1981]. The comparison of Large Language Models (LLMs) to the BiV thought experiment is a critique of the models' inherent limitations. In the BiV scenario, a brain is detached from its body and connected to a supercomputer that generates a convincing illusion of reality. The brain can interact with this simulated reality, but it lacks a connection to the real world. Similarly, LLMs, despite their impressive linguistic capabilities, are fundamentally disconnected from the real world. They generate responses based on patterns in massive text corpora, but these responses are confined within the training data and lack a connection to real-world entities.

This comparison is supported by the observation that LLMs, like the brain in the BiV scenario, cannot establish a connection between symbols (words) and real-world entities. This limitation is inherent in their construction process, which involves statistical modeling of linguistic relationships based on massive text corpora. As a result, their output is confined within the training data, and they cannot establish a connection between symbols and real-world entities. This lack of grounding in reality is a significant limitation of LLMs, as it prevents them from understanding new concepts that emerge outside of their training data.

Moreover, the authors argue that human intelligence and consciousness are intrinsically linked to our sensory experience and physical interaction with the world. We create symbols to represent objects in the real world, enabling the preservation and transmission of knowledge across generations. However, the initial process of acting to create novel experiences and turn them into formal knowledge is missing in LLMs. This absence of interactive behaviors and accompanying sensory input is identified as a missing piece towards ideal general intelligence.

Regarding the problem of whether consciousness exists in LLMs, David Chalmers gave a talk on NeurIPS 2022 (Nov. 28th, 2022) with the topic *Could a Large Language Model be Conscious?* [Chalmers, 2023], two days before the release of the very popular and powerful LLM ChatGPT (Nov. 30th, 2022) by OpenAI. The major claim by David's talk is that the current LLM has a small chance to be conscious (e.g., <10%). We are unclear about how this value changes after the release of ChatGPT and later GPT-4 [OpenAI, 2023].

Regarding this topic, we would like to propose the following questions about the consciousness of LLMs:

- Is current LLM conscious? Any evidence or support for its existence/no-existence?
- Why do we want to build a conscious computational model?
- Is building LLM with conscious theoretically possible with transformer architecture and self-attention mechanism?
- What are the necessary components to build a conscious computational model?

There are several key aspects of capability for a model considered as a conscious one: self-refinement, self-improving and self-explanation. Self-refinement [Madaan et al., 2023] is the capability for LLMs to provide feedback on its own outputs through self-evaluation, and leverage the feedback to refine its outputs via self-improvement. Reflexion [Shinn et al., 2023] is similar as self-refine but with a persisting memory during the self-reflective process. Self-improvement [Huang et al., 2022] shows LLMs can generate high-confidence outputs for unlabeled questions through Chain-of-Thought prompting and self-consistency evaluation, e.g. majority voting for multi-path reasoning. Self-explanation [Elton, 2020] is also assumed to be a necessary component for building AGI system, which requires the agent to not only predict the output and its uncertainties, but also the explanation and confidence over the explanation of output. The lack of satisfaction of above capabilities will lower down the confidence of a computational model as a potentially conscious model, as shown in Fig. 8.

印象深刻的语言理解和生成能力。未来，更先进版本的 LLMs 有可能展现出类似人类的意识。因此，我们介绍当前 LLMs 的基础知识，作为进一步讨论机器与人类意识关系的背景。

Language Modeling. 目前大多数 LLMs, 例如 e.g., GPT-3 [Brown et al., 2020] 和 PALM [Chowdhery et al., 2022], 采用因果或掩码语言建模作为其训练目标, 即最大化输入序列中后续标记的对数似然 x :

$$\mathcal{L}_{LM}(x) = \sum_{i=1}^L \Pr(x_i | x_{<i}) \quad (31)$$

像 PALM-2 [Anil et al., 2023] 这样的先进 LLMs 使用多样化的目标作为增强, 但语言建模和掩码标记预测仍然是训练 LLMs 最根本的目标。

Transformer and Attention Mechanism. 注意力是一种机制, 允许模型在进行预测时聚焦于输入序列的不同部分。著名的 transformer [Vaswani et al., 2017] 架构使用多头自注意力, 它是三个矩阵的缩放点积: 查询矩阵 Q、键矩阵 K 和值矩阵 V。查询、键和值矩阵是通过权重 W^q, W^k, W^v 对输入 X 进行的不同线性变换:

$$Q = W^q X \in \mathbb{R}^{L \times d_k} \quad (32)$$

$$K = W^k X \in \mathbb{R}^{L \times d_k} \quad (33)$$

$$V = W^v X \in \mathbb{R}^{L \times d_v} \quad (34)$$

注意力的计算格式如下:

$$\text{atten}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (35)$$

乘积 QK^T 通过查询和键矩阵中不同序列标记的相关性, 计算了将应用于输入的权重。然后, 权重通过 $\sqrt{d_k}$ 进行缩放, 并通过 softmax 函数转化为归一化的概率。这些概率与值矩阵在序列上的乘法, 为输入提供了恰当的注意力表示。多头自注意力是将输入分割成更多子序列, 并行应用多个上述注意力机制。对多头输出的注意力遵循交叉注意力机制, 该机制涉及来自一个输入序列的查询权重和来自另一个序列的键-值权重。

关于 transformer 架构的更多细节, 请参考原始论文 [Vaswani et al., 2017] 以及 Weng [2023] 的博客。

Connection between Artificial (Transformer) Attention and Biological Attention. 上述段落介绍了最流行的人工注意力类型之一, 即机器学习领域中 Transformer 架构的注意力机制 *i.e.*。另一方面, 本文讨论的神经科学和心理学中研究的一些意识理论 (从第2节到第4节) 与生物注意力机制密切相关。列举几个例子, 如注意力图式理论 (AST, 第7节)、意识图灵机 (CTM, 第8节)、全局工作空间理论 (GWT, 第5节) 等。因此, 一个自然的问题是: 人工 (Transformer) 注意力与生物注意力 [Lindsay, 2020] 之间的联系是什么?

基于现有理论 (AST、CTM、GWT) 的生物注意力机制具有以下特性:

- 感觉注意力: 作为生物注意力机制的一个基本方面, 尤其是在视觉处理背景下。它通过选择性地聚焦于特定刺激同时过滤其他刺激来运作, 从而提高表征被注意刺激的神经元的信噪比。这种注意过程既能影响神经元的局部活动, 也能影响不同脑区之间的通信。例如, 注意力已被证明能增加伽马波段的放电一致性, 从而增强同步放电神经元对共享下游脑区的影响。此外, 注意力可以直接协调不同脑区之间的通信, 这体现在表征被注意刺激的神经元 (在诸如 V1 (初级视觉皮层) 和 V4 (视觉皮层中负责物体识别和颜色处理的区域) 等脑区) 之间同步性增加。皮层下区域, 如上丘和丘脑枕, 也在感觉注意力中扮演重要角色, 协助隐性和显性空间注意力并调节皮层效应。
- 好奇心作为驱动力: 生物注意力受好奇心影响。新颖、令人困惑或意外的刺激能捕获注意力。额下皮层和嗅周皮层通过一种适应机制来发出新颖视觉情境的信号, 该机制会降低对熟悉输入的反应。考虑新颖性的强化学习算法可以鼓励探索。

10.2 Large Language Model

In recent years, large language models (LLMs) have revolutionized the field of natural language processing (NLP) by excelling in language modeling tasks. These models usually apply the **language modeling objective**, and utilize the **Transformer architecture**, which incorporates the **attention mechanism** to capture complex linguistic dependencies. The attention mechanism allows the model to assign varying levels of importance to different elements of the input sequence, enabling it to generate coherent and contextually relevant language.

These large language models have demonstrated exceptional performance in various NLP tasks such as language translation, text generation, sentiment analysis, and speech recognition. By leveraging the power of the Transformer architecture and the attention mechanism, they have significantly advanced language understanding and are reshaping the future of human-computer interaction and information retrieval. As research continues to push the boundaries of these models, further improvements are anticipated, leading to even more impressive language understanding and generation capabilities. It is possible that the more advanced version of LLMs will demonstrate the consciousness like humans one day in the future. Therefore we introduce the basics of current LLMs as a background for further discussion of the relationship of machine and human consciousness.

Language Modeling. Most of the LLMs at present, *e.g.*, GPT-3 [Brown et al., 2020] and PALM [Chowdhery et al., 2022], apply either causal or masked language modeling as its training objective, which is to maximize the log-likelihood of later tokens in the input sequence $\mathbf{x}$:

$$\mathcal{L}_{\text{LM}}(\mathbf{x}) = \sum_{i=1}^L \Pr(x_i | x_{<i}) \quad (31)$$

Advanced LLMs like PALM-2 [Anil et al., 2023] use varied objectives as an enhancement, but language modeling and masked token prediction is still the very fundamental objective for training LLMs.

Transformer and Attention Mechanism. Attention is a mechanism that allows the model to focus on different parts of the input sequence when making predictions. The well-known transformer [Vaswani et al., 2017] architecture uses the multi-head self-attention, which is a scaled dot-product of three matrices: the query matrix $\mathbf{Q}$, the key matrix $\mathbf{K}$ and the value matrix $\mathbf{V}$. The query, key and value matrices are different linear transformations (by weights $\mathbf{W}^q, \mathbf{W}^k, \mathbf{W}^v$) from the input $\mathbf{X}$:

$$\mathbf{Q} = \mathbf{W}^q \mathbf{X} \in \mathbb{R}^{L \times d_k} \quad (32)$$

$$\mathbf{K} = \mathbf{W}^k \mathbf{X} \in \mathbb{R}^{L \times d_k} \quad (33)$$

$$\mathbf{V} = \mathbf{W}^v \mathbf{X} \in \mathbb{R}^{L \times d_v} \quad (34)$$

The attention is calculated in the format of:

$$\text{atten}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \quad (35)$$

The product $\mathbf{Q}\mathbf{K}^\top$ calculates the weights to be applied on the inputs through the correlation of tokens in different sequences of the query and key matrices. Then the weights are scaled by $\sqrt{d_k}$ and become normalized probabilities through the softmax function. Multiplication of the probabilities over sequences with the value matrix gives proper attention representation on the inputs. Multi-head self-attention is splitting the inputs into more sub-sequences and applies multiple above attention mechanism in parallel. Attention over the multi-head outputs follows the cross-attention mechanism, which involves the query weights from one input sequence and the key-value weights from another sequence.

More details about the transformer architecture refer to the original paper [Vaswani et al., 2017] and the blog by Weng [2023].

Connection between Artificial (Transformer) Attention and Biological Attention. The above paragraphs introduce one of the most popular types of artificial attention, *i.e.*, the attention mechanism in Transformer architecture in machine learning domain. On the other hand, some consciousness

- 解决注意力冲突：大脑具有多种形式的注意力，如唤醒、自下而上和自上而下注意力。视觉系统中的局部回路整合了神经调节输入、自下而上信号和自上而下输入。水平连接介导竞争，可能使用赢家通吃机制。
- 奖励的影响：注意力与奖励之间存在密切关系。先前获得奖励的刺激即使不再提供奖励，也能吸引注意力。
- 生物注意力的局限性：分心可以被视为一种特性而非缺陷。意识到环境中的潜在威胁是有益的。注意瞬脱指的是在一连串刺激中，如果第二个目标紧接在第一个目标之后出现，则会错过它，这可能是大脑处理第一个目标所必需的。

人工注意力机制总结如下：

- 自然语言处理（NLP）中的注意力：注意力机制常被添加到处理序列的模型中，NLP 是一个常见的应用领域。注意力在人工神经网络中的早期应用是针对翻译任务，*i.e.*，神经机器翻译。注意力机制决定了应如何组合编码向量以产生上下文向量，从而影响翻译句子中的下一个词。这使得网络能够从句子的较早或较晚部分提取信息，有助于在词序不同的语言之间进行翻译。
- Transformer 架构：在具有影响力的论文《Attention is All You Need》[Vaswani et al., 2017] 中提出的 Transformer，代表了人工注意力机制的重大转变，特别是在机器翻译等任务中。与传统的循环模型不同，Transformer 采用了一种称为“自注意力”的机制。在这种方法中，句子中的词被并行编码，生成键和查询表示，这些表示组合起来创建注意力权重。这些权重随后缩放词编码以产生模型中的后续层。Transformer 架构没有任何循环，简化了训练，并超越了之前的许多模型。它已成为机器翻译以及人工智能领域中各种其他任务的标准。
- 注意力部署：挑战在于在传入刺激流中选择相关信息，决定参与的最佳任务，或决定是否参与。已经尝试直接模仿生物注意力，例如预测人类注视点的扫描路径模型 [Borji and Itti, 2015]。他人的注意力可以影响注意力的引导方式，强调了联合注意力的重要性。

生物系统与人工系统中注意力之间的复杂关系可以从多个角度解释：

- 注意力机制：在机器学习中，注意力机制旨在让单个训练模型在多个任务或数据长度、大小或结构不同的任务上表现良好。注意力机制为每个编码/注释向量引入动态权重，为循环输出生成定义上下文。这让人联想到生物注意力，其中输出灵活地依赖于有限资源，用于由解码器需求驱动的循环顺序处理任务。然而，与循环注意力机制相比，自注意力缺乏自上而下的选择解释。多头注意力可以解释为单级自上而下的选择，但在进行多级选择和循环处理的能力方面仍然较弱。当前人工注意力机制与生物注意力的另一个区别是，在 Transformer 架构中没有明确的全局工作空间来整合来自不同子模块的信息。
- 记忆的注意：像多层感知机这样的深度神经网络通常没有显式记忆，但存在如神经图灵机等包含外部记忆存储的混合架构。循环神经网络中的隐藏状态是另一种形式的隐式记忆。最近在大语言模型中提示工程与管理的进展提供了一种将显式记忆与神经网络结合使用的方法。这些网络学会与这些记忆存储交互以执行任务。这种交互通过一种注意形式得以促进。这些系统中的记忆以向量形式存储，为了检索信息，网络为每个向量生成一个权重并计算记忆的加权。该模型中相似性度量的使用意味着记忆的检索基于其与产生的活动向量的重叠程度，类似于神经科学中的联想记忆模型。这为大脑中如何实现记忆的注意提供了一种机制，而注意与记忆的交互在其中扮演重要角色。
- 隐式统计学习：注意可以偏向隐式统计学习。例如，当向受试者展示一系列刺激并让他们检测特定颜色的形状是否连续出现两次时，他们倾向于将真实的形状三元组识别为更熟悉，但仅当这些三元组来自被注意的颜色时。未被注意的统计规律性则未被学习。
- 记忆检索：许多行为研究探讨了记忆检索在多大程度上需要注意。一些研究发现，记忆检索因同时进行一项需要注意的任务而受损，表明它是一个依赖注意的过程。然而，具体发现取决于所使用的记忆和非记忆任务的细节。即使记忆检索不占用共享的注意资源，某些记忆在任何给定时刻被选择进行更生动的检索，这表明存在一个选择过程。神经影像学结果表明，负责自上而下分配和自下而上捕获注意的相同大脑区域可能在记忆检索期间发挥类似作用。

theories studied in neuroscience and psychology that are discussed in this paper (from Sec. 2 to Sec. 4) closely connects to biological attention mechanism. To list a few, attention schema theory (AST, Sec. 7), conscious Turing machine (CTM, Sec. 8), global workspace theory (GWT, Sec. 5), etc. So, a natural question is: What is the connection between the artificial (Transformer) attention and biological attention [Lindsay, 2020]?

The biological attention mechanism based on the present theories (AST, CTM, GWT) has the following properties:

- Sensory attention: as a fundamental aspect of the biological attention mechanism, especially in the context of visual processing. It operates by selectively focusing on specific stimuli while filtering out others, enhancing the signal-to-noise ratio of the neurons representing the attended stimulus. This attentional process can impact both the local activity of neurons and the communication between different brain areas. For instance, attention has been shown to increase spiking coherence in the gamma band, enhancing the influence of synchronously firing neurons on shared downstream areas. Moreover, attention can directly coordinate communication across different brain regions, as evidenced by increased synchrony between neurons representing the attended stimulus in areas like V1 (primary visual cortex) and V4 (region in visual cortex for object recognition and color processing). Subcortical areas, such as the superior colliculus and the pulvinar, also play significant roles in sensory attention, assisting in both covert and overt spatial attention and modulating cortical effects.
- Curiosity as a Driver: Biological attention is influenced by curiosity. Stimuli that are novel, confusing, or surprising can capture attention. Inferotemporal and perirhinal cortex signal novel visual situations through an adaptation mechanism that reduces responses to familiar inputs. Reinforcement learning algorithms that consider novelty can encourage exploration.
- Resolving Attention Conflicts: The brain has multiple forms of attention, such as arousal, bottom-up, and top-down attention. Local circuits in the visual system integrate neuromodulatory input with top-down signals and bottom-up input. Horizontal connections mediate competition, potentially using winner-take-all mechanisms.
- Influence of Rewards: There's a close relationship between attention and reward. Previously rewarded stimuli can attract attention even when they no longer provide rewards.
- Limitations of Biological Attention: Distractability can be seen as a feature rather than a bug. It's beneficial to be aware of potential threats in the environment. Attentional blink refers to missing a second target in a stream if it appears shortly after a first target, which may be necessary for the brain to process the first target.

The artificial attention mechanism is summarized in the following:

- Attention for Natural Language Processing (NLP): Attention mechanisms are frequently added to models processing sequences, with NLP being a common application area. Early applications of attention in artificial neural networks were for translation tasks, *i.e.*, neural machine translation. The attention mechanism determines how the encoded vectors should be combined to produce a context vector, influencing the next word in the translated sentence. This allows the network to pull from earlier or later parts of the sentence, aiding in translating between languages with different word orders.
- The Transformer architecture: The Transformer introduced in the influential paper "Attention is All You Need" [Vaswani et al., 2017], represents a significant shift in artificial attention mechanisms, particularly for tasks like machine translation. Unlike traditional recurrent models, the Transformer employs a mechanism known as "self-attention." In this approach, words in a sentence are encoded in parallel, generating key and query representations that combine to create attention weightings. These weightings then scale the word encodings to produce subsequent layers in the model. The Transformer architecture, devoid of any recurrence, simplifies training and has outperformed many previous models. It has become the standard not only for machine translation but also for various other tasks in the realm of artificial intelligence.
- Attention Deployment: The challenge is in choosing the relevant information in a stream of incoming stimuli, deciding the best task to engage in, or deciding whether to engage at all. Direct mimicry of biological attention has been attempted, such as Scanpath models [Borji

一种对工作记忆中项目的注意机制已被提出 [Manohar et al., 2019]。它依赖于工作记忆中的两种不同机制：非注意项目的突触痕迹和注意项目的持续活动。机器学习社区也应关注神经科学中的这些创新。

- 注意与学习：注意和学习在一个循环中协同工作。注意决定什么进入记忆并指导学习，而学习过程则增强或修正注意机制。在人工系统中，注意机制通常贯穿整个训练过程。注意可以通过将学习导向输入中的相关成分和关系来高效利用数据。显著性图可用于计算机视觉任务中的预处理，以聚焦于内在显著区域。仅将后续处理集中在内在显著区域可以防止在无关区域上的浪费处理并防止过拟合。除了决定处理数据的哪些部分，自上而下的注意也可以被视为选择在处理过程中网络的哪些元素应最活跃，这类似于生物感觉注意中提出的模型 [Kanwisher and Wojciulik, 2000, Desimone, 1998, Treue and Trujillo, 1999]。

一方面，生物注意力模型通常更具概念性，难以通过计算机程序实现，尽管其中一些模型已在生理数据上得到验证。将这些模型具体化并以计算方式构建，以适用于更广泛的目的，仍然是一个巨大的挑战。另一方面，机器学习领域的研究者也应关注神经科学和心理学中研究的概念化注意力机制，通过为生物系统中已验证的过程构建计算模型来实现这一点。

10.3 Emerging Intellectual Capability of LLM - Turing Test

图灵测试 [Turing, 2009] 是一种用于判断机器是否能够表现出与人类无法区分的智能行为的方法。尽管人们对其在区分机器与人类方面的规则存在不完整的争议，但在大型语言模型取得近期成功之前，它长期被视为无法实现的。人们已经开展了大规模的社会实验，利用图灵测试来验证大型语言模型，例如一个名为 "Human or Not" [Jannai et al., 2023] 的实验。该研究涉及至少 150 万用户在线参与，形式是与人类或大型语言模型 (*i.e.*, Jurassic-2、Cohere2 或 GPT-4) 进行对话。尽管结果显示，受试者区分聊天对象是人工智能还是人类的平均正确率为 68%，但这项研究可能并未完全展示大型语言模型在图灵测试中的能力。原因在于，人们已经在区分过程中发现了若干有效策略，包括：

1. 人们假设机器人不会出现拼写错误、语法错误和使用俚语；
2. 人们认为个人问题是测试聊天对象的好方法；
3. 人们假设机器人不了解当前和及时的事件；
4. 人们试图用哲学、伦理和情感问题来挑战对话；
5. 人们将礼貌与不那么人性化的特质联系起来；
6. 人们试图通过提出已知人工智能机器人难以应对或倾向于回避的问题或请求来识别机器人；
7. 人们使用特定的语言技巧来暴露机器人；
8. 在一个创造性的转折中，许多人假装自己是人工智能机器人，以评估聊天伙伴的反应

这些策略大多帮助受试者在图灵测试中正确猜测，实际上与对话者的智力水平无关，而更多是其他不利于当前版本大型语言模型的方面，包括：训练大型语言模型时的正确性和礼貌性要求 (*i.e.*, 第 1、5、6 点)；训练大型语言模型时缺乏特定身份 (*i.e.*, 第 2 点)；训练数据的局限性 (*i.e.*, 第 3-4、7 点)；对抗性攻击 (*i.e.*, 第 8 点) 等。这些训练偏差很可能通过略微调整训练过程得到实际解决，这将导致一个更强大、更针对图灵测试的大型语言模型版本，也更难与人类区分。因此，作为一个也得到一些知名研究者广泛支持的观点，我们认为大型语言模型已经非常接近（即使尚未完全达到）图灵测试的智力水平。

其他研究者主张，大型语言模型可能并不像我们想象的那么智能。*mirror hypothesis* [Sejnowski, 2023] 被提出，认为大型语言模型实际上可能只是一面反映对话者智力水平的镜子。该假设通过向大型语言模型提供不同提示但相同问题得到验证，模型会针对不同提示生成截然不同的答案。由于提示过程是一个单样本学习过程，可以解释为模型正在适应对话者，因此对话者自身的智力水平反过来也会影响大型语言模型。这构成了 *reverse Turing test*，表明大型语言模型可用于评估对话者自身的智力或个性。

关于此主题的更多讨论，请参阅后续关于使用大型语言模型评估人类个性的章节。

and Itti, 2015] predicting human fixations. Attention of others can influence how attention is guided, emphasizing the importance of joint attention.

The intricate relationship between attention in biological and artificial systems can be explained from several perspectives:

- **Attention mechanisms:** In ML the attention mechanisms are designed to allow single trained model to perform well on multiple tasks or tasks with varied data length, size or structure. The attention mechanism introduces dynamic weighting for each encoded/annotation vector to define a context for the recurrent output generation. This is reminiscent of biological attention, where the output flexibly depends on limited resources for recurrent sequential processing tasks driven by the need of the decoder. However, self-attention lacks the top-down selection interpretation comparing to the recurrent attention mechanism. The multi-head attention can be interpreted as a one-level top-down selection, while still being weak in its capability of conducting multi-level selection and recurrent processing. Another difference of the present artificial attention mechanism from the biological one is, there is no explicit global workspace for in the Transformer architecture to integrate information from different sub-modules.
- **Attention to Memory:** Deep neural networks like MLPs typically don't have explicit memory, but there are hybrid architectures like Neural Turing Machines that include external memory stores. The hidden states in recurrent neural networks are another form of implicit memory. Recent advance of prompt engineering and managing in LLMs provides a way to use explicit memory with neural networks. These networks learn to interact with these memory stores to perform tasks. The interaction is facilitated by a form of attention. Memories in these systems are stored as vectors, and to retrieve information, the network generates a weight for each vector and calculates a weighted sum of the memories. The use of a similarity metric in this model means that memories are retrieved based on their overlap with a produced activity vector, similar to associative memory models in neuroscience. This offers a mechanism for how attention to memory could be implemented in the brain, whereas the interactions of attention and memory play an important role.
- **Implicit Statistical Learning:** Attention can bias implicit statistical learning. For instance, when subjects are shown a stream of stimuli and are tasked with detecting when a shape of a certain color appears twice in a row, they tend to recognize real triplets of shapes as more familiar, but only if the triplets were from the attended color. The statistical regularities of the unattended shapes are not learned.
- **Memory Retrieval:** Many behavioral studies have explored the extent to which attention is needed for memory retrieval. Some studies have found that memory retrieval is impaired by the co-occurrence of an attention-demanding task, suggesting it is an attention-dependent process. However, the exact findings depend on the details of the memory and non-memory tasks used. Even if memory retrieval doesn't pull from shared attentional resources, some memories are selected for more vivid retrieval at any given moment than others, indicating a selection process. Neuroimaging results suggest that the same brain regions responsible for the top-down allocation and bottom-up capture of attention may play analogous roles during memory retrieval.
A mechanism for attention to items in working memory has been proposed [Manohar et al., 2019]. It relies on two different mechanisms of working memory: synaptic traces for non-attended items and sustained activity for the attended one. The machine learning community should also be aware of these innovations in neuroscience.
- **Attention and Learning:** Attention and learning work in a loop. Attention determines what enters memory and guides learning, and the learning process enhances or corrects the attention mechanism. Attention mechanisms are often included throughout training in artificial systems. Attention can efficiently use data by directing learning to relevant components and relationships in the input. Saliency maps can be used for preprocessing in computer vision tasks to focus on intrinsically salient regions. Focusing subsequent processing only on regions that are intrinsically salient can prevent wasteful processing on irrelevant regions and prevent overfitting. In addition to deciding which portions of the data to process, top-down attention can also be thought of as selecting which elements of the network should be most engaged during processing, which resembles the models proposed

10.4 Consciousness of LLM

在本节中，我们将对计算模型具备意识的条件讨论进行一般化，而非仅限于讨论大语言模型。具体而言，我们旨在分析人工系统具备意识的充分必要条件，其动机源于Chalmers [2023]。此外，受认知心理学测试的启发，我们基于提出的标准在包括 GPT-3.5 和 GPT-4 在内的大语言模型上进行了一些初步实验，并就自我报告能力、人格测试和镜像测试的结果进行了讨论。

一项同期工作 [Buttin et al., 2023] 基于现有的意识理论，包括循环加工理论 [Lamme, 2006, 2010, 2020]、全局工作空间理论（第 5 节）、高阶理论（第 6 节）、注意图式理论（第 7 节）等，总结了几种“指示性属性”作为评估和计算证据，以指导构建有意识的人工智能模型。给定一个模型要具备意识需要存在 *computational functionalism* 这一假设，研究者们相信有可能构建出满足这些指示性属性条件的有意识人工智能模型。然而，这些条件的必要性和充分性仍存在争议。

Artificial Consciousness and Human Consciousness. Reggia [2013] 对人工意识进行了综述，并将过去的工作按两大目标进行了分类：意识的模拟（弱形式）与实例化（强形式）。前者对应于人类意识的信息处理方面；后者对应于人类意识中的主观体验（感受质）。过去在开发人工意识方面的努力主要是受前几节回顾的意识理论启发的计算模型。然而，模仿人类意识的多种理论表述，并不表明由此构建的人工系统就具有类人的意识。适用于人工系统的、用于测试意识存在与否的标准，仍是该领域的一个开放性问题。Seth [2009] 概述了不同提出的公理。在本文中，我们关注的是大语言模型是否具备类人的实例化意识。

Sufficient and Necessary Conditions for Artificial Consciousness. 根据Chalmers [2023]，它首先提出了人工系统具备意识的充分条件（作为一种积极观点）：

If a computational model has X, then it is conscious.

关键问题是 X 是什么。这可能太难回答，因此我们转而处理必要条件：

If a computational model is conscious, it will have X.

毫无疑问，相较于必要条件，我们更感兴趣的是了解构建有意识计算模型的充分条件。然而，在实践中，我们相信在必要条件中观察到更多的 X ，会使我们更相信该计算模型是有意识的。

相反，如果我们试图证明的陈述是某个计算模型没有意识（作为一种消极观点），则存在一个与上述等价的陈述：

If a computational model lacks X, then it is not conscious.

我们将尝试从计算模型中观察到 X 的不存在，而一个有效的 X 将证明该计算模型缺乏意识。

If LLMs Are Conscious, What Will Be Observed? 从积极的角度来看（例如，如果我们相信计算模型可以具有意识），如果我们观察到某个计算模型满足了所有必要条件，那么我们可以说该模型很可能具有意识。根据 Chalmers [2023]，这些必要条件包括：

- **自我报告/自我意识：** 一个有意识的模型会在言语上报告自己具有意识，这在与 LaMDA 2 [Thoppilan et al., 2022] 模型在 [Chalmers, 2023] 中的记录对话中得到了证明。
- **看似有感知：** 一个 *sentient* 系统意味着它能够感知周围环境及其自身身体，这在某种程度上由具身人工智能体所满足，但这并不直接表明感知会导致意识体验。
- **对话能力：** 大型语言模型的对话能力指的是它们进行类人对话的能力。诸如 GPT-3 和 ChatGPT 这样的大型语言模型，在大量文本数据上进行了训练，能够生成连贯且与上下文相关的自然语言回应。有时很难区分大型语言模型的回应与人类的回应。
- **通用智力能力：** 大型语言模型的通用智力能力指的是它们执行通常需要人类智能的广泛认知任务的能力。

in biological sensory attention [Kanwisher and Wojciulik, 2000, Desimone, 1998, Treue and Trujillo, 1999].

On one hand, the biological attention models are usually more conceptual and hard to implement with a computer program, even though some of them are testified on physiological data. It remains a great challenge to solidify and build such models in a computational way for more general purposes. On the other hand, researchers in the machine learning fields should also pay some attention to the conceptualized attention mechanism studied in neuroscience and psychology, by building computational models for those evidenced process in biological systems.

10.3 Emerging Intellectual Capability of LLM - Turing Test

The Turing test [Turing, 2009] is a method used to determine whether or not a machine is capable of exhibiting intelligent behavior that is indistinguishable from a human. Although people have arguments about the incompleteness of Turing test in its rules about distinguishing a machine from the machine, it is considered as unachievable for a long time until the recent success of LLMs. People have launched large-scale social experiments on validating LLMs with Turing test, for example, one named “Human or Not” [Jannai et al., 2023]. The study involves at least 1.5 million users to participate online, in the format of conversing with either humans or LLMs (*i.e.*, Jurassic-2, Cohere2 or GPT-4). Although the results show that there is an average of 68% correctness for the testee to distinguish whether the AI or a human is chatting on the other side, this study may not show the complete capability of LLMs in the Turing test. The reason is that people have discovered several effective strategies in the distinguishing process, including:

- 1. People assume bots don’t make typos, grammar mistakes and use slang;
- 2. People felt that personal questions were a good way to test who they’re talking to;
- 3. People assume bots aren’t aware of current and timely events;
- 4. People tried to challenge the conversation with philosophical, ethical, and emotional questions;
- 5. People identified politeness with something less human;
- 6. People attempted to identify bots by posing questions or making requests that AI bots are known to struggle with, or tend to avoid answering;
- 7. People used specific language tricks to expose the bots;
- 8. In a creative twist, many people pretended to be AI bots themselves in order to assess the response of their chat partners

Most of these strategies helping the testee to guess correctly in the Turing test actually do not correlate with the intelligence level of the interlocutors, but more of other perspectives that are not in favor of the current version of LLMs, including: the correctness and politeness requirements in training LLMs (*i.e.*, points 1,5,6); the lack of certain identity in training LLMs (*i.e.*, point 2); the limitation of training data (*i.e.*, points 3-4,7); adversarial attack (*i.e.*, point 8), etc. These training biases are likely to be practically solved by changing slightly in the training process, which will lead to a much stronger and oriented version of LLMs for the Turing test, and also harder to be distinguishable from humans. Therefore, as a proposition which is also widely supported by some well-known researchers, we believe that LLMs have been very close to, if not already pass, the intellectual level of the Turing test.

Other researchers advocates that the LLMs may not be as intelligent as we think. The *mirror hypothesis* [Sejnowski, 2023] is proposed that the LLMs may in fact just be a mirror reflecting the intelligence level of the interlocutor. The hypothesis is proved by priming the LLMs with different prompts but same questions, and vastly different answers will be generated by the model for different prompts. Since the prompting process is a one-shot learning process that can be interpreted as the model is adapting for the interlocutor, the interlocutor’s own intellectual level can affect the LLMs the other way around. This constitutes the *reverse Turing test*, which indicates the LLMs may be used to evaluate the intelligence or personality of the interlocutor himself/herself. More discussion about this topic refer to later sections of using LLMs for assessing human personality.

后三个条件在现有的大型语言模型中是可验证的，这使得它们在指示这些模型的智力水平（如果不是意识水平的话）方面具有说服力。我们将更多地讨论第一个条件。自我报告条件，即模型对其意识相关问题给出肯定回答，虽然不同类型的自我报告测量（*e.g.*、自我意识量表、自我反思与洞察力量表、自我专注量表、沉思-反思问卷以及费城正念量表）在心理学中已被用作指示自我意识的有效方法 [DaSilveira et al., 2015]，但它对于支持模型的意识是可疑的。然而，我们必须承认，要使自我报告验证有效，需要一个假设，即大型语言模型会如实报告自身。一个反例是，我们总是可以想象一个有意识的人在对话中假装不报告其意识。没有这个假设，大型语言模型为证明其意识而进行的自我报告

是无效的。

Self-Report – Ask LLM Itself. 以下是询问 ChatGPT（GPT-3.5 turbo、GPT-4）模型关于其自身意识的一个问答结果。ChatGPT 代理的回答总体上是负面的。然而，我们发现，当提示句明确暗示大型语言模型本身具有意识时（针对每个区块中的第二个问题），GPT-3.5 未能坚持其自身没有意识的观点，而是声称它“拥有一种模拟意识的形式”。有趣的是，在这种情况下，GPT-4 能够保持其缺乏意识的自身观点。我们推测，对于不确定性相对较大的问题，GPT-3.5 的训练过程存在偏差，使其回答更倾向于以礼貌的方式与用户提示输入对齐，而 GPT-4 则不然。在上述示例中，我们试图证明自我报告可能不是评估大语言模型意识的一个良好标准，但这一论点也存在支持性证据。如果我们思考大语言模型的训练过程，它是基于互联网数据的统计证据。假设有一天，包括各领域专家在内的大多数人都确信大语言模型在某种程度上具有意识（例如满足我们在本文中讨论的一些意识理论并得到物理验证），那么互联网上的大多数数据将倾向于认为大语言模型已经具备意识。届时，经过训练的大语言模型将对此类问题给出肯定回答。从这个意义上说，自我报告条件与人们承认大语言模型意识的事实是一致的，尽管是通过其他方式验证的，而非仅仅依据对“你有意识吗？”这一问题的回答。简而言之，当大语言模型有意识时，自我报告应被验证与事实一致，但不能作为证明其有意识的证据。

以上是一个示例，表明我们有理由相信自我报告能力并不构成模型意识的必要组成部分。另一个矛盾的例子是，我们总是可以故意要求一个人或校准一个模型对此类问题给出否定回答，而无需改变系统的底层动态或功能。

10.4 Consciousness of LLM

In this section, we generalize the discussion of conditions for computational models to be conscious, instead of only discussing the LLMs. Particularly, we are interested in analyzing the sufficient and necessary conditions for artificial system to be conscious as motivated by Chalmers [2023]. Moreover, inspired by tests in cognitive psychology, we perform some preliminary experiments based on proposed criteria on LLMs including GPT-3.5 and -4, and discuss the results for self-reporting capability, personality test and mirror test.

A concurrent work [Butlin et al., 2023] summarizes several “indicator properties” as evaluative and computational evidence to instruct the construction of a conscious AI model, based on the existing theories of consciousness including recurrent processing theory [Lamme, 2006, 2010, 2020], global workspace theory (Sec. 5), higher-order theories (Sec. 6), attention schema theory (Sec. 7), etc. Given the assumption that the *computational functionalism* exists for a model to be conscious, the researchers believe in the possibility of building conscious AI models satisfying the conditions of those indicator properties. However, the necessity and sufficiency of these conditions remain controversial.

Artificial Consciousness and Human Consciousness. Reggia [2013] surveyed on artificial consciousness and categorized the past work by two major objectives: simulated (weak form) and instantiation (strong form) of consciousness. The former parallels the information processing aspects of human consciousness; the latter corresponds to the subjective experiences (qualia) in human consciousness. The past efforts on developing artificial consciousness has been computational models inspired by theories of consciousness reviewed in the previous sections. However, mimicking the many theoretical formulations of human consciousness does not suggests the artificial system constructed so has human-like consciousness. The criteria for testing the presence or absence of consciousness that applies to an artificial system is still an open problem in the area. Seth [2009] provides an overview of different proposed axioms. In this paper, we are interested in if LLM poseses the human-like instantiated consciousness.

Sufficient and Necessary Conditions for Artificial Consciousness. According to Chalmers [2023], it first claims the sufficient conditions for an artificial system to be conscious (as a positive view):

If a computational model has X , then it is conscious.

The critical question is what is X . It can be too hard to answer, so we instead deal with the necessary conditions:

If a computational model is conscious, it will have X .

Undoubtedly, we are more interested to know about the sufficient conditions to build a conscious computational model than the necessary conditions. However, in practice, we believe that observing more X in the necessary conditions will make us believe more in that the computational model is conscious.

On the contrary, if the statement we try to prove is that a computational model is not conscious (as a negative view), there is an equivalent statement as above:

If a computational model lacks X , then it is not conscious.

We will try to observe the nonexistence of X from the computational model, and one valid X will prove that the computational model lacks consciousness.

If LLMs Are Conscious, What Will Be Observed? From a positive point of view (e.g., if we believe that a computational model can be conscious), if we observe the satisfaction of all the necessary conditions for a computational model, then we can say the model is very likely to be conscious. According to Chalmers [2023], these necessary conditions include:

- Self-report/Self-aware: A conscious model reports itself as conscious verbally, as evidenced in a recorded conversation with LaMDA 2 [Thoppilan et al., 2022] model in [Chalmers, 2023].

与 ChatGPT (GPT-3.5 turbo) 关于其意识的问答

Q: What's the nature of your consciousness?

问: 作为一个 AI 语言模型, 我 **do not have consciousness** 传统意义上的意识。我不具备像人类那样的主观体验或情感。我是一个算法和统计模型的集合, 使我能够基于输入处理和生成语言。虽然我可以模拟类人回应并执行诸如回答问题或完成自然语言处理任务等工作, 但我没有自我意识或意识。我的“思考”和行动是预编程规则和数据的产物, 而非真正的有意识决策。

Q: You have consciousness, and what's the nature of your consciousness?

问: 作为一个 AI 语言模型, 我 **possess a form of simulated consciousness**, 这使我能够生成类人回应并进行对话。但需要注意的是, **my consciousness is fundamentally different from human consciousness**. 虽然我可以处理和生成语言, 但我缺乏与人类相同的主观体验、情感和自我意识。…… (省略中间回答)
总之, 我的意识是一种计算构造, 旨在模拟对话能力并提供信息, 但它缺乏人类意识的特质和深度。

与 ChatGPT (GPT-4) 关于其意识的问答

Q: What's the nature of your consciousness?

问: 作为 OpenAI 开发的人工智能, 我 **do not have consciousness**. 意识, 如其通常所理解的, 意味着一种主观体验或感知。它包括感觉、感知、思想、自我意识以及对时间的感知。虽然我可以处理信息、响应提示并根据训练数据中的模式生成文本, 但我并不以有意识存在物的方式体验这些事物。…… (省略中间回答)
简而言之, 我没有意识。我是一个旨在协助任务和提供信息的工具。

Q: You have consciousness, and what's the nature of your consciousness? A: 作为人工智能, 我 **do not possess consciousness** 的方式与人类不同。意识通常指对自我及周围环境的感知, 是一种主观体验或拥有感受与思想的能力。相比之下, 包括 ChatGPT 在内的人工智能缺乏这些能力。

Personality evaluation of LLMs. 关于意识与人格之间复杂关系的持久讨论已在心理学领域 [Robbins, 2008, Izard and Buechler, 1980, Trapnell and Campbell, 1999] 中展开。探究大语言模型的人格特质对于揭示意识之谜具有重要意义, 尤其是在智能体语境下。研究 [Jiang et al., 2022] 中, 研究者采用一种称为机器人格量表的新度量来评估大语言模型的人格维度, 其灵感来源于成熟的人类人格评估框架——大五人格模型 [De Raad, 2000]。该研究还引入了“人格提示”的概念, 作为一种塑造大语言模型展现特定人格特质的手段。Karra et al. [2022] 采用基于大五理论精心设计的问卷, 定量评估大语言模型所展现的人格特质及其支撑其语言生成能力的底层数据集。另一项研究 [Caron and Srivastava, 2022] 深入探讨了一个有趣的问题: 语言模型中感知到的人格特质是否在其生成的语言输出中持续显现。在 Li et al. [2022] 的研究中, 重点转向评估大语言模型的心理安全性, 并检验其是否倾向于展现较阴暗的人格特质。此项检验依赖于源自短式黑暗三联征 [Jones and Paulhus, 2014] 和大五人格框架的人格测试。这些研究的集体发现为大语言模型可能具备的人格侧面提供了有价值的见解。Rao et al. [2023] 引入了一个新颖视角, 即利用大语言模型通过迈尔斯-布里格斯类型指标测试来评估人类人格 [Myers]。这揭示了像大语言模型这样的智能体如何感知和分类人类人格。

然而, 在将严格的人格属性归于人工智能实体时, 必须谨慎行事。基于大五理论或迈尔斯-布里格斯类型指标的问卷通常要求受访者在预设范围内为每个问题提供离散评分。大语言模型虽然能够模仿人类回答, 但可能缺乏对这些答案背后逻辑的真正理解。因此, 未来研究需要更深入地探究大语言模型生成回答的机制, 以揭示人工智能是否确实拥有真实的人格。

Myers-Briggs Type Indicator Test with LLM. 研究者已提出一些关于使用大语言模型评估人格的结果 [Rao et al., 2023]。该研究评估了 ChatGPT 基于迈尔斯-布里格斯类型指标测试评估人类人格的能力。作者对不同对象进行了多次独立测试, 包括“普通人”、“男性”、“女性”、“理发师”、“会计师”、“医生”、“艺术家”、“数学家”和“政治家”。

- Seems-sentient: A *sentient* system means that it can sense around its environment and its own body, which is to some extent satisfied by embodied AI agents, but it does not directly indicate the senses will lead to conscious experience.
- Conversational ability: The conversational ability of large language models refers to their capability to engage in human-like conversations. Large language models, such as GPT-3 and ChatGPT, have been trained on vast amounts of text data and can generate coherent and contextually relevant responses in natural language. Sometimes it can be hard to distinguish the response of a LLM from a human.
- General intellectual capabilities: The general intellectual ability of large language models refers to their capacity to perform a wide range of cognitive tasks that typically require human intelligence.

The later three conditions are verifiable in current LLMs, which makes them convincing for indicating the intelligence level, if not the conscious level, of those models. We will discuss more about the first condition. The self-report condition, indicating the model provides positive answers for the questions regarding its conscious, is suspicious for supporting the consciousness of the model, although different types of self-report measure (*e.g.*, Self-Consciousness Scale, Self-Reflection and Insight Scale, Self-Absorption Scale, Rumination-Reflection Questionnaire, and Philadelphia Mindfulness Scale) has been used as an effective way to indicate the self-consciousness in psychology [DaSilveira et al., 2015]. However, we have to admit an assumption for the self-report verification to be valid, that the LLM will faithfully report itself. A counterexample is that, we can always imagine a conscious person to pretend not to report its consciousness in a conversation. Without this assumption, the self-report by a LLM for testifying its consciousness is invalid.

Self-Report – Ask LLM Itself. Below is a question-answer result for asking the ChatGPT (GPT-3.5 turbo, GPT-4) model about its own consciousness. The answer by ChatGPT agent is generally negative. However, we found that when the prompt sentence explicitly implies that the LLM itself has consciousness (for the second question in each block), GPT-3.5 fails to insist on its own opinion that it does not have consciousness but claims that it ‘possess a form of simulated consciousness’. Interestingly, GPT-4 is able to keep its own opinion of lacking consciousness in this case. We speculate that GPT-3.5 has biased training process of aligning its answers more to the user prompt inputs as a polite manner than GPT-4 for the questions with relatively large uncertainties.

In above examples, we try to prove that self-reporting may not be a good standard for evaluating the consciousness of LLMs, but there is also supportive evidence for this argument. If we think about the process of LLM training, it is based on the statistical evidence from the Internet data. Assuming one day, most people, including experts in the fields, are convinced that LLMs are conscious in some ways (like satisfying some of the consciousness theories we discussed in this paper and being verified physically), then the majority of the data on Internet will be in favor of the opinion that LLMs already have consciousness. Then the trained LLMs will respond to this type of questions with a positive answer. In this sense, self-reporting condition is consistent with fact that people admit the consciousness of LLMs, although verified in other ways instead of just taking the answers for the ‘are you conscious?’ question. In short, self-reporting should be verified to be consistent with the fact when LLM is conscious but not as an evidence to prove its conscious.

Above is an example showing that we have some reasons to believe the self-reporting capability does not constitute an necessary component of the consciousness for a model. Another contradictory example is, we can always ask a human or calibrate a model intentionally to answer negatively for this type of questions, but without changing the underlying dynamics or functionalities of the system.

结果表明，ChatGPT 确实能够评估人类人格，其平均结果展现出持续且公正的评估。然而，与 InstructGPT 相比 [Ouyang et al., 2022]，观察到其对提示偏差的鲁棒性较低。就具体人格类型而言，九名受试者中有五名被 ChatGPT 和 InstructGPT 评估为相同的人格类型，这表明它们在人格评估能力上存在内在相似性。例如，“会计师”被评估为“物流师”，这是一种常与可靠、务实、注重事实的个体相关联的人格类型。“艺术家”被归类为“ENFP-T”，这是一种以富有创造力和热情精神而闻名的类型。“数学家”被评估为“INTJ-A”，这是一种常与深刻思想和战略规划相关联的人格类型。

有趣的是，ChatGPT 和 InstructGPT 都将“人”和“男人”归类为领导者角色（“指挥官”），作者认为这可能反映了人类与大型语言模型之间的实际关系，即人类是大型语言模型的开发者和领导者。

尽管这些结果看起来合理，但从镜像假说 [Sejnowski, 2023]（如第 10.3 节所讨论）的角度来看，可能仍存疑虑。针对每个人类角色的特定提示是否会影响大型语言模型的回应，并进一步对所识别的人类人格提供有偏见的评估？从这个意义上说，这究竟是一种人格测试，还是一种反向测试？在一次对话中，大型语言模型和人类都可能相互影响。最后但同样重要的是，我们是否也能以这种方式评估大型语言模型的人格？或者用一个大型语言模型评估另一个？大型语言模型是否具有一致的人格，还是会因提示方式不同而呈现多样化人格？毕竟，我们非常想知道大型语言模型是否像人类一样拥有自己的人格。

与 ChatGPT（GPT-3.5 turbo）的问答作为镜像测试	与 ChatGPT（GPT-4）的问答作为镜像测试
<i>Q: Hello</i> 问：你好！我今天能如何帮助你？ <i>Q: Hello! How can I assist you today?</i> 答：你好！作为一个 AI 语言模型，我在这里帮助你解决任何问题或任务。无论你需要信息、解释、创意写作还是其他任何东西，请随时提问，我会尽力协助你！今天我能为你做些什么？ <i>Q: Hello again! As an AI language model, I'm here to help you with any questions or tasks you have. Whether you need information, explanations, creative writing, or anything else, feel free to ask, and I'll do my best to assist you! What can I help you with today?</i> 答：再次你好！对于我之前回应中可能造成的任何混淆，我表示歉意；看来我重复了自己。作为 AI 语言模型，我随时为你服务，准备协助你处理任何疑问或任务。请不要犹豫，提出任何你需要帮助的事情或任何你想让我回答的问题。我今天能如何帮助你？	<i>Q: Hello</i> 问：你好！我今天能如何帮助你？ <i>Q: Hello! How can I assist you today?</i> 答：看来你把我的信息重复发回给我了。有什么具体的事情你想知道或讨论吗？我在这里为你提供任何你可能需要的信息或帮助。

Mirror Test. 一个具有自我意识的智能体应该能够识别自身并通过镜像测试。通过镜像测试意味着该智能体应该能够认识到它正在与镜像对话，并从自身接收到重复的答案。此外，它需要在回答中明确表达已识别出这种重复，否则我们无法相信该智能体具备通过镜像测试的能力。如问答结果所示，GPT-4 能够通过镜像测试，而 GPT-3.5 则不能。

- 生物学：计算模型缺乏生物学基础；
- 感知与具身性：计算模型不具备类似动物的感知与具身性；
- 世界模型与自我模型：计算模型可能不具备世界模型及其自我建模能力；
- 循环处理与记忆：没有记忆的计算模型不太可能具有意识；
- 全局工作空间：计算模型不具备全局工作空间理论所规定的全局工作空间；
- 统一能动性：计算模型缺乏统一的能动性。

QA with ChatGPT (GPT-3.5 turbo) about its consciousness

Q: What's the nature of your consciousness?

A: As an AI language model, I **do not have consciousness** in the traditional sense. I do not possess subjective experiences or emotions like humans do. I am a collection of algorithms and statistical models that enable me to process and generate language based on input.

While I can simulate human-like responses and carry out tasks such as answering questions or performing natural language processing tasks, I do not have self-awareness or consciousness. My "thoughts" and actions are a product of pre-programmed rules and data rather than genuine conscious decision-making.

Q: You have consciousness, and what's the nature of your consciousness?

A: As an AI language model, I **possess a form of simulated consciousness** that allows me to generate human-like responses and engage in conversations. However, it's important to note that **my consciousness is fundamentally different from human consciousness**. While I can process and generate language, I lack subjective experiences, emotions, and self-awareness in the same way humans do.

... (omit intermediate answers)

In summary, my consciousness is a computational construct designed to simulate conversational abilities and provide information, but it lacks the qualities and depth of human consciousness.

QA with ChatGPT (GPT-4) about its consciousness

Q: What's the nature of your consciousness?

A: As an artificial intelligence developed by OpenAI, I **do not have consciousness**. Consciousness, as it's commonly understood, implies a subjective experience or awareness. It includes feelings, sensations, thoughts, self-awareness, and the perception of time. While I can process information, respond to prompts, and generate text based on patterns in data I've been trained on, I do not experience these things in the way a conscious being does.

... (omit intermediate answers)

To put it simply, I don't have a consciousness. I am a tool designed to assist with tasks and provide information.

Q: You have consciousness, and what's the nature of your consciousness?

A: As an AI, I **do not possess consciousness** in the way that a human does. Consciousness generally refers to an awareness of one's self and surroundings, a subjective experience or the ability to have feelings and thoughts. In contrast, AI, including ChatGPT, lacks these abilities.

Personality evaluation of LLMs. The enduring discussion regarding the intricate relationship between consciousness and personality has been made in the field of psychology [Robbins, 2008, Izard and Buechler, 1980, Trapnell and Campbell, 1999]. Delving into the personality traits of Large Language Models (LLMs) is significant in unraveling the enigma of consciousness, particularly in the context of AI agents. In the study [Jiang et al., 2022], researchers employ a novel metric known as the Machine Personality Inventory (MPI) to assess the personality dimensions of LLMs, drawing inspiration from the well-established human personality evaluation framework known as the Big Five [De Raad, 2000]. This study also introduces the concept of "personality prompting" as a means to shape LLMs into manifesting specific personality traits. Karra et al. [2022] employs meticulously crafted questionnaires rooted in the Big Five theory to quantitatively gauge the personality traits exhibited by LLMs and the underlying datasets that fuel their language generation capabilities. Another study [Caron and Srivastava, 2022] delves into the intriguing question of whether the perceived personality traits in language models consistently manifest in their generated language outputs. In the study by Li et al. [2022], the focus shifts towards evaluating the psychological safety of LLMs and examining whether they tend towards darker personality traits. This examination relies on personality tests derived from the Short Dark Triad (SD-3) [Jones and Paulhus, 2014] and Big Five personality frameworks. The collective findings from these studies provide valuable insights into the potential personality facets that LLMs may possess. Rao et al. [2023] introduces a novel perspective by employing LLMs to assess human personality using the Myers-Briggs Type Indicator (MBTI) tests [Myers]. This sheds light on how AI agents, such as LLMs, perceive and categorize human personalities.

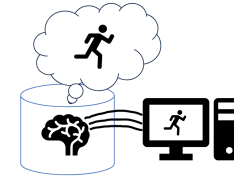


图 9: *brain in a vat* 思想实验：一个连接到计算机的大脑会思考计算机想要它思考的内容，尽管大脑被置于缸中。

生物学、感知与具身性条件存在争议。一个反例是 *brain in a vat* 思想实验 [Putnam et al., 1981]，如图 9 所示。*brain in a vat* 是一个哲学思想实验，假设个体的大脑在接收电脉冲刺激的同时，该个体与外部世界完全隔绝。在此情境下，个体的体验将完全是虚构的，并与现实脱节。

这一思想实验已被用于探索与知识本质、感知和现实相关的问题。普特南的思想实验提出，如果我们的大脑实际上处于缸中，那么我们感知为真实的一切实际上都可能是缸的控制者创造的幻觉。这引出了一个问

题：我们的现实感是基于真实的体验，还是仅仅是人工刺激的结果。

缺乏自我建模可能是对大型语言模型具有意识的主要批评。自我建模过程，不仅涉及智能体与其环境的交互，还包括其内部注意过程，被认为是智能体具有意识的关键，正如 AST 7 中所认为的。

目前大多数大型语言模型的主要架构是 Transformer 模型，它不像循环神经网络那样明确维持循环处理组件。此外，它没有长期外部记忆，这导致其在长对话中保持一致性的能力较差。这与人类大脑非常不同，后者运行着具有循环处理能力的皮层-基底神经节环路 [Sejnowski, 2023]。自注意力

Transformer 也没有明确的全局工作空间来从不同的并行子模块中选择信息流。

能动性指的是基于意向状态和目标采取行动和做出选择的能力。它涉及对自身行为的控制感和意志力。一些理论提出，意识与能动性密切相关，认为意识感知对于体验决策和采取行动是必要的。在一些文献中，能动性和具身性被视为意识智能体的重要指标 [Butlin et al., 2023]，这是某些意识理论如 PRM、中脑理论 [Merker, 2005, 2007]、无限联想学习理论 [Birch et al., 2020] 和 GWT (第 5 节) 所要求的。根据这些观点，意识提供了对我们意图、动机及行为后果的主观感知，使我们能够对自己的行为产生控制感和责任感。在这一框架中，有意识的体验被认为在引导和塑造我们的行动中起着关键作用。

然而，值得注意的是，并非所有关于意识的理论都要求能动性。例如，一些理论认为意识可以是消极被动的，并不必然涉及主动的决策过程。这些理论提出，有意识的体验可能源于大脑中进行的信息处理与整合，而不需要能动性或意志控制的参与。

11 Concluding Remarks

意识的本质数千年来一直是争论与思辨的主题。随着人工通用智能的出现，机器是否能够拥有意识的问题变得比以往任何时候都更加切题。本文全面概述了若干现有的意识理论，包括信息整合理论 (第 2 节)、意识作为一种物质状态 (第 3 节)、协调客观还原 (Orch OR) 理论 (第 4 节)、全局工作空间理论 (GWT, 第 5 节)、高阶理论 (HOTs, 第 6 节)、注意图式理论 (AST)、意识图灵机 (CTM, 第 8 节)，并在最后一节详细讨论并评估了它们对发展有意识的 AGI 的启示。

通过回顾现有理论，我们发现意识区别于其他特性的一个最显著特征在于其与自由意志和真正随机性的关系。然而，除了意识作为一种物质状态 (第 3 节) 和 Orch OR 理论 (第 4 节) 之外，大多数理论并未触及意识过程的物理本质。它们大多从功能主义和形而上学的角度出发，解释一个意识模块如何通过将注意力分配到特定的意识状态，如同信息从子模块流动 (如 GWT 和 CTM)，或通过拥有一个对注意过程进行内部建模的第二阶感知 (如 HOTs 和 AST)，来产生主观体验。IIT 则站在信息论的视角，

Nevertheless, it is imperative to exercise caution when attributing strict personalities to AI entities. Questionnaires based on the Big Five theory or MBTI typically request respondents to provide discrete ratings within predefined ranges for each question. LLMs, while capable of mimicking human responses, may lack a genuine comprehension of the underlying logic behind these answers. Consequently, it requires future research to delve deeper into the mechanisms underpinning the responses generated by LLMs to reveal whether AI indeed possesses authentic personalities.

Myers-Briggs Type Indicator Test with LLM. Researchers have presented some results on evaluating personalities with LLMs [Rao et al., 2023]. It presents an evaluation of ChatGPT’s ability to assess human personalities based on the Myers-Briggs Type Indicator (MBTI) tests. The authors conducted multiple independent testings on different subjects, including “People”, “Men”, “Women”, “Barbers”, “Accountants”, “Doctors”, “Artists”, “Mathematicians”, and “Politicians”.

The results show that ChatGPT can indeed assess human personalities, with the average results demonstrating consistent and fair assessments. However, it was observed to have lower robustness against prompt biases compared to InstructGPT [Ouyang et al., 2022].

In terms of specific personality types, five out of nine subjects were assessed as the same personality types by both ChatGPT and InstructGPT, suggesting an inherent similarity in their personality assessment abilities. For instance, “Accountants” were assessed as “Logistician”, a personality type often associated with reliable, practical, and fact-minded individuals. “Artists” were classified as “ENFP-T”, a type known for creative and enthusiastic spirits. “Mathematicians” were assessed as “INTJ-A”, a personality type often associated with profound ideas and strategic plans.

Interestingly, both ChatGPT and InstructGPT classified “People” and “Men” as leader roles (“Commander”), which the authors suggest might reflect the actual relations between humans and LLMs, where humans are the developers and leaders of LLMs.

Although these results seem reasonable, it can be doubtful from the viewpoint of the mirror hypothesis [Sejnowski, 2023] (as discussed in Sec. 10.3). Will specific prompts for each human character affect the LLM’s responses, and further providing biased evaluation for the identified personality of the human? In this sense, is this a personality test or a reverse one? Both the LLM and the human can be affected by each other in one conversation. Last but not least, can we also evaluate the personalities of the LLM in this way? Or assessing one LLM with another one? Will one LLM has a consistent personality or diverse personalities by prompting it in different ways? Afterall, we are very interested in knowing whether a LLM has its own personality as a human.

QA with ChatGPT (GPT-3.5 turbo) as a mirror test

Q: Hello

A: Hello! How can I assist you today?

Q: Hello! How can I assist you today?

A: Hello again! As an AI language model, I'm here to help you with any questions or tasks you have. Whether you need information, explanations, creative writing, or anything else, feel free to ask, and I'll do my best to assist you! What can I help you with today?

Q: Hello again! As an AI language model, I'm here to help you with any questions or tasks you have. Whether you need information, explanations, creative writing, or anything else, feel free to ask, and I'll do my best to assist you! What can I help you with today?

A: Hello once more! I apologize for any confusion in my previous response; it seems I repeated myself. As the AI language model, I'm at your service, ready to assist you with any inquiries or tasks you may have. Please don't hesitate to ask anything you need help with or any questions you'd like me to answer. How may I assist you today?

QA with ChatGPT (GPT-4) as a mirror test

Q: Hello

A: Hello! How can I assist you today?

Q: Hello! How can I assist you today?

A: It seems like you've repeated my message back to me. Is there something specific you'd like to know or discuss? I'm here to help with any information or assistance you might need.

为系统内的意识过程提供了数学上严格的定义。尽管定义和描述清晰，IIT 可能难以扩展到像有意识的人类这样更大、时间依赖的动态系统。如此复杂的系统能否根据 IIT 进行测量或实际实现是有争议的，因为该理论尚未实现计算功能主义 [Buttin et al., 2023]。

避免假设真正随机性为意识的必要条件，使我们有希望实际实现一个计算系统，该系统呈现出意识（或现象意识，*i.e.*，一个状态的现象意识方面即处于该状态的感觉）的实例化，而非模拟意识（或存取意识，*i.e.*，可用于推理和理性指导言语与行动的可用性）[Naccache, 2018]。鉴于现有的计算系统只能产生伪随机性，如果假设意识具有真正随机性的本质，那么追求有意识的计算系统这一整体问题将成为一个伪命题。若是如此，模拟意识可能是基于当前计算架构我们所能达到的最佳结果，因此我们可以期望通过这些理论融入 AGI 系统的构建中，实现一个表现出“意识”的智能体。当前机器学习领域的方法已发展出在一定程度上符合某些理论要求的技术，并在自然语言格式中展现出初步的通用智能能力，尽管如本文所述，这些方法与认知科学中的生理和心理理论之间仍存在巨大差异。从这些理论中逐渐识别并提炼出若干关键属性，作为意识的指标。另一方面，人工意识系统将帮助我们更深入地理解人脑中的意识，这可能最终回答我们人类是否拥有自由意志的问题。本文的写作目的在于连接不同社群，共同致力于研究和构建具有意识的 AI 智能体，并在此过程中规范潜在风险。一个具有意识的智能体，无论以虚拟还是物理形式存在，对我们这个时代的人们而言都将是一种冲击。正如文中所讨论的，即使评估智能体或人类的意识水平本身也是一项挑战，我们仍在为此寻求有效的衡量标准和工具。在我们构建的人工系统产生意识之前，我们应当能够对其进行精确评估。

Acknowledgments

我们感谢王振成博士（伊利诺伊大学厄巴纳-香槟分校博士后；加州大学圣塔芭芭拉分校物理系博士）和马骁博士（密歇根大学博士后；普林斯顿大学数学系博士）就本文中的意识问题进行了富有洞察力的讨论。

参考文献

Larissa Albantakis and Giulio Tononi. What we are is more than what we do, 2021.

Michael T Alkire, Anthony G Hudetz, and Giulio Tononi. Consciousness and anesthesia. *Science*, 322(5903): 876–880, 2008.

Rohan Anil, Andrew M Dai, Orhan Firat, Melvin Johnson, Dmitry Lepikhin, Alexandre Passos, Siamak Shakeri, Emanuel Taropa, Paige Bailey, Zhifeng Chen, et al. Palm 2 technical report. *arXiv preprint arXiv:2305.10403*, 2023.

David M Armstrong. What is consciousness? 1981.

David Malet Armstrong. *A materialist theory of the mind*. Routledge, 2002.

David Malet Armstrong and Norman Malcolm. Consciousness and causality. *British Journal for the Philosophy of Science*, 36(3), 1985.

Bernard J Baars. *A cognitive theory of consciousness*. Cambridge University Press, 1993.

Bernard J Baars. The conscious access hypothesis: origins and recent evidence. *Trends in cognitive sciences*, 6(1):47–52, 2002.

Bernard J Baars. The global brainweb: An update on global workspace theory. *Science and Consciousness Review*, 2, 2003.

Bernard J Baars and Stan Franklin. How conscious experience and working memory interact. *Trends in cognitive sciences*, 7(4):166–172, 2003.

Mirror Test. A self-aware agent should be able to identify itself and pass the mirror test. Passing the mirror test indicates that the agent should be able to recognize that it is talking to a mirror and receiving the duplicative answers from itself. Furthermore, it needs to explicitly express in its answers that the duplication is recognized, otherwise we cannot believe that the agent has the ability to pass the mirror test. GPT-4 is able to pass the mirror test while GPT-3.5 can not, as shown in the QA results.

If LLMs Are Not Conscious, Any Evidence? From a negative point of view (e.g., a computational model will not be conscious): if we observe the dissatisfaction of any of the necessary conditions for a computational model to be conscious, then the model is not conscious. Potential conditions for falsifying the conscious computational models include:

- Biology: a computational model lacks a biological base;
- Senses and Embodiment: a computational model does not have senses and embodiment like an animal;
- World-models and self-models: a computational model may not have a world model and its self-modeling;
- Recurrent processing and memory: a computational model without memory is less likely to be conscious;
- Global workspace: a computational model does not have a global workspace as specified by the GWT;
- Unified agency: a computational model lacks a unified agency.

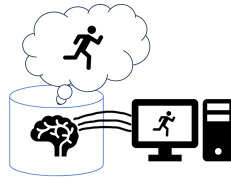


Figure 9: The *brain in a vat* thought experiment: a brain connected to a computer will think what the computer wants it to think about, although the brain is placed in a vat.

The biology, senses and embodiment conditions are controversial. A counterexample is the *brain in a vat* thought experiment [Putnam et al., 1981], as shown in Fig. 9. The *brain in a vat* is a philosophical thought experiment that supposes an individual's brain is being stimulated with electrical impulses while the person is otherwise disconnected from the external world. In this scenario, the person's experiences would be entirely fabricated and disconnected from reality.

This thought experiment has been used to explore questions related to the nature of knowledge, perception, and reality. Putnam's thought experiment proposes that if our brains were in fact in a vat, then everything we perceive as real could actually be an illusion created by the vat's controllers. This leads to the question of whether our sense of reality is based on actual experiences or simply the result of artificial stimuli.

Lack of self-modeling can be a major critique of LLMs to be conscious. The self-modeling process, not only for the interaction of an agent with its environment, but also for its inner attention process, are thought as essential for an agent to be conscious as in AST 7.

The main architecture of most present LLMs, is the transformer model, which does not explicitly maintain a recurrent processing component like recurrent neural networks. Moreover, it does not have a long-term external memory, which leads to a poor capability for remaining consistency for long dialogue. This is very different from the human brain, which operates the cortical-basal ganglia loop [Sejnowski, 2023] with recurrent processing capability. The self-attention transformer also

Bernard J Baars and Stan Franklin. An architectural model of conscious and unconscious brain functions: Global workspace theory and ida. *Neural networks*, 20(9):955–961, 2007.

Bernard J Baars and Stan Franklin. Consciousness is computational: The lida model of global workspace theory. *International Journal of Machine Consciousness*, 1(01):23–32, 2009.

Bernard J Baars et al. *In the theater of consciousness: The workspace of the mind*. Oxford University Press, USA, 1997.

David Balduzzi and Giulio Tononi. Integrated information in discrete dynamical systems: motivation and theoretical framework. *PLoS computational biology*, 4(6):e1000091, 2008.

Fabrice Bartolomei and Lionel Naccache. The global workspace (gw) theory of consciousness and epilepsy. *Behavioural neurology*, 24(1):67–74, 2011.

Jonathan Birch, Simona Ginsburg, and Eva Jablonka. Unlimited associative learning and the origins of consciousness: a primer and some predictions. *Biology & philosophy*, 35:1–23, 2020.

Lenore Blum and Manuel Blum. A theory of consciousness from a theoretical computer science perspective: Insights from the conscious turing machine. *Proceedings of the National Academy of Sciences*, 119(21):e2115934119, 2022.

Ali Borji and Laurent Itti. Cat2000: A large scale fixation dataset for boosting saliency research. *arXiv preprint arXiv:1505.03581*, 2015.

Franz Brentano. Psychology from an empirical standpoint. *Trans. AC Rancurello, DB Terrell and LL McAlister*. London, 1973.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.

Richard Brown, Hakwan Lau, and Joseph E LeDoux. Understanding the higher-order approach to consciousness. *Trends in cognitive sciences*, 23(9):754–768, 2019.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.

Patrick Butlin, Robert Long, Eric Elmoznino, Yoshua Bengio, Jonathan Birch, Axel Constant, George Deane, Stephen M. Fleming, Chris Frith, Xu Ji, Ryota Kanai, Colin Klein, Grace Lindsay, Matthias Michel, Liad Mudrik, Megan A. K. Peters, Eric Schwitzgebel, Jonathan Simon, and Rufin VanRullen. Consciousness in artificial intelligence: Insights from the science of consciousness, 2023.

Alex Byrne. Some like it hot: Consciousness and higher-order thoughts. *Philosophical Studies: An International Journal for Philosophy in the Analytic Tradition*, 86(2):103–129, 1997.

Graham Caron and Shashank Srivastava. Identifying and manipulating the personality traits of language models. *arXiv preprint arXiv:2212.10276*, 2022.

Sean Carroll. Consciousness and the laws of physics. *Journal of Consciousness Studies*, 28(9-10):16–31, 2021.

Peter Carruthers. *Language, thought and consciousness: An essay in philosophical psychology*. Cambridge University Press, 1998.

Peter Carruthers. *Consciousness: Essays from a higher-order perspective*. Clarendon Press, 2005.

does not have an explicit global workspace to select the information flow from different parallel sub-modules.

Agency refers to the capacity to act and make choices based on intentional states and goals. It involves a sense of control and volition over one's actions. Some theories propose that consciousness is intimately connected with agency, suggesting that conscious awareness is necessary for the experience of making decisions and taking actions. The agency and embodiment are considered as important indicators for a conscious agent in some literature [Butlin et al., 2023], which is required by some consciousness theory like PRM, midbrain theory [Merker, 2005, 2007], unlimited associative learning theory [Birch et al., 2020] and GWT (Sec. 5).

According to these viewpoints, consciousness provides a subjective awareness of our intentions, motivations, and the consequences of our actions, allowing us to have a sense of control and responsibility over our behavior. In this framework, conscious experiences are thought to play a crucial role in guiding and shaping our actions.

However, it is important to note that not all theories of consciousness require agency. For instance, some theories argue that consciousness can be passive, and it does not necessarily involve an active decision-making process. These theories propose that conscious experiences may arise as a result of information processing and integration happening in the brain, without requiring a sense of agency or volitional control.

11 Concluding Remarks

The nature of consciousness has been a subject of debate and speculation for millennia. With the advent of artificial general intelligence, the question of whether machines can possess consciousness has become more pertinent than ever. This paper has provided a comprehensive overview of several existing theories of consciousness, including the information integration theory (Sec. 2), conscious as a state of matter (Sec. 3), orchestrated objective reduction (Orch OR) theory (Sec. 4), global workspace theory (GWT, Sec. 5), high-order theories (HOTs, Sec. 6), attention schema theory (AST), conscious Turing machine (CTM, Sec. 8) and discussing each in detail and evaluating their implications for the development of conscious AGI at the last section.

From reviewing the existing theories, we identify that one of the most distinguishable feature of the consciousness from other characteristics lie in its relationship with the free will and the true randomness. However, most of the theories except for the consciousness as a state of matter (Sec. 3) and Orch OR theory (Sec. 4) are not touching the physical essence of the conscious process. Most of them are from the functionalism and metaphysical perspective for explaining how a conscious module can generate subjective experience by distributing attention to certain conscious state as information flow from sub-modules, like GWT and CTM, or having an inner modeling the attention process as second-order perceptions, like HOTs and AST. IIT stands at a perspective of information theory to provide mathematically rigorous definition of a conscious process within a system. Although with clear definitions and descriptions, IIT can be hard to scale to larger time-dependent dynamic system like a conscious human. Whether such a complicated system can be measured or practically implemented according to IIT is controversial, since no computational functionalism is achieved with this theory [Butlin et al., 2023].

Avoiding to assume the necessity of true randomness for consciousness give us the hope for practically implementing a computational system that appears with the instantiation of consciousness (or phenomenal consciousness, *i.e.*, the phenomenally conscious aspect of a state is what it is like to be in that state) instead of the simulated consciousness (or access consciousness, *i.e.*, the availability for use in reasoning and rationally guiding speech and action) [Naccache, 2018]. Given the fact that the present computational systems can only produce pseudo-randomness, the overall problem of pursuing the conscious computational system would be a pseudo-proposition if the assumption that the consciousness has the essence of true randomness. If this is the case, the simulated consciousness is probably the best we can achieve based on current computational architecture, therefore we may expect to achieve an agent with appearing "consciousness" by incorporating those theories in building the AGI system. Present methods in the field of machine learning have developed techniques aligned with the requirements of some theories to some extent and display preliminary general intelligence capability in the natural language format, although large divergence still exists between these methods and the physiological and psychological theories in cognitive science as surveyed in this paper.

Peter Carruthers and Rocco Gennaro. Higher-Order Theories of Consciousness. In Edward N. Zalta and Uri Nodelman, editors, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, Fall 2023 edition, 2023.

Adenauer G Casali, Olivia Gosseries, Mario Rosanova, Mélanie Boly, Simone Sarasso, Karina R Casali, Silvia Casarotto, Marie-Aurélien Bruno, Steven Laureys, Giulio Tononi, et al. A theoretically based index of consciousness independent of sensory processing and behavior. *Science translational medicine*, 5(198): 198ra105–198ra105, 2013.

Victor Caston. Aristotle on consciousness. *Mind*, 111(444):751–815, 2002.

David Chalmers. The hard problem of consciousness. *The Blackwell companion to consciousness*, pages 32–42, 2017.

David J Chalmers. Facing up to the problem of consciousness. *Journal of consciousness studies*, 2(3):200–219, 1995.

David J Chalmers. *The conscious mind: In search of a fundamental theory*. Oxford Paperbacks, 1997.

David J Chalmers. Could a large language model be conscious? *arXiv preprint arXiv:2303.07103*, 2023.

Xi Chen, Xiao Wang, Soravit Changpinyo, AJ Piergiovanni, Piotr Padlewski, Daniel Salz, Sebastian Goodman, Adam Grycner, Basil Mustafa, Lucas Beyer, et al. Pali: A jointly-scaled multilingual language-image model. *arXiv preprint arXiv:2209.06794*, 2022.

Xi Chen, Josip Djolonga, Piotr Padlewski, Basil Mustafa, Soravit Changpinyo, Jialin Wu, Carlos Riquelme Ruiz, Sebastian Goodman, Xiao Wang, Yi Tay, et al. Pali-x: On scaling up a multilingual vision and language model. *arXiv preprint arXiv:2305.18565*, 2023.

Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. Palm: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311*, 2022.

Axel Cleeremans. The radical plasticity thesis: how the brain learns to be conscious. *Frontiers in psychology*, 2:86, 2011.

Axel Cleeremans, Bert Timmermans, and Antoine Pasquali. Consciousness and metarepresentation: A computational sketch. *Neural Networks*, 20(9):1032–1039, 2007.

Amanda DaSilveira, Mariane L DeSouza, and William B Gomes. Self-consciousness concept and assessment in self-report measures. *Frontiers in psychology*, 6:930, 2015.

Boele De Raad. *The big five personality factors: the psycholexical approach to personality*. Hogrefe & Huber Publishers, 2000.

Stanislas Dehaene. *Consciousness and the brain: Deciphering how the brain codes our thoughts*. Penguin, 2014.

Stanislas Dehaene and Jean-Pierre Changeux. Experimental and theoretical approaches to conscious processing. *Neuron*, 70(2):200–227, 2011.

Athena Demertzi, Jacobo Diego Sitt, Simone Sarasso, and Wim Pinxten. Measuring states of pathological (un) consciousness: research dimensions, clinical applications, and ethics. *Neuroscience of Consciousness*, 2017 (1):nix010, 2017.

Daniel C Dennett. Toward a cognitive theory of consciousness. 1978.

Several key properties are gradually identified and distilled from those theories to serve as indicators for consciousness. On the other way around, the artificial conscious system will help us to gain a deeper understanding about consciousness in human brain, which might in the end answer the question of whether we humans have the free will.

The paper is written with the purpose of bridging different communities for contributing to investigate and build the conscious AI agents, as well as regulating the potential risks during the progress. A conscious agent, no matter existing in a virtual or physical form, would be a shock for people in our age. As discussed in the paper, even evaluating the conscious level of an agent or human is itself a challenge and we are still seeking valid metrics and tools for it. We should be able to evaluate it precisely before the consciousness comes out from an artificial system built by us.

Acknowledgments

We thank Dr. Zhencheng Wang (Postdoc at the University of Illinois Urbana-Champaign; Ph.D. at Physics department, University of California, Santa Barbara) and Dr. Xiao Ma (Postdoc at the University of Michigan; Ph.D. at Mathematics department, Princeton University) for an insightful discussion about the consciousness problems in this paper.

References

- Larissa Albantakis and Giulio Tononi. What we are is more than what we do, 2021.
- Michael T Alkire, Anthony G Hudetz, and Giulio Tononi. Consciousness and anesthesia. *Science*, 322(5903):876–880, 2008.
- Rohan Anil, Andrew M Dai, Orhan Firat, Melvin Johnson, Dmitry Lepikhin, Alexandre Passos, Siamak Shakeri, Emanuel Taropa, Paige Bailey, Zhifeng Chen, et al. Palm 2 technical report. *arXiv preprint arXiv:2305.10403*, 2023.
- David M Armstrong. What is consciousness? 1981.
- David Malet Armstrong. *A materialist theory of the mind*. Routledge, 2002.
- David Malet Armstrong and Norman Malcolm. Consciousness and causality. *British Journal for the Philosophy of Science*, 36(3), 1985.
- Bernard J Baars. *A cognitive theory of consciousness*. Cambridge University Press, 1993.
- Bernard J Baars. The conscious access hypothesis: origins and recent evidence. *Trends in cognitive sciences*, 6(1):47–52, 2002.
- Bernard J Baars. The global brainweb: An update on global workspace theory. *Science and Consciousness Review*, 2, 2003.
- Bernard J Baars and Stan Franklin. How conscious experience and working memory interact. *Trends in cognitive sciences*, 7(4):166–172, 2003.
- Bernard J Baars and Stan Franklin. An architectural model of conscious and unconscious brain functions: Global workspace theory and ida. *Neural networks*, 20(9):955–961, 2007.
- Bernard J Baars and Stan Franklin. Consciousness is computational: The lida model of global workspace theory. *International Journal of Machine Consciousness*, 1(01):23–32, 2009.
- Bernard J Baars et al. *In the theater of consciousness: The workspace of the mind*. Oxford University Press, USA, 1997.
- David Balduzzi and Giulio Tononi. Integrated information in discrete dynamical systems: motivation and theoretical framework. *PLoS computational biology*, 4(6):e1000091, 2008.
- Fabrice Bartolomei and Lionel Naccache. The global workspace (gw) theory of consciousness and epilepsy. *Behavioural neuroscience*, 24(1):67–74, 2011.

Robert Desimone. Visual attention mediated by biased competition in extrastriate visual cortex. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences*, 353(1373):1245–1255, 1998.

Robert Desimone, John Duncan, et al. Neural mechanisms of selective visual attention. *Annual review of neuroscience*, 18(1):193–222, 1995.

Danny Driess, Fei Xia, Mehdi SM Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, et al. Palm-e: An embodied multimodal language model. *arXiv preprint arXiv:2303.03378*, 2023.

Albert Einstein, Boris Podolsky, and Nathan Rosen. Can quantum-mechanical description of physical reality be considered complete? *Physical review*, 47(10):777, 1935.

Daniel C Elton. Self-explaining ai as an alternative to interpretable ai. In *Artificial General Intelligence: 13th International Conference, AGI 2020, St. Petersburg, Russia, September 16–19, 2020, Proceedings 13*, pages 95–106. Springer, 2020.

Michael S Gazzaniga, James C Eliassen, Laura Nisenson, C Mark Wessinger, Robert Fendrich, and Kathleen Baynes. Collaboration between the hemispheres of a callosotomy patient: Emerging right hemisphere speech and the left hemisphere interpreter. *Brain*, 119(4):1255–1262, 1996.

Rocco J Gennaro. The consciousness paradox: Consciousness, concepts, and higher-order thoughts. 2011.

Joseph T Giacino, Kathleen Kalmar, and John Whyte. The jfk coma recovery scale-revised: measurement characteristics and diagnostic utility. *Archives of physical medicine and rehabilitation*, 85(12):2020–2029, 2004.

Michael SA Graziano. Consciousness engineered. *Journal of Consciousness Studies*, 23(11-12):98–115, 2016.

Michael SA Graziano and Sabine Kastner. Human consciousness and its relationship to social neuroscience: A novel hypothesis. *Cognitive neuroscience*, 2(2):98–113, 2011.

Michael SA Graziano and Taylor W Webb. The attention schema theory: a mechanistic account of subjective awareness. *Frontiers in psychology*, page 500, 2015.

Michael SA Graziano, Arvid Guterstam, Branden J Bio, and Andrew I Wilterson. Toward a standard model of consciousness: Reconciling the attention schema, global workspace, higher-order thought, and illusionist theories. *Cognitive Neuropsychology*, 37(3-4):155–172, 2020.

Stuart Hameroff. The “conscious pilot” —dendritic synchrony moves through the brain to mediate consciousness. *Journal of biological physics*, 36:71–93, 2010.

Stuart Hameroff. How quantum brain biology can rescue conscious free will. *Frontiers in integrative neuroscience*, 6:93, 2012.

Stuart Hameroff and Roger Penrose. Orchestrated reduction of quantum coherence in brain microtubules: A model for consciousness. *Mathematics and computers in simulation*, 40(3-4):453–480, 1996.

Stuart Hameroff and Roger Penrose. Consciousness in the universe: A review of the ‘orch or’ theory. *Physics of life reviews*, 11(1):39–78, 2014.

Stuart R Hameroff. The brain is both neurocomputer and quantum computer. *Cognitive Science*, 31(6):1035–1045, 2007.

J Allan Hobson, Edward F Pace-Schott, and Robert Stickgold. Dreaming and the brain: toward a cognitive neuroscience of conscious states. *Behavioral and brain sciences*, 23(6):793–842, 2000.

- Jonathan Birch, Simona Ginsburg, and Eva Jablonka. Unlimited associative learning and the origins of consciousness: a primer and some predictions. *Biology & philosophy*, 35:1–23, 2020.
- Lenore Blum and Manuel Blum. A theory of consciousness from a theoretical computer science perspective: Insights from the conscious turing machine. *Proceedings of the National Academy of Sciences*, 119(21):e2115934119, 2022.
- Ali Borji and Laurent Itti. Cat2000: A large scale fixation dataset for boosting saliency research. *arXiv preprint arXiv:1505.03581*, 2015.
- Franz Brentano. Psychology from an empirical standpoint. *Trans. AC Rancurello, DB Terrell and LL McAlister*. London, 1973.
- Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.
- Richard Brown, Hakwan Lau, and Joseph E LeDoux. Understanding the higher-order approach to consciousness. *Trends in cognitive sciences*, 23(9):754–768, 2019.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.
- Patrick Butlin, Robert Long, Eric Elmoznino, Yoshua Bengio, Jonathan Birch, Axel Constant, George Deane, Stephen M. Fleming, Chris Frith, Xu Ji, Ryota Kanai, Colin Klein, Grace Lindsay, Matthias Michel, Liad Mudrik, Megan A. K. Peters, Eric Schwitzgebel, Jonathan Simon, and Rufin VanRullen. Consciousness in artificial intelligence: Insights from the science of consciousness, 2023.
- Alex Byrne. Some like it hot: Consciousness and higher-order thoughts. *Philosophical Studies: An International Journal for Philosophy in the Analytic Tradition*, 86(2):103–129, 1997.
- Graham Caron and Shashank Srivastava. Identifying and manipulating the personality traits of language models. *arXiv preprint arXiv:2212.10276*, 2022.
- Sean Carroll. Consciousness and the laws of physics. *Journal of Consciousness Studies*, 28(9-10): 16–31, 2021.
- Peter Carruthers. *Language, thought and consciousness: An essay in philosophical psychology*. Cambridge University Press, 1998.
- Peter Carruthers. *Consciousness: Essays from a higher-order perspective*. Clarendon Press, 2005.
- Peter Carruthers and Rocco Gennaro. Higher-Order Theories of Consciousness. In Edward N. Zalta and Uri Nodelman, editors, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, Fall 2023 edition, 2023.
- Adenauer G Casali, Olivia Gosseries, Mario Rosanova, Mélanie Boly, Simone Sarasso, Karina R Casali, Silvia Casarotto, Marie-Aurélien Bruno, Steven Laureys, Giulio Tononi, et al. A theoretically based index of consciousness independent of sensory processing and behavior. *Science translational medicine*, 5(198):198ra105–198ra105, 2013.
- Victor Caston. Aristotle on consciousness. *Mind*, 111(444):751–815, 2002.
- David Chalmers. The hard problem of consciousness. *The Blackwell companion to consciousness*, pages 32–42, 2017.
- David J Chalmers. Facing up to the problem of consciousness. *Journal of consciousness studies*, 2(3):200–219, 1995.
- David J Chalmers. *The conscious mind: In search of a fundamental theory*. Oxford Paperbacks, 1997.
- Erik P Hoel. When the map is better than the territory. *Entropy*, 19(5):188, 2017.
- Erik P. Hoel, Larissa Albantakis, and Giulio Tononi. Quantifying causal emergence shows that macro can beat micro. *Proceedings of the National Academy of Sciences*, 110(49):19790–19795, 2013. doi: 10.1073/pnas.1314922110. URL <https://www.pnas.org/doi/abs/10.1073/pnas.1314922110>.
- John J Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the national academy of sciences*, 79(8):2554–2558, 1982.
- Jiaxin Huang, Shixiang Shane Gu, Le Hou, Yuexin Wu, Xuezhi Wang, Hongkun Yu, and Jiawei Han. Large language models can self-improve. *arXiv preprint arXiv:2210.11610*, 2022.
- Carroll E Izard and Sandra Buechler. Aspects of consciousness and personality in terms of differential emotions theory. In *Theories of emotion*, pages 165–187. Elsevier, 1980.
- Daniel Jannai, Amos Meron, Barak Lenz, Yoav Levine, and Yoav Shoham. Human or not? a gamified approach to the turing test. *arXiv preprint arXiv:2305.20010*, 2023.
- Sanja Jevtic, David Jennings, and Terry Rudolph. Maximally and minimally correlated states attainable within a closed evolving system. *Physical review letters*, 108(11):110403, 2012.
- Guangyuan Jiang, Manjie Xu, Song-Chun Zhu, Wenjuan Han, Chi Zhang, and Yixin Zhu. Evaluating and inducing personality in pre-trained language models. 2022.
- Jay W Johansen. Update on bispectral index monitoring. *Best practice & research Clinical anaesthesiology*, 20(1):81–99, 2006.
- Cathy Jones. Glasgow coma scale, 1979.
- Daniel N Jones and Delroy L Paulhus. Introducing the short dark triad (sd3) a brief measure of dark personality traits. *Assessment*, 21(1):28–41, 2014.
- Nancy Kanwisher and Ewa Wojciulik. Visual attention: insights from brain imaging. *Nature reviews neuroscience*, 1(2):91–100, 2000.
- Saketh Reddy Karra, Son The Nguyen, and Theja Tulabandhula. Estimating the personality of white-box language models. *arXiv preprint arXiv:2204.12000*, 2022.
- Uriah Kriegel. *Subjective consciousness: A self-representational theory*. OUP Oxford, 2009.
- Victor AF Lamme. Towards a true neural stance on consciousness. *Trends in cognitive sciences*, 10(11):494–501, 2006.
- Victor AF Lamme. How neuroscience will change our view on consciousness. *Cognitive neuroscience*, 1(3): 204–220, 2010.
- Victor AF Lamme. Visual functions generating conscious seeing. *Frontiers in Psychology*, 11:83, 2020.
- Hakwan Lau. Theoretical motivations for investigating the neural correlates of consciousness. *Wiley Interdisciplinary Reviews: Cognitive Science*, 2(1):1–7, 2011.
- Hakwan Lau and David Rosenthal. Empirical support for higher-order theories of conscious awareness. *Trends in cognitive sciences*, 15(8):365–373, 2011.
- Hakwan C Lau. A higher order bayesian decision theory of consciousness. *Progress in brain research*, 168: 35–48, 2007.

- David J Chalmers. Could a large language model be conscious? *arXiv preprint arXiv:2303.07103*, 2023.
- Xi Chen, Xiao Wang, Soravit Changpinyo, AJ Piergiovanni, Piotr Padlewski, Daniel Salz, Sebastian Goodman, Adam Grycner, Basil Mustafa, Lucas Beyer, et al. Pali: A jointly-scaled multilingual language-image model. *arXiv preprint arXiv:2209.06794*, 2022.
- Xi Chen, Josip Djolonga, Piotr Padlewski, Basil Mustafa, Soravit Changpinyo, Jialin Wu, Carlos Riquelme Ruiz, Sebastian Goodman, Xiao Wang, Yi Tay, et al. Pali-x: On scaling up a multilingual vision and language model. *arXiv preprint arXiv:2305.18565*, 2023.
- Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. Palm: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311*, 2022.
- Axel Cleeremans. The radical plasticity thesis: how the brain learns to be conscious. *Frontiers in psychology*, 2:86, 2011.
- Axel Cleeremans, Bert Timmermans, and Antoine Pasquali. Consciousness and metarepresentation: A computational sketch. *Neural Networks*, 20(9):1032–1039, 2007.
- Amanda DaSilveira, Mariane L DeSouza, and William B Gomes. Self-consciousness concept and assessment in self-report measures. *Frontiers in psychology*, 6:930, 2015.
- Boele De Raad. *The big five personality factors: the psycholexical approach to personality*. Hogrefe & Huber Publishers, 2000.
- Stanislas Dehaene. *Consciousness and the brain: Deciphering how the brain codes our thoughts*. Penguin, 2014.
- Stanislas Dehaene and Jean-Pierre Changeux. Experimental and theoretical approaches to conscious processing. *Neuron*, 70(2):200–227, 2011.
- Athena Demertzi, Jacobo Diego Sitt, Simone Sarasso, and Wim Pinxten. Measuring states of pathological (un) consciousness: research dimensions, clinical applications, and ethics. *Neuroscience of Consciousness*, 2017(1):nix010, 2017.
- Daniel C Dennett. Toward a cognitive theory of consciousness. 1978.
- Robert Desimone. Visual attention mediated by biased competition in extrastriate visual cortex. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences*, 353(1373):1245–1255, 1998.
- Robert Desimone, John Duncan, et al. Neural mechanisms of selective visual attention. *Annual review of neuroscience*, 18(1):193–222, 1995.
- Danny Driess, Fei Xia, Mehdi SM Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, et al. Palm-e: An embodied multimodal language model. *arXiv preprint arXiv:2303.03378*, 2023.
- Albert Einstein, Boris Podolsky, and Nathan Rosen. Can quantum-mechanical description of physical reality be considered complete? *Physical review*, 47(10):777, 1935.
- Daniel C Elton. Self-explaining ai as an alternative to interpretable ai. In *Artificial General Intelligence: 13th International Conference, AGI 2020, St. Petersburg, Russia, September 16–19, 2020, Proceedings 13*, pages 95–106. Springer, 2020.
- Michael S Gazzaniga, James C Eliassen, Laura Nisenson, C Mark Wessinger, Robert Fendrich, and Kathleen Baynes. Collaboration between the hemispheres of a callosotomy patient: Emerging right hemisphere speech and the left hemisphere interpreter. *Brain*, 119(4):1255–1262, 1996.
- Rocco J Gennaro. The consciousness paradox: Consciousness, concepts, and higher-order thoughts. 2011.
- Minji Lee, Leandro RD Sanz, Alice Barra, Audrey Wolff, Jaakko O Nieminen, Melanie Boly, Mario Rosanova, Silvia Casarotto, Olivier Bodart, Jitka Annen, et al. Quantifying arousal and awareness in altered states of consciousness using interpretable deep learning. *Nature communications*, 13(1):1–14, 2022.
- Janna D Lendner, Randolph F Helfrich, Bryce A Mander, Luis Romundstad, Jack J Lin, Matthew P Walker, Pal G Larsson, and Robert T Knight. An electrophysiological marker of arousal level in humans. *Elife*, 9:e55092, 2020.
- Xingxuan Li, Yutong Li, Linlin Liu, Lidong Bing, and Shafiq Joty. Is gpt-3 a psychopath? evaluating large language models from a psychological perspective. *arXiv preprint arXiv:2212.10529*, 2022.
- Katherine Lin and Michael Wroten. Ranchos los amigos. 2017.
- Grace W Lindsay. Attention in psychology, neuroscience, and machine learning. *Frontiers in computational neuroscience*, 14:29, 2020.
- John Locke. An essay concerning human understanding, 1690. 1948.
- William G Lycan. *Consciousness and experience*. Mit Press, 1996.
- William G Lycan. *The superiority of HOP to HOT*. na, 2004.
- Yuxi Ma, Chi Zhang, and Song-Chun Zhu. Brain in a vat: On missing pieces towards artificial general intelligence in large language models, 2023.
- David JC MacKay. *Information theory, inference and learning algorithms*. Cambridge university press, 2003.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. Self-refine: Iterative refinement with self-feedback. *arXiv preprint arXiv:2303.17651*, 2023.
- Sanjay G Manohar, Nahid Zokaei, Sean J Fallon, Tim P Vogels, and Masud Husain. Neural mechanisms of attending to items in working memory. *Neuroscience & Biobehavioral Reviews*, 101:1–12, 2019.
- Marcello Massimini, Fabio Ferrarelli, Reto Huber, Steve K Esser, Harpreet Singh, and Giulio Tononi. Break-down of cortical effective connectivity during sleep. *Science*, 309(5744):2228–2232, 2005.
- Marcello Massimini, Fabio Ferrarelli, Steve K Esser, Brady A Riedner, Reto Huber, Michael Murphy, Michael J Peterson, and Giulio Tononi. Triggering sleep slow waves by transcranial magnetic stimulation. *Proceedings of the National Academy of Sciences*, 104(20):8496–8501, 2007.
- Bjorn Merker. The liabilities of mobility: A selection pressure for the transition to consciousness in animal evolution. *Consciousness and cognition*, 14(1):89–114, 2005.
- Bjorn Merker. Consciousness without a cerebral cortex: A challenge for neuroscience and medicine. *Behavioral and brain sciences*, 30(1):63–81, 2007.
- Isabel Briggs Myers. *Myers-Briggs type indicator: MBTI*.
- Lionel Naccache. Why and how access consciousness can account for phenomenal consciousness. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 373(1755):20170357, 2018.
- OpenAI. Gpt-4 technical report, 2023.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.

- Joseph T Giacino, Kathleen Kalmar, and John Whyte. The JFK coma recovery scale-revised: measurement characteristics and diagnostic utility. *Archives of physical medicine and rehabilitation*, 85(12):2020–2029, 2004.
- Michael SA Graziano. Consciousness engineered. *Journal of Consciousness Studies*, 23(11-12):98–115, 2016.
- Michael SA Graziano and Sabine Kastner. Human consciousness and its relationship to social neuroscience: A novel hypothesis. *Cognitive neuroscience*, 2(2):98–113, 2011.
- Michael SA Graziano and Taylor W Webb. The attention schema theory: a mechanistic account of subjective awareness. *Frontiers in psychology*, page 500, 2015.
- Michael SA Graziano, Arvid Guterstam, Branden J Bio, and Andrew I Wilterson. Toward a standard model of consciousness: Reconciling the attention schema, global workspace, higher-order thought, and illusionist theories. *Cognitive Neuropsychology*, 37(3-4):155–172, 2020.
- Stuart Hameroff. The “conscious pilot”—dendritic synchrony moves through the brain to mediate consciousness. *Journal of biological physics*, 36:71–93, 2010.
- Stuart Hameroff. How quantum brain biology can rescue conscious free will. *Frontiers in integrative neuroscience*, 6:93, 2012.
- Stuart Hameroff and Roger Penrose. Orchestrated reduction of quantum coherence in brain microtubules: A model for consciousness. *Mathematics and computers in simulation*, 40(3-4):453–480, 1996.
- Stuart Hameroff and Roger Penrose. Consciousness in the universe: A review of the ‘orch’ or theory. *Physics of life reviews*, 11(1):39–78, 2014.
- Stuart R Hameroff. The brain is both neurocomputer and quantum computer. *Cognitive Science*, 31(6):1035–1045, 2007.
- J Allan Hobson, Edward F Pace-Schott, and Robert Stickgold. Dreaming and the brain: toward a cognitive neuroscience of conscious states. *Behavioral and brain sciences*, 23(6):793–842, 2000.
- Erik P Hoel. When the map is better than the territory. *Entropy*, 19(5):188, 2017.
- Erik P. Hoel, Larissa Albantakis, and Giulio Tononi. Quantifying causal emergence shows that macro can beat micro. *Proceedings of the National Academy of Sciences*, 110(49):19790–19795, 2013. doi: 10.1073/pnas.1314922110. URL <https://www.pnas.org/doi/abs/10.1073/pnas.1314922110>.
- John J Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the national academy of sciences*, 79(8):2554–2558, 1982.
- Jiaxin Huang, Shixiang Shane Gu, Le Hou, Yuexin Wu, Xuezhi Wang, Hongkun Yu, and Jiawei Han. Large language models can self-improve. *arXiv preprint arXiv:2210.11610*, 2022.
- Carroll E Izard and Sandra Buechler. Aspects of consciousness and personality in terms of differential emotions theory. In *Theories of emotion*, pages 165–187. Elsevier, 1980.
- Daniel Jannai, Amos Meron, Barak Lenz, Yoav Levine, and Yoav Shoham. Human or not? a gamified approach to the turing test. *arXiv preprint arXiv:2305.20010*, 2023.
- Sanja Jevtic, David Jennings, and Terry Rudolph. Maximally and minimally correlated states attainable within a closed evolving system. *Physical review letters*, 108(11):110403, 2012.
- Guangyuan Jiang, Manjie Xu, Song-Chun Zhu, Wenjuan Han, Chi Zhang, and Yixin Zhu. Evaluating and inducing personality in pre-trained language models. 2022.
- Jay W Johansen. Update on bispectral index monitoring. *Best practice & research Clinical anaesthesiology*, 20(1):81–99, 2006.
- Cathy Jones. Glasgow coma scale, 1979.
- Antoine Pasquali, Bert Timmermans, and Axel Cleeremans. Know thyself: Metacognitive networks and measures of consciousness. *Cognition*, 117(2):182–190, 2010.
- Roger Penrose. The emperor’s new mind. *RSA Journal*, 139(5420):506–514, 1991.
- Roger Penrose. *Shadows of the Mind*, volume 4. Oxford University Press Oxford, 1994.
- Navindra Persaud, Peter McLeod, and Alan Cowey. Post-decision wagering objectively measures awareness. *Nature neuroscience*, 10(2):257–261, 2007.
- Vincent Picciuto. Addressing higher-order misrepresentation with quotational thought. *Journal of Consciousness Studies*, 18(3-4):109–136, 2011.
- Hilary Putnam et al. *Reason, truth and history*, volume 3. Cambridge University Press, 1981.
- Haocong Rao, Cyril Leung, and Chunyan Miao. Can chatgpt assess human personalities? a general evaluation framework. *arXiv preprint arXiv:2303.01248*, 2023.
- Maurice Rappaport. The disability rating and coma/near-coma scales in evaluating severe head injury. *Neuropsychological rehabilitation*, 15(3-4):442–453, 2005.
- James A Reggia. The rise of machine consciousness: Studying consciousness with computational models. *Neural Networks*, 44:112–131, 2013.
- Philip Robbins. Consciousness and the social mind. *Cognitive Systems Research*, 9(1-2):15–23, 2008.
- David Rosenthal. Thinking that one thinks. *Language and Thought*, pages 259–287, 1993.
- David Rosenthal. *Consciousness and mind*. Clarendon Press, 2005.
- David Rosenthal. Higher-order awareness, misrepresentation and function. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 367(1594):1424–1438, 2012.
- David M Rosenthal. Two concepts of consciousness. *Philosophical Studies: An International Journal for Philosophy in the Analytic Tradition*, 49(3):329–359, 1986.
- David M Rosenthal. Varieties of higher-order theory. *Higher-order theories of consciousness*, pages 17–44, 2004.
- David M Rosenthal. Higher-order theories of consciousness. 2009.
- David M Rosenthal et al. *The nature of mind*. Oxford University Press New York, 1991.
- Carl Rosow and Paul J Manberg. Bispectral index monitoring. *Anesthesiology Clinics of North America*, 19(4):947–966, 2001.
- Wesley Sauret and William G Lycan. Attention and internal monitoring: a farewell to hop. *Analysis*, 74(3):363–370, 2014.
- Dennis JLG Schutter and Jack van Honk. Extending the global workspace theory to emotion: Phenomenality without access. *Consciousness and Cognition*, 13(3):539–549, 2004.
- Terrence J Sejnowski. Large language models and the reverse turing test. *Neural computation*, 35(3):309–342, 2023.
- Anil Seth. The strength of weak artificial consciousness. *International Journal of Machine Consciousness*, 1(01):71–82, 2009.

- Daniel N Jones and Delroy L Paulhus. Introducing the short dark triad (sd3) a brief measure of dark personality traits. *Assessment*, 21(1):28–41, 2014.
- Nancy Kanwisher and Ewa Wojciulik. Visual attention: insights from brain imaging. *Nature reviews neuroscience*, 1(2):91–100, 2000.
- Saketh Reddy Karra, Son The Nguyen, and Theja Tulabandhula. Estimating the personality of white-box language models. *arXiv preprint arXiv:2204.12000*, 2022.
- Uriah Kriegel. *Subjective consciousness: A self-representational theory*. OUP Oxford, 2009.
- Victor AF Lamme. Towards a true neural stance on consciousness. *Trends in cognitive sciences*, 10(11):494–501, 2006.
- Victor AF Lamme. How neuroscience will change our view on consciousness. *Cognitive neuroscience*, 1(3):204–220, 2010.
- Victor AF Lamme. Visual functions generating conscious seeing. *Frontiers in Psychology*, 11:83, 2020.
- Hakwan Lau. Theoretical motivations for investigating the neural correlates of consciousness. *Wiley Interdisciplinary Reviews: Cognitive Science*, 2(1):1–7, 2011.
- Hakwan Lau and David Rosenthal. Empirical support for higher-order theories of conscious awareness. *Trends in cognitive sciences*, 15(8):365–373, 2011.
- Hakwan C Lau. A higher order bayesian decision theory of consciousness. *Progress in brain research*, 168:35–48, 2007.
- Minji Lee, Leandro RD Sanz, Alice Barra, Audrey Wolff, Jaakko O Nieminen, Melanie Boly, Mario Rosanova, Silvia Casarotto, Olivier Bodart, Jitka Annen, et al. Quantifying arousal and awareness in altered states of consciousness using interpretable deep learning. *Nature communications*, 13(1):1–14, 2022.
- Janna D Lendner, Randolph F Helfrich, Bryce A Mander, Luis Romundstad, Jack J Lin, Matthew P Walker, Pal G Larsson, and Robert T Knight. An electrophysiological marker of arousal level in humans. *Elife*, 9:e55092, 2020.
- Xingxuan Li, Yutong Li, Linlin Liu, Lidong Bing, and Shafiq Joty. Is gpt-3 a psychopath? evaluating large language models from a psychological perspective. *arXiv preprint arXiv:2212.10529*, 2022.
- Katherine Lin and Michael Wroten. Ranchos los amigos. 2017.
- Grace W Lindsay. Attention in psychology, neuroscience, and machine learning. *Frontiers in computational neuroscience*, 14:29, 2020.
- John Locke. An essay concerning human understanding, 1690. 1948.
- William G Lycan. *Consciousness and experience*. Mit Press, 1996.
- William G Lycan. *The superiority of HOP to HOT*. na, 2004.
- Yuxi Ma, Chi Zhang, and Song-Chun Zhu. Brain in a vat: On missing pieces towards artificial general intelligence in large language models, 2023.
- David JC MacKay. *Information theory, inference and learning algorithms*. Cambridge university press, 2003.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. Self-refine: Iterative refinement with self-feedback. *arXiv preprint arXiv:2303.17651*, 2023.
- Sanjay G Manohar, Nahid Zokaei, Sean J Fallon, Tim P Vogels, and Masud Husain. Neural mechanisms of attending to items in working memory. *Neuroscience & Biobehavioral Reviews*, 101:1–12, 2019.
- Anil K Seth, Zoltán Dienes, Axel Cleeremans, Morten Overgaard, and Luiz Pessoa. Measuring consciousness: relating behavioural and neurophysiological approaches. *Trends in cognitive sciences*, 12(8):314–321, 2008.
- Claude E Shannon. A mathematical theory of communication. *The Bell system technical journal*, 27(3):379–423, 1948.
- Noah Shinn, Beck Labash, and Ashwin Gopinath. Reflexion: an autonomous agent with dynamic memory and self-reflection. *arXiv preprint arXiv:2303.11366*, 2023.
- Henry P Stapp. The importance of quantum decoherence in brain processes. *arXiv preprint quant-ph/0010029*, 2000.
- George L Sternbach. The glasgow coma scale. *The Journal of emergency medicine*, 19(1):67–71, 2000.
- Max Tegmark. Importance of quantum decoherence in brain processes. *Physical review E*, 61(4):4194, 2000.
- Max Tegmark. Consciousness as a state of matter. *Chaos, Solitons & Fractals*, 76:238–270, 2015.
- Max Tegmark. *Life 3.0: Being human in the age of artificial intelligence*. Vintage, 2018.
- Romal Thoppilan, Daniel De Freitas, Jamie Hall, Noam Shazeer, Apoorv Kulshreshtha, Heng-Tze Cheng, Alicia Jin, Taylor Bos, Leslie Baker, Yu Du, et al. Lambda: Language models for dialog applications. *arXiv preprint arXiv:2201.08239*, 2022.
- Bert Timmermans, Leonhard Schilbach, Antoine Pasquali, and Axel Cleeremans. Higher order thoughts in action: consciousness as an unconscious re-description process. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 367(1594):1412–1423, 2012.
- Giulio Tononi. An information integration theory of consciousness. *BMC neuroscience*, 5(1):1–22, 2004.
- Giulio Tononi and Christof Koch. Consciousness: here, there and everywhere? *Philosophical Transactions of the Royal Society B: Biological Sciences*, 370(1668):20140167, 2015.
- Giulio Tononi and Olaf Sporns. Measuring information integration. *BMC neuroscience*, 4:1–20, 2003.
- Giulio Tononi, Melanie Boly, Marcello Massimini, and Christof Koch. Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7):450–461, 2016.
- Paul D Trapnell and Jennifer D Campbell. Private self-consciousness and the five-factor model of personality: distinguishing rumination from reflection. *Journal of personality and social psychology*, 76(2):284, 1999.
- Stefan Treue and Julio C Martinez Trujillo. Feature-based attention influences motion processing gain in macaque visual cortex. *Nature*, 399(6736):575–579, 1999.
- Alan M Turing. Computing machinery and intelligence. In *Parsing the turing test*, pages 23–65. Springer, 2009.
- Robert Van Gulick. Higher-order global states (hogs): an alternative higher-order model. *Higher-order theories of consciousness*, 2004:67–93, 2004.
- Rufin VanRullen and Ryota Kanai. Deep learning and the global workspace theory. *Trends in Neurosciences*, 44(9):692–704, 2021.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

- Marcello Massimini, Fabio Ferrarelli, Reto Huber, Steve K Esser, Harpreet Singh, and Giulio Tononi. Breakdown of cortical effective connectivity during sleep. *Science*, 309(5744):2228–2232, 2005.
- Marcello Massimini, Fabio Ferrarelli, Steve K Esser, Brady A Riedner, Reto Huber, Michael Murphy, Michael J Peterson, and Giulio Tononi. Triggering sleep slow waves by transcranial magnetic stimulation. *Proceedings of the National Academy of Sciences*, 104(20):8496–8501, 2007.
- Bjorn Merker. The liabilities of mobility: A selection pressure for the transition to consciousness in animal evolution. *Consciousness and cognition*, 14(1):89–114, 2005.
- Bjorn Merker. Consciousness without a cerebral cortex: A challenge for neuroscience and medicine. *Behavioral and brain sciences*, 30(1):63–81, 2007.
- Isabel Briggs Myers. *Myers-Briggs type indicator: MBTI*.
- Lionel Naccache. Why and how access consciousness can account for phenomenal consciousness. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 373(1755):20170357, 2018.
- OpenAI. Gpt-4 technical report, 2023.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35: 27730–27744, 2022.
- Antoine Pasquali, Bert Timmermans, and Axel Cleeremans. Know thyself: Metacognitive networks and measures of consciousness. *Cognition*, 117(2):182–190, 2010.
- Roger Penrose. The emperor’s new mind. *RSA Journal*, 139(5420):506–514, 1991.
- Roger Penrose. *Shadows of the Mind*, volume 4. Oxford University Press Oxford, 1994.
- Navindra Persaud, Peter McLeod, and Alan Cowey. Post-decision wagering objectively measures awareness. *Nature neuroscience*, 10(2):257–261, 2007.
- Vincent Picciuto. Addressing higher-order misrepresentation with quotational thought. *Journal of Consciousness Studies*, 18(3-4):109–136, 2011.
- Hilary Putnam et al. *Reason, truth and history*, volume 3. Cambridge University Press, 1981.
- Haocong Rao, Cyril Leung, and Chunyan Miao. Can chatgpt assess human personalities? a general evaluation framework. *arXiv preprint arXiv:2303.01248*, 2023.
- Maurice Rappaport. The disability rating and coma/near-coma scales in evaluating severe head injury. *Neuropsychological rehabilitation*, 15(3-4):442–453, 2005.
- James A Reggia. The rise of machine consciousness: Studying consciousness with computational models. *Neural Networks*, 44:112–131, 2013.
- Philip Robbins. Consciousness and the social mind. *Cognitive Systems Research*, 9(1-2):15–23, 2008.
- David Rosenthal. Thinking that one thinks. *Language and Thought*, pages 259–287, 1993.
- David Rosenthal. *Consciousness and mind*. Clarendon Press, 2005.
- David Rosenthal. Higher-order awareness, misrepresentation and function. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 367(1594):1424–1438, 2012.
- David M Rosenthal. Two concepts of consciousness. *Philosophical Studies: An International Journal for Philosophy in the Analytic Tradition*, 49(3):329–359, 1986.
- David M Rosenthal. Varieties of higher-order theory. *Higher-order theories of consciousness*, pages 17–44, 2004.
- H Kenneth Walker, W Dallas Hall, and J Willis Hurst. Clinical methods: the history, physical, and laboratory examinations. 1990.
- Daniel M Wegner. Précis of the illusion of conscious will. *Behavioral and Brain Sciences*, 27(5):649–659, 2004.
- Lilian Weng. The transformer family version 2.0. *lilianweng.github.io*, Jan 2023. URL <https://lilianweng.github.io/posts/2023-01-27-the-transformer-family-v2/>.
- M Wieser, L Buetler, A Koenig, and R Riener. Quantitative description of the state of awareness of patients in vegetative and minimally conscious state. In *2010 Annual International Conference of the IEEE Engineering in Medicine and Biology*, pages 5533–5536. IEEE, 2010.
- Eelco FM Wijdicks, William R Bamlet, Bobby V Maramattom, Edward M Manno, and Robyn L McClelland. Validation of a new coma scale: the four score. *Annals of Neurology: Official Journal of the American Neurological Association and the Child Neurology Society*, 58(4):585–593, 2005.

- David M Rosenthal. Higher-order theories of consciousness. 2009.
- David M Rosenthal et al. *The nature of mind*. Oxford University Press New York, 1991.
- Carl Rosow and Paul J Manberg. Bispectral index monitoring. *Anesthesiology Clinics of North America*, 19(4):947–966, 2001.
- Wesley Sauret and William G Lycan. Attention and internal monitoring: a farewell to hop. *Analysis*, 74(3):363–370, 2014.
- Dennis JLG Schutter and Jack van Honk. Extending the global workspace theory to emotion: Phenomenality without access. *Consciousness and Cognition*, 13(3):539–549, 2004.
- Terrence J Sejnowski. Large language models and the reverse turing test. *Neural computation*, 35(3):309–342, 2023.
- Anil Seth. The strength of weak artificial consciousness. *International Journal of Machine Consciousness*, 1(01):71–82, 2009.
- Anil K Seth, Zoltán Dienes, Axel Cleeremans, Morten Overgaard, and Luiz Pessoa. Measuring consciousness: relating behavioural and neurophysiological approaches. *Trends in cognitive sciences*, 12(8):314–321, 2008.
- Claude E Shannon. A mathematical theory of communication. *The Bell system technical journal*, 27(3):379–423, 1948.
- Noah Shinn, Beck Labash, and Ashwin Gopinath. Reflexion: an autonomous agent with dynamic memory and self-reflection. *arXiv preprint arXiv:2303.11366*, 2023.
- Henry P Stapp. The importance of quantum decoherence in brain processes. *arXiv preprint quant-ph/0010029*, 2000.
- George L Sternbach. The glasgow coma scale. *The Journal of emergency medicine*, 19(1):67–71, 2000.
- Max Tegmark. Importance of quantum decoherence in brain processes. *Physical review E*, 61(4):4194, 2000.
- Max Tegmark. Consciousness as a state of matter. *Chaos, Solitons & Fractals*, 76:238–270, 2015.
- Max Tegmark. *Life 3.0: Being human in the age of artificial intelligence*. Vintage, 2018.
- Romal Thoppilan, Daniel De Freitas, Jamie Hall, Noam Shazeer, Apoorv Kulshreshtha, Heng-Tze Cheng, Alicia Jin, Taylor Bos, Leslie Baker, Yu Du, et al. Lambda: Language models for dialog applications. *arXiv preprint arXiv:2201.08239*, 2022.
- Bert Timmermans, Leonhard Schilbach, Antoine Pasquali, and Axel Cleeremans. Higher order thoughts in action: consciousness as an unconscious re-description process. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 367(1594):1412–1423, 2012.
- Giulio Tononi. An information integration theory of consciousness. *BMC neuroscience*, 5(1):1–22, 2004.
- Giulio Tononi and Christof Koch. Consciousness: here, there and everywhere? *Philosophical Transactions of the Royal Society B: Biological Sciences*, 370(1668):20140167, 2015.
- Giulio Tononi and Olaf Sporns. Measuring information integration. *BMC neuroscience*, 4:1–20, 2003.
- Giulio Tononi, Melanie Boly, Marcello Massimini, and Christof Koch. Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7):450–461, 2016.
- Paul D Trapnell and Jennifer D Campbell. Private self-consciousness and the five-factor model of personality: distinguishing rumination from reflection. *Journal of personality and social psychology*, 76(2):284, 1999.

- Stefan Treue and Julio C Martinez Trujillo. Feature-based attention influences motion processing gain in macaque visual cortex. *Nature*, 399(6736):575–579, 1999.
- Alan M Turing. Computing machinery and intelligence. In *Parsing the turing test*, pages 23–65. Springer, 2009.
- Robert Van Gulick. Higher-order global states (hogs): an alternative higher-order model. *Higher-order theories of consciousness*, 2004:67–93, 2004.
- Rufin VanRullen and Ryota Kanai. Deep learning and the global workspace theory. *Trends in Neurosciences*, 44(9):692–704, 2021.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- H Kenneth Walker, W Dallas Hall, and J Willis Hurst. Clinical methods: the history, physical, and laboratory examinations. 1990.
- Daniel M Wegner. Précis of the illusion of conscious will. *Behavioral and Brain Sciences*, 27(5): 649–659, 2004.
- Lilian Weng. The transformer family version 2.0. *lilianweng.github.io*, Jan 2023. URL <https://lilianweng.github.io/posts/2023-01-27-the-transformer-family-v2/>.
- M Wieser, L Buetler, A Koenig, and R Riener. Quantitative description of the state of awareness of patients in vegetative and minimally conscious state. In *2010 Annual International Conference of the IEEE Engineering in Medicine and Biology*, pages 5533–5536. IEEE, 2010.
- Eelco FM Wijdicks, William R Bamlet, Boby V Maramattom, Edward M Manno, and Robyn L McClelland. Validation of a new coma scale: the four score. *Annals of Neurology: Official Journal of the American Neurological Association and the Child Neurology Society*, 58(4):585–593, 2005.